AMENDMENT Reference No AMD-887648-6

Grant Agreement number: 887648 — Development of innovative biotic symbiosis for plastic biodegradation and synthesis to solve their end of life challenges in the agriculture and food industries (RECOVER)

The parties agree to amend the Grant Agreement as follows ('**Amendment**'):

1 . Removal of a beneficiary whose participation was terminated (for other reasons)

The participation of the following beneficiary has been terminated:

- ORGANIC WASTE SYSTEMS NV (OWS) - on the day after the notification of termination

This implies the **following changes** to the Grant Agreement:

- The 'termination date' is added, for the beneficiary, in the **Preamble**:

  ORGANIC WASTE SYSTEMS NV (OWS), established in DOK NOORD 5, GENT 9000, Belgium, VAT number: BE0433270195, - until termination date

In accordance with Article 50 of the Grant Agreement, the beneficiary's obligations continue to apply after termination.

2 . Addition of a new beneficiary

The following new beneficiary is added:

- OWS RESEARCH FOUNDATION PS (OWS RF) — as from 15 June 2021

This implies the **following changes** to the Grant Agreement:

- The new beneficiary and the 'accession date' is added to the **Preamble**:

  *18. "**OWS RESEARCH FOUNDATION PS (OWS RF)**, established in EIKELDREEF 29, SINT-MARTENS-LATEM 9830, Belgium, — as from 15 June 2021"*

3. Change of Annex 1 (description of the action)

Annex 1 is changed and replaced by the Annex 1 attached to this Amendment.

4 . Changes of Annex 2 (estimated budget)

Annex 2 is changed and replaced by the Annex 2 attached to this Amendment.

All other provisions of the Grant Agreement and its Annexes remain unchanged.

This Amendment **enters into force** on the day of the last signature.

This Amendment **takes effect** on the date on which the amendment enters into force, except where a different date has been agreed by the parties (for one or more changes).

Please inform the other members of the consortium of the Amendment.

SIGNATURES

For the coordinator

For the JU

Enclosures:

Annex 2
Annex 1

ANNEX 1 (part A)

Bio-based Industries Research and Innovation action

NUMBER — 887648 — RECOVER

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1.1. The project summary

Project Number [1]	887648	Project Acronym [2]	RECOVER

One form per project

General information	
Project title [3]	Development of innovative biotic symbiosis for plastic biodegradation and synthesis to solve their end of life challenges in the agriculture and food industries
Starting date [4]	01/06/2020
Duration in months [5]	48
Call (part) identifier [6]	H2020-BBI-JTI-2019
Topic	BBI-2019-SO2-R3 Apply microorganisms and/or enzymes to resolve end-of-life issues of plastics
Fixed EC Keywords	Circular economy, Environmental biotechnology, bioremediation, biodegradation, Microbiology, Enzymology
Free keywords	Agri-food waste plastics; microplastics; microorganisms; enzymes; insects; earthworms; biodegradation modeling; chitin-based biopolymer; eBeam; compost; biofertilizer; mulching film

Abstract [7]

RECOVER is a disruptive RIA proposal built upon the actual needs to solve the contamination of agro-fields with non-biodegradable agro-plastics as well as to improve municipal waste handling by decreasing drastically the packaging fraction going to unsustainable waste management routes.
We intend to reach our ambitious objectives applying biotech solutions that will include symbiotic bioprocesses between previously reported high powered microorganisms through microbiotic systems vectored by several insect species. Indeed, one of the main innovations in our proposal is the combination of endogenous and exogenous microorganisms and their inoculation on insects. This will allow achieving higher levels of biodegradation and direct conversion (at the same time) of agri-food plastic wastes in insect by-products such as chitin, that can be converted in chitosan (high value as bioplastics raw materials and other interesting industrial applications due to anti-microbial activity) that will be used in active packaging, enhanced mulching films and biofertilizers applications.
RECOVER will help providing novel biotechnological solutions applying microorganisms, enzymes and insects to degrade conventional plastic packaging and agricultural films waste streams but also result in new feedstocks for the bio-based industries. The process will be optimized and performed either ex-situ in composting reactors (preferred route for non-recyclable plastics from municipal solid waste or for agricultural films that can be collected easily) or in-situ in the case of bioremediating soil pollution, by e.g. mulching films, further contributing to the long-term removal of non-biodegradable polymers from the environment. In addition, RECOVER can solve the microplastics pollution at the level of both industrial composting and soil.

1.2. List of Beneficiaries

Project Number [1]	887648	Project Acronym [2]	RECOVER

List of Beneficiaries

No	Name	Short name	Country	Project entry date[8]	Project exit date
1	UNIVERSIDAD DE ALMERIA	UAL	Spain		
2	UNIVERSITA DI PISA	UNIPI	Italy		
3	ASOCIACIÓN AGRARIA DE JÓVENES AGRICULTORES	ASAJA	Spain		
4	BRUNEL UNIVERSITY LONDON	BRUNEL	United Kingdom		
5	UNIVERSIDAD MIGUEL HERNANDEZ DE ELCHE	UMH	Spain		
6	FACHHOCHSCHULE ALBSTADT-SIGMARINGEN	ASU	Germany		
7	NATURPLAS PLASTICOS AGRICOLAS SL	NATURPLAS	Spain		
8	ASA SPEZIALENZYME GMBH	ASA	Germany		
9	NUTRINSECT SRL	NUTRINSECT	Italy		
10	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	IRIS	Spain		
11	ORGANIC WASTE SYSTEMS NV	OWS	Belgium		the day after the notification of termination
12	IDELUX ENVIRONNEMENT	IDELUX	Belgium		
13	FEMTO ENGINEERING SRL	FEMTO	Italy		
14	INGREDIENT ODYSSEY SA	ENTO	Portugal		
15	CARTON BROS	MANOR	Ireland		
16	ENCO SRL	ENCO	Italy		
17	SOCIEDAD ANONIMA AGRICULTORES DE LAVEGA DE VALENCIA	SAV	Spain		
18	OWS RESEARCH FOUNDATION PS	OWS RF	Belgium	15/06/2021	

1.3. Workplan Tables - Detailed implementation

1.3.1. WT1 List of work packages

WP Number[9]	WP Title	Lead beneficiary[10]		Start month[12]	End month[13]
WP1	Agro-plastic waste (APW) mapping, characterization, collection strategy and pre-treatment	12 - IDELUX	[illegible]	1	12
WP2	Selection of biotech tools for AWP biodegradation	1 - UAL	[illegible]	1	24
WP3	Up-scaled production of biotech tools for APW biodegradation	5 - UMH	[illegible]	12	24
WP4	Pilot scale ex and in situ degradation of municipal and rural APW	18 - OWS RF	[illegible]	18	36
WP5	Validation of value-added downstream products	2 - UNIPI	[illegible]	24	48
WP6	Safety, environmental, social and economic sustainability assessments	3 - ASAJA	[illegible]	1	48
WP7	Cross-cutting activities and Innovation impats	16 - ENCO	[illegible]	1	48
WP8	Project Management	1 - UAL	[illegible]	1	48
WP9	Ethics requirements	1 - UAL	[illegible]	1	48
		Total	[illegible]		

1.3.2. WT2 list of deliverables

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Mapping of AWP and their characteristics	WP1	12 - IDELUX	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D1.2	Report on AWP collection and monitoring strategies	WP1	2 - UNIPI	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D1.3	Report on AWP pre-treatment options to enhance biodegradation of AWP	WP1	5 - UMH	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D2.1	Report on microorganisms and enzymes for AWP degradation	WP2	1 - UAL	Report	Confidential, only for members of the consortium (including the Commission Services)	20
D2.2	Report on novel synthetic enzymes	WP2	4 - BRUNEL	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D2.3	Report on selection and pre-compatibility test of earthworm/insects-micro consortia	WP2	5 - UMH	Report	Confidential, only for members of the consortium (including the Commission Services)	20
D3.1	Formulations of PMC to be used as enhancers of AWP degradation	WP3	8 - ASA	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D3.2	Protocol for the production of insect and earthworm plus PMC with enhanced AWP degrading capability	WP3	5 - UMH	Report	Confidential, only for members of the consortium (including the Commission Services)	24

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D4.1	Report on operational conditions for enhance bioremediation process	WP4	18 - OWS RF	Report	Confidential, only for members of the consortium (including the Commission Services)	30
D4.2	Report on upscaled ex situ and in situ AWP biodegradation process	WP4	12 - IDELUX	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D4.3	Biodegradation monitoring and modelling	WP4	10 - IRIS	Other	Confidential, only for members of the consortium (including the Commission Services)	36
D5.1	Report on up-graded biofertilizer products	WP5	14 - ENTO	Report	Confidential, only for members of the consortium (including the Commission Services)	26
D5.2	Report on eBeam-related optimizations	WP5	6 - ASU	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D5.3	Report on chitin-based bioplastics	WP5	2 - UNIPI	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	42
D5.4	Optimized mulching films	WP5	2 - UNIPI	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D6.1	Report of the mass, energy balance and techno-economic studies	WP6	3 - ASAJA	Report	Confidential, only for members of the consortium (including the Commission Services)	30

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D6.2	Interim report of the carbon foot- print and LCA	WP6	18 - OWS RF	Report	Confidential, only for members of the consortium (including the Commission Services)	20
D6.3	Final report of the carbon foot- print and LCA	WP6	18 - OWS RF	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D6.4	Report on risk assessment including social acceptance assessments	WP6	3 - ASAJA	Report	Confidential, only for members of the consortium (including the Commission Services)	32
D6.5	RECOVER logistic approach	WP6	2 - UNIPI	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D7.1	Project website	WP7	16 - ENCO	Websites, patents filing, etc.	Public	4
D7.2	Communication Plan	WP7	3 - ASAJA	Report	Public	6
D7.3	Draft Plan for the Exploitation and Dissemination of the Results	WP7	16 - ENCO	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D7.4	Report on RECOVER value chain	WP7	16 - ENCO	Report	Public	12
D7.5	Final Plan for the Exploitation and Dissemination of the Results	WP7	16 - ENCO	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D7.6	Industry, stakeholders and policy guidelines regarding RECOVER waste management	WP7	3 - ASAJA	Report	Public	48
D8.1	Project Management Tool	WP8	16 - ENCO	Other	Confidential, only for members	1

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
					of the consortium (including the Commission Services)	
D8.2	Project management Plan	WP8	1 - UAL	Report	Confidential, only for members of the consortium (including the Commission Services)	3
D8.3	Report on National Ethics Councils (NEC) requirements	WP8	1 - UAL	Report	Confidential, only for members of the consortium (including the Commission Services)	3
D8.4	Report on Ethic Position Questionnaire (EPQ) requirements	WP8	1 - UAL	Report	Confidential, only for members of the consortium (including the Commission Services)	3
D8.5	Data management plan	WP8	16 - ENCO	ORDP: Open Research Data Pilot	Confidential, only for members of the consortium (including the Commission Services)	6
D8.6	First Report from Advisory Board	WP8	16 - ENCO	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D8.7	Second report from Advisory Board	WP8	16 - ENCO	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D8.8	KPI & Impact Questionnaire 1	WP8	1 - UAL	Other	Confidential, only for members of the consortium (including the Commission Services)	16
D8.9	KPI & Impact Questionnaire 2	WP8	1 - UAL	Other	Confidential, only for members of the consortium (including the	28

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
					Commission Services)	
D8.10	KPI & Impact Questionnaire 3	WP8	1 - UAL	Other	Confidential, only for members of the consortium (including the Commission Services)	40
D8.11	KPI & Impact Questionnaire 4	WP8	1 - UAL	Other	Confidential, only for members of the consortium (including the Commission Services)	48
D9.1	H - Requirement No. 1	WP9	1 - UAL	Ethics	Confidential, only for members of the consortium (including the Commission Services)	6
D9.2	POPD - Requirement No. 2	WP9	1 - UAL	Ethics	Confidential, only for members of the consortium (including the Commission Services)	6

1.3.3. WT3 Work package descriptions

Work package number [9]	WP1	**Lead beneficiary** [10]	12 - IDELUX
Work package title	Agro-plastic waste (APW) mapping, characterization, collection strategy and pre-treatment		
Start month	1	**End month**	12

Objectives

● To quantify the currently non recovered Agri-food plastic waste (AWP) depending on regional distribution, AWP nature, application.
● To characterize the different types of plastics used in agriculture and agri-food sector and the effect of ageing in AWP
● To select the proper AWP types for the project
● To diagnose the current collection and waste management approach and propose viable new ones in the context of RECOVER value chains
● To study and validate feasible pre-treatments to enhance AWP degradation.

Description of work and role of partners

WP1 - Agro-plastic waste (APW) mapping, characterization, collection strategy and pre-treatment [Months: 1-12]
IDELUX, UAL, UNIPI, ASAJA, UMH, ASU, NATURPLAS, IRIS, FEMTO, ENTO, MANOR, ENCO, SAV
Task 1.1. AWP inventory and characterisation (M1-M6)
A wide range of plastics is commonly used in agriculture and agri-food sectors. First of all, the literature data will be reviewed to establish a mapping of the agricultural films use vs. country. In terms of the food plastic packaging, the rough composition of the currently non recycled mixed plastic packaging fractions among the relevant municipal solid waste streams will be inventoried for the most representative EU countries.
A complete collection of samples (>30) will be developed by IDELUX, NATURPLAS, UNIPI, and ASAJA and analyses will be carried out to characterise the main variables that are expected to affect the biodegradation capacities (polymer type, polymerization grade, presence of plasticizers and other chemical additives, elemental composition, functional groups on the surface, hydrophobicity/hydrophilicity, crystallinity, surface topography, mechanical properties, molecular weight distribution and mass balance, etc.). Plastic samples will be classified depending on plastic type (PE, PS), plastic use (mulching, green-housing, packaging) and ageing (raw & wasted samples, AWP).
Leader: IDELUX, Partners involved: UNIPI, ASAJA, NATURPLAS, MANOR, SAV, ENTO

Task 1.2. Selection of AWP to be used into the project (M3-M12)
In this subtask, a selection of 4-6 types of most representative fossil plastics (including in each case virgin plastics and AWP counterparts picked individually from MSW or agricultural films) will be made in order to provide normalized samples to all the WPs. Based on the results of the previous task, the selection criteria will be centred to cover the most significant plastic in agricultural and food packaging environments, the trends, the review in terms of state of the art for the research on the use of micro/macro organisms for their degradation, etc.
Leader: UNIPI. Partners involved: NATURPLAS, SAV, ENCO, UAL, UMH, FEMTO

Task 1.3. Strategy for AWP collection and sorting (M3-M12)
The current practices in terms of end of life handling of the selected AWP will be reviewed under the lead of ASAJA in case of agricultural plastics and of IDELUX in case of food packaging. A strategy for the targeted plastic waste collection and delivery to the appropriate degradation plant will be proposed. The cases where a collection is not seen viable and an in situ (soil, compost) degradation is preferable will be highlighted.
For the selected AWP types, it will be developed, by UNIPI, a logistic chain analysis, determining costs and involved stakeholders for the phases of collection, transport, treatment and re-use. From this Base Scenario analysis a general strategy for the reconstruction of a Project Scenario to be better delineated in WP6 will be designed.
In view of a possible automatic sorting of the AWP compatible with the RECOVER end of life management , the previous samples will be supplied to IRIS for near infrared spectroscopy (NIR), hyper spectral imaging (HSI) in visible (VIS), and short wave infra-red (SWIR) spectral ranges to test the accuracy of the distinction among different materials thanks to digital image processing and chemometric analysis of the obtained spectra. Contaminated streams which could affect the biodegradation, e.g. with pesticides, will also be analysed to determine the accuracy of their detection. The

combination of the results obtained from several techniques may also be tested. The most suitable approach to implement will be recommended for post project implementation. The outcome information will help correlating the composition of the waste stream with the efficiency of the treatment.
Leader: IRIS. Partners involved: ASAJA, IDELUX, UNIPI, UAL, FEMTO

Task 1.4. Pre-treatment of AWP (M3-M12)
In this task, several abiotic pretreatment (fragmentation, micronization, thermal treatments, UV irradiation, oxidizing agents, pH, irradiation by eBeam to initiate chain scission) will be done to enhance AWP biodegradation of selected samples from Task 1.2. One approach in abiotic pre-treatment of AWP will be accomplished by eBeam treatment (performed by ASU). Considering that when polymers are irradiated material degradation and cross-linking occur simultaneously, and that application of lower doses results in enhanced cross-linking of the material whereas the use of higher doses leads to material degradation AWP from Task 1.2 will be irradiated with a wide range of doses (50 – 200 kGy) to initiate chain scission and subsequent degradation. The effect of the pre-treatments will be quantified using techniques from Task 1.1. An efficiency and cost-benefit pre-analyses will be done to select the pre-treatment procedures to be applied on the collected samples used in the following WPs.
Leader: UMH. Partners involved: UNIPI, UAL, ASU, ENTO

Participation per Partner

Partner number and short name	WP1 effort
1 - UAL	
2 - UNIPI	
3 - ASAJA	
5 - UMH	
6 - ASU	
7 - NATURPLAS	
10 - IRIS	
12 - IDELUX	
13 - FEMTO	
14 - ENTO	
15 - MANOR	
16 - ENCO	
17 - SAV	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Mapping of AWP and their characteristics	12 - IDELUX	Report	Confidential, only for members of the consortium (including the Commission Services)	6

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.2	Report on AWP collection and monitoring strategies	2 - UNIPI	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D1.3	Report on AWP pre-treatment options to enhance biodegradation of AWP	5 - UMH	Report	Confidential, only for members of the consortium (including the Commission Services)	12

Description of deliverables

D1.1 : Mapping of AWP and their characteristics [6]

Samples and Written report linked to tasks 1.1 and 1.2 including the specifications of AWPs and their availability as well as a selection matrix of 4-6 AWP for later project activities

D1.2 : Report on AWP collection and monitoring strategies [12]

Written report on the outcome of task 1.3

D1.3 : Report on AWP pre-treatment options to enhance biodegradation of AWP [12]

Written report on the outcome of task 1.4 and samples of 4-6 selected AWP after pre-treatment

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS1	Mapping of AWP and their characteristics	12 - IDELUX	6	AWP inventory and characterization ready as a basis for the selection of samples of 4-6 AWP types (including in each case virgin plastics and AWP counterparts picked individually from MSW or agricultural films) for later project activities.
MS2	Strategy for AWP collection, monitoring & sorting	2 - UNIPI	12	Matrix for the definition of the classification, characterization and procedure for obtaining AWP available (will feed in WP6).
MS3	Pre-treatment rules and procedures basing on composition of mixtures of AWP	10 - IRIS	12	Abiotic pre-treatment to be used depending on the mixture of AWP to biodegrade (fragmentation, micronization, thermal treatments, UV irradiation,

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
				oxidizing agents, pH and irradiation) defined and samples of 4-6 selected AWP after pretreatment available

Work package number [9]	WP2	Lead beneficiary [10]	1 - UAL
Work package title	Selection of biotech tools for AWP biodegradation		
Start month	1	End month	24

Objectives

- To select the microbial communities for AWP biodegradation
- To develop novel synthetic enzymes for AWP biodegradation
- To select the most suitable earthworms and insects for AWP biodegradation
- To demonstrate pre-compatibility of micro and earthworm/insect consortia for AWP biodegradation

Description of work and role of partners

WP2 - Selection of biotech tools for AWP biodegradation [Months: 1-24]
UAL, UNIPI, BRUNEL, UMH, ASA, NUTRINSECT, IRIS, OWS, IDELUX, ENTO, OWS RF
Task 2.1. Selection and building of microbial communities for AWP degradation (M1-20)
Microorganisms for the biodegradation of AWP (from WP1) will be selected among the strains available at UAL and SynbiSTRAIN BRUNEL microbial collections. SynbiSTRAIN collection includes over 400 Pseudomonas strains (from 3 species), tens of composting Geobacillus strains, and Bacillus species; each of which are documented to contain plastic degradative properties. These sequenced collections are built to maximally represent species evolutionary and behavioural diversity. The potential candidates from UAL microbial collection will be also sequenced. Combinations of strains will be selected and tested. Because the strains to be tested are or will be fully sequenced, it will be possible to study population enrichment of mixed cultures to identify strains with greater capacity to use AWP as carbon source; and those that are associated with greater degradative performance. These will be identified using pre- and post-culture re-sequencing and comparisons of the starting and finishing sub-populations; in parallel cultures using the selected plastic substrates will be performed. From this, strains will be selected for iterative rounds of selection and consortia performance optimization according to their final use in WP4.
In parallel, the dominant microbiomes of the earthworms and insect larvae determined in Task 2.4 and strains isolated and enriched from AWP-enriched soils will be characterized by sequencing. The species will be used to inform selection of strains for consortia building with macro-organisms, and new strains from enrichment will be combined with existing strain resources based upon comparisons of performance, and associated enzymes used to inform new enzyme development (Task 2.2).
UAL will be in charge of community development and strain selection. BRUNEL will perform sequencing and genomic analysis. UMH and UAL will provide earthworms/vermicompost and insect samples from Task 2.3, respectively. The extent of AWP modification/mineralization will be analysed by means of techniques described in WP1 and WP4 by IRIS. UNIPI will perform morphological (SEM), thermal (TGA, DSC), and chemical (FTIR, GPC) analysis on AWP.
Leader: UAL. Partners involved: BRUNEL, UMH, ASAJA, IDELUX, UNIPI

Task 2.2. Design and development of novel synthetic enzymes for AWP biodegradation (M1-M24)
The BRUNEL SynbiSTRAIN collection sequences, and public sequence resources for single strain and metagenomic datasets will be screened for all members of enzyme families with known plastic-degradative properties (targets led by selection process in WP4). From these a new strategy will be developed and implemented for selected / directed evolution based upon synthetic DNA templates. These will be designed to simultaneously optimize the in-vitro recombination of templates and represent the naturally occurring variation within enzymes that affect resulting proteins. Using this strategy the coding space of the enzyme families will be explored in a focused fashion to combine independently evolved enzyme optimizations from different lineages of the same species, and from independent lineages that would have no capabilities to recombine and co-evolve enzymes in nature; through the use of metagenomic data it will also include genes and alleles that are derived from non-cultivable strains and species.
The growth of strains (of E. coli) carrying novel enzymes will be determined on media containing indicators, plastic-based carbon sources, and on plastic surfaces (as determined by the AWP substrates selected in WP1) to select and enrich for novel enzymes and variants with high activity. These will variably be present in single colonies or mixtures. The optimum enzymes will be identified by population enrichment on these substrates, followed by population re-sequencing of plasmid-extracts. The most abundant enzyme versions will be identified within these populations (through an in-house method of 'in silico subtractive hybridization' that first identifies the variants within a population, and then determines their proportions in the population), this process will also be used to identify secondary adaptive mutation occurring within the enzymes that contribute to their adaptation to the combination of previously separately occurring

variation. Based upon the results of this one of two or both strategies will follow: 1. New synthetic templates will be generated that preferentially contain variations that have been selected through selection and recombination to attempt further enzyme improvement, including secondary internal adaptations; and 2. Final enzymes present in mixtures will be specifically synthesized for pure enzyme production.
At a final stage, it is envisaged that a re-coded version of the enzymes optimized for production in commercial strains / species (e.g. Bacillus will be generated). This or other process optimization steps will be selected based upon progress and needs determined through the larger program of research. Additional stages of optimization for pH, temperature, or other process / production requirements may also be pursued in WP3 depending upon the wider needs and progress of the project. Partners UNIPI, OWS, IRIS will perform analysis on AWP before and after degradation.
Leader: BRUNEL. Partners involved: UAL, ASA, OWS, IRIS, UNIPI
Task 2.3. Preselection of earthworm and insect larvae AWP-degraders (M1-M20)
The potential polymer-degraders in composting earthworms (Eisenia andrei) and soil-dwelling earthworms (e.g., Lumbricus terrestris) will be promoted by incubation in plastic-spiked feedstock in microcosm trials. The materials deployed by L. terrestris (middens, burrow linings and casts) and E. andrei (vermicast and gut symbionts) and their gastrointestinal tract will be used for microbiome analysis in Task 2.4. UMH will monitor earthworm activity by the measurement of key enzymatic activities and in vitro degradation of plastics.
Preselection of insects will be done at coupling bibliographic assessment and degradation tests of AWP by WP1 quantification biodegradation methods. Specific varieties initially considered are: Acheta domestica (Orthoptera: Gryllidae), Galleria mellonella (Lepidoptera: Pyralidae), Ephestia kuehniella (Lepidoptera: Pyralidae), Plodia interpunctella (Lepidoptera: Pyralidae). Sitotroga cerealella (Lepidoptera: Gelechiidae), Tenebrio molitor (Coleoptera: Tenebrionidae),Tribolium castaneum (Coleoptera: Tenebrionidae), and Sitophilus granarius (Coleoptera: Dryophthoridae). At least 4-6 species of insects will be selected for trials to be performed in Task 2.4 and 2.5 according to their capability to biodegrade AWP and management aspects (life-cycle, specific conditions etc.).
UMH and UAL will perform lab scale experiments with earthworms and insect larvae, respectively. ASAJA and NUTRINSECT will provide insects. ASAJA and IDELUX will provide AWP. UNIPI will perform analysis of AWP as defined in WP1, for FTIR GPC, SEM, TGA etc.
Leader: UMH. Partners involved: BRUNEL, UAL, ASAJA, IDELUX, NUTRINSECT, IRIS

Task 2.4. Analysis of microbial community associated with earthworm and insect (M1-M20)
The microbiomes of the earthworms and insect larvae fed on AWP (Task 2.3) will be determined in order to know which are the most abundant species in their natural colonizing populations and thus identify the ones most bio-compatible with them. Earthworms and insect larvae will be placed on sterile operating board to allow collection of their evacuated gut contents. In addition, earthworms and insect larvae will be dissected under sterile conditions. DNA of samples (gastrointestinal tract and excreta) will be extracted and sequenced. The information provided will be used for microorganisms selection & building microbial communities in Task 2.1.
Leader: BRUNEL. Partners involved: UAL, UMH

Task 2.5. Pre-compatibility analysis of earthworm/insect-microorganisms (M1-M20)
Microorganisms and enzyme pools selected in Task 2.1 and 2.2 will be added to earthworms (Eisenia andrei and Lumbricus terrestris) and insects larvae selected in Task 2.3 and their compatibility will be tested by monitoring insect larvae, earthworms, and microorganism survival as well as enzyme activity stability. Microorganisms aimed at enriching gastrointestinal tract of earthworms or insect (endo-micro) will be used to feed them. Microorganisms/enzymes aimed at adding exogenous abilities (exo-micro/enz) will be externally added to earthworms and insects. These experiments will be performed at lab-scale under specific culture conditions for each earthworm and insect species. For endo-micro, samples of digestive content will be periodically checked to ensure symbiosis of the selected microbiota in the luminal microenvironment of both earthworms and insect larvae. This task will provide a final set of 2-3 compatible microbial/ enzyme pool for each earthworm or insect that will be used to obtain insect/earthworms with fortified plastic degrading capabilities (FP insect/earthworms) in WP3. UMH and UAL will perform earthworms/insect incubation, and survival tests. UAL and BRUNEL will provide microorganisms and enzymes and will monitor microbial survival and enzyme stability and UMH will test key enzymes in earthworm/insect excreta. ASAJA and NUTRINSECT will provide insects.
Leader: UMH. Partners involved: UAL, BRUNEL, NUTRINSECT, ASAJA, ENTO.

Participation per Partner

Partner number and short name	WP2 effort
1 - UAL	26.00

Partner number and short name	WP2 effort
2 - UNIPI	
4 - BRUNEL	
5 - UMH	
8 - ASA	
9 - NUTRINSECT	
10 - IRIS	
11 - OWS	
12 - IDELUX	
14 - ENTO	
18 - OWS RF	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D2.1	Report on microorganisms and enzymes for AWP degradation	1 - UAL	Report	Confidential, only for members of the consortium (including the Commission Services)	20
D2.2	Report on novel synthetic enzymes	4 - BRUNEL	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D2.3	Report on selection and pre-compatibility test of earthworm/insects-micro consortia	5 - UMH	Report	Confidential, only for members of the consortium (including the Commission Services)	20

Description of deliverables

D2.1 : Report on microorganisms and enzymes for AWP degradation [20]

Written report linked to tasks 2.1, partially 2.2 and 2.4 including identification and AWP biodegradation capabilities of microbial consortia and enzymes as well as a set of 3-4 microbial consortia and a preliminary selection of at least 2-3 enzymes for later project activities.

D2.2 : Report on novel synthetic enzymes [24]

Written report linked to task 2.2 including description of new synthetic enzyme strategies, the genomics-selection processes for design, the enzyme generation and selection processes, and the generation of increased activity and properties of the new enzymes, with assessments of their of their activity and performance related to degradative modification of plastics.

D2.3 : Report on selection and pre-compatibility test of earthworm/insects-micro consortia [20]

Written report linked to tasks 2.3 and 2.5 including earthworm and insect selected for further project activities as well as 2-3 microbial/enzyme pool for each earthworm and insect according to pre-compatibility results.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS4	Microorganisms and enzymes for AWP degradation	1 - UAL	12	Biodegradation capabilities of microbial consortia identified as well as a set of 3-4 microbial consortia and a preliminary selection of at least 2-3 enzymes for later project activities.
MS5	Novel synthetic enzymes	4 - BRUNEL	24	Novel enzymes samples available and new strategy for selected / directed evolution
MS6	Selection and pre-compatibility test of macro-meso-micro consortia	5 - UMH	18	Earthworm and insect selected and 2-3 microbial/ enzyme pool for each earthworm and insect according to pre-compatibility results (available for activities in WP3&WP4).

Work package number [9]	WP3	Lead beneficiary [10]	5 - UMH
Work package title	Up-scaled production of biotech tools for APW biodegradation		
Start month	12	End month	24

Objectives

● To produce stable plastic-degrading microbial/enzyme consortia (PMC).
● To obtain earthworms and insects larvae fortified with PMC for enhanced AWP degradation capability (FP-earthworm/insect).

Description of work and role of partners

WP3 - Up-scaled production of biotech tools for APW biodegradation [Months: 12-24]
UMH, UAL, UNIPI, ASAJA, BRUNEL, ASA, NUTRINSECT, IRIS, ENTO
This WPs aims to obtain stable microbial/enzyme inoculums (PMC) and the worms and insects larvae fortified with the microbial consortia selected in WP2. For this purpose microorganisms/enzymes will be prepared in suitable formulations for the intended application, that is, to be added to soil or compost or used to feed earthworms/insects. Establishment of microorganisms on digestive tract of earthworm/insects larvae will permit them to improve AWP biodegradation. The approach will allow also establishing key times in insects and worms live cycle where the PMC incorporation will be more effective.

Task 3.1. Production of stable plastic-degrading microbial/enzyme consortia (PMC) (M12-M24)
Subtask 3.1.1. Production of microorganisms and enzymes
The microorganisms and enzymes selected in WP2 will be produced by ASA in bioreactor up to 25 L reactor in enough quantity for the following tasks in WP3 and WP4. The culture medium and protocol will be replicated in 10 L scale. The culture conditions will be adjusted according to the information provided from WP2 for each microorganism and to the further optimization for scaling-up. After that, the physical-chemical parameters of the fermentation process will be optimized. Furthermore, the carbon and nitrogen sources of the fermentation media will be optimized in order to establish fed batch processes. Growth rate and maintenance of AWP degrading activity will be checked and culture time will be selected according to production peaks.
Then downstream processes for the enzymes and microorganisms will be developed. The fermentation procedures and purification processes will be scaled up to a fermentation volume of 25 L. Finally, the selected enzymes and microorganisms will be characterized i.e. with regard to their AWP degrading capability.
Leader: ASA. Partners involved: UAL, BRUNEL
Subtask 3.1.2. Preparation of stable PMC formulations
PMC stability upon spray drying, lyophilizing and/or storage at different temperatures and times will be tested by taking samples periodically after treatment and assayed for microorganism survival and maintenance of AWP biodegradation capability. Treatment will be selected according to the easiest way to store, transport and manipulate PMC without losing AWP biodegrading capabilities.
PMC formulations intended for adding to soil or compost will be prepared by adding microbial biomass or enzymes to inorganic or organic carriers, or a combination of both. Inorganic matrix clays such calcium and sodium bentonite, and organic matrix such as alginate, polyethylene (PEG) spent coffee, biotic-inert compost, biochar or extract of compost will be tested.
PMC formulations for feeding earthworm/insects will be prepared by adding microbial biomass or enzymes to pre-composted organic materials such as agroindustry residues for earthworms and food suitable for insect larvae. Formulations will be re-designed and re-evaluated according to results obtained in Task 3.2.
Microorganisms and enzymes stability and maintenance of AWP degrading activity will be monitored by using the techniques described in WP2. UNIPI will analyse the AWP after degradation.
This task will provide at least 2-4 PMC stable formulations for use in WP4.
Leader: UAL Partners involved: ASAJA, ASA, BRUNEL, NUTRINSECT, UMH, UAL, UNIPI

Task 3.2. Production of Insects/Earthworms fortified with PMC with enhanced AWP biodegrading capabilities (M12-M24)
Organic matrix produced in subtask 3.1.2 will be used to feed the insects and earthworms selected in WP2. The PMC will be incorporated into the organic matrix at different doses and at different growing cycle of insects and earthworms in order to establish the optimal one. The need for PMC re-inoculation during rearing will be also evaluated. The insects and

earthworms will be analysed in order to establish the grade of PMC absorbed by measuring them using the techniques proposed in WP2 (UAL, BRUNEL) and the enzymatic activities (UMH). Digestive enzyme activities, preferentially those involved with potential AWP breakdown (esterases), will be measured in both the gastrointestinal tract and luminal content of insects and worm to provide evidences for enzymatic hydrolysis derived from PMC inoculums. Catalytic performance (Vmax and apparent Km) will be compared between esterase enzymes from the gastrointestinal tract of insect larvae and worms, and from the microbial consortia selected in WP2.
This task will provide at least 1-2 FP-earthworm and 2-5 FP-insects formulations for use in WP4.
Leader: UMH. Partners involved: ASAJA, NUTRINSECT, UAL, BRUNEL

Task 3.3. Development of plastic-degrader formulations (M12-M24)
Plastic-degrader microorganisms, enzymes, microorganism-enzyme-fortified earthworms (FPew) and insects (FPinsect) developed in tasks 3.1 and 3.2 will be formulated in several formats, depending on their purposes (addition to soil or compost) and AWP degradation capacities, and tested for their potential up-scaling formulation for eventual commercialization. On the basis on lab scale experiments in task 4.1 and 2, up-scale production will be performed including maintenance/depletion of plastic-degrading capacities in time. Finally, field application requirements for the final formulations will included the outputs from in situ experiments (task 4.3).
Leader: NUTRINSECT. Partners involved: ASAJA, UAL, BRUNEL, UMH, IRIS, ENTO

Participation per Partner

Partner number and short name	WP3 effort
1 - UAL	
2 - UNIPI	
3 - ASAJA	
4 - BRUNEL	
5 - UMH	
8 - ASA	
9 - NUTRINSECT	
10 - IRIS	
14 - ENTO	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D3.1	Formulations of PMC to be used as enhancers of AWP degradation	8 - ASA	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D3.2	Protocol for the production of insect and earthworm plus PMC with enhanced AWP degrading capability	5 - UMH	Report	Confidential, only for members of the consortium (including the Commission Services)	24

Description of deliverables

D3.1 : Formulations of PMC to be used as enhancers of AWP degradation [24]

Written report linked to Task 3.1 and 3.3 including protocol for PMC formulations preparation and a set of at least 2-4 PMC formulations for use in WP4

D3.2 : Protocol for the production of insect and earthworm plus PMC with enhanced AWP degrading capability [24]

Written report linked to Task 3.2 and 3.3 including protocol for fortified insect (FP-insect) and earthworm (FP-ew) formulations preparation and a set of at least 1-2 FP-ew and 2-5 FP-insects formulations for use in WP4.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS7	Formulations and production of stable plastic-degrading microbial/enzyme consortia (PMC)	8 - ASA	24	Set of at least 2-4 stable PMC formulations to be used as enhancers of AWP degradation consortia available for WP4
MS8	Insect and earthworm plus PMC with enhanced AWP degrading capability	5 - UMH	24	Protocol for fortified insect (FP-insect) and earthworm (FP-earthworm) formulations preparation validated and a set of at least 1-2 FP-earthworm and 2-5 FP-insects formulations for use in WP4.

Work package number [9]	WP4	Lead beneficiary [10]	18 - OWS RF
Work package title	Pilot scale ex and in situ degradation of municipal and rural APW		
Start month	18	End month	36

Objectives

- To test the biodegradation capacities of the fortified microbial consortia (FMC) from previous WP both ex and in situ.
- To examine the best operational conditions.
- To upscale the ex situ biodegradation process.
- To monitor biodegradation with the aim to reach over 80% degradation of the selected AWP setting up novel characterisation approach.
- To model the biodegradation for further optimisation of the approach vs. AWP streams to degrade

Description of work and role of partners

WP4 - Pilot scale ex and in situ degradation of municipal and rural APW [Months: 18-36]
OWS RF, UAL, UNIPI, UMH, ASU, NATURPLAS, NUTRINSECT, IRIS, IDELUX, FEMTO, ENTO, SAV
Task 4.1. Development of biodegradation processes (M18-M30)
Subtask 4.1.1. Examination of operational conditions to optimize AWP degradation of the selected PMC
The effectiveness in the biodegradation capability of the selected PMC for complete mineralization of the AWP will be examined at medium scale (4 litre reactors). Tests will be run (1) under the conditions of a limited competition and biodiversity (with small amounts of organic matter and partial sterilization) and (2) in the presence of a full bio-diversity and competition. The biodegradation will be monitored on the basis of ISO 14855 (compost) and ISO 17556 (soil) and the growth conditions defined in WP3. The biodegradation will be determined by measuring the final conversion of the organic carbon of plastics into CO2. In a first stage one well-defined type of plastic will be used to examine the biodegradation potential of the selected PMC, followed by the treatment of mixed AWP. In order to ensure a fast as possible result the plastic will be milled into powder. When positive results are obtained the test conditions (temperature, moisture, aeration,…) will be adjusted to maximize the degradation rate and examine the operational boundaries.
Subtask 4.1.2. Examination of operational conditions to optimize AWP degradation of the selected FP-earthworms and FP-insects
Based on the information of WP3 with regards to growth conditions of FP-earthworms and FP-insects and knowledge obtained in subtask 4.1.1, the biodegradation capability for complete mineralization of AWP by these organisms will be examined. The biodegradation will be monitored on the basis of ISO 14855 (compost) and ISO 17556 (soil) (CO2 measurements). In a first phase milled single stream plastic will be examined. Subsequently plastic films and mixtures of AWP will be examined. The operational conditions will be optimized to obtain high degradation rate.
Leader: OWS RF. Partners involved: UMH, UAL
Task 4.2. Set up of pilot-scale ex-situ plastic degradation process (M24-M36)
Subtask 4.2.1. Ex-situ AWP degradation by composting and vermicomposting
In the strategy to manage biodegradation of plastic residues ex situ, a biodegradation reactor of at least 10 kg aimed at using the PMC and/or FPBug-worms produced in the previous WP3 will be set up. The optimal conditions will be based on the information of Task 4.1. The organic waste will be optimised and blended with defined proportions (ranging from 10 up to 50% (v/v) of pre-treated AWP. Temperature and moisture of the substrates will be kept in optimal conditions for successful composting and further vermicomposting process. Once bio-oxidative phase of composting is finished FP-earthworms will be added so the vermicomposting phase will start. Periodically, samples will be taken for assessing the efficiency of the biodegradation process in the following tasks. Mass balance, mass flow, etc will be monitored.
Final compost/vermicompost will be checked for the presence of microplastics (Task 4.4) after screening out large size plastics and an additional trial will be performed with the aim to remove those materials by means of treatment with suitable PMC. Samples will be taken periodically and analysed as above.
Also with plastic contaminated waste of real-life waste streams of IDELUX (compost – digestate) will be treated by the developed bioremediation process of specialised vermicomposting.
Leader: OWS RF. Partners involved: UMH, IDELUX
Subtask 4.2.2. Ex-situ AWP degradation by insects rearing
In this strategy, selected FP-insects from WP3, will be used in a specific reactor to develop AWP degradation using a raw mix of AWP and feedstock (10-50% AWP). Rearing conditions and final micro/enzyme cocktail will be settled according to results obtained in WP3-Task 3.2 and in agreement with results from Task 4.1. Periodically, samples will be taken for assessing the efficiency of the biodegradation process in the following task 4.4. Mass balance, mass flow,

etc. will be monitored as well as insect larvae performance and enzyme activity as in WP2. After the end of the insect-AWP degrading cycle (at insect adult stage), insect will be provided for chitin extraction in Task 5.2 while remains in the rearing chamber will be used for biofertilizers preparation (Task 5.1). In addition, the potential use of insects produced for biological control will be considered mainly focusing on two areas: the use of insect eggs as "non-pests" food in the open air and in greenhouses to improve and increase the management systems of insect pest natural enemies and the "in vivo" production of natural enemies.
Leader: NUTRINSECT. Partners involved: ASAJA, UAL, ENTO

Task 4.3. Set up of in-situ bioremediation of AWP in soil (M24-M36)
Based on the information of task 4.1 the dynamics of plastic-degrading activities in plastic-contaminated agricultural soils using compost and vermicompost amended with PMC produced in WP3 as well as fortified soil-dwelling earthworms will be analysed. UAL will monitor microbial load and identity. For monitoring the earthworm health and the process, ecotoxicological biomarkers will be used as an indicator of soil bioturbation.
Even soil contaminated with real-life waste streams and mulching films will be treated by the developed bioremediation process. Fortified earthworms will be applied in field trials (mesocosm). Persistence of plastic debris will be monitored using physical and optical methods, whereas the dynamic of earthworm, microbial activity and enzyme production will be determined as reported above WPs 2 and 3.
Leader: UMH. Partners involved: ASAJA, UAL, NATURPLAS, SAV.

Task 4.4. Monitoring of AWP degradation
Subtask 4.4.1 Chemical decomposition assessment
The biodegradation of AWP will be followed by its chemical decomposition: monitoring of markers residues (phthalate esters such as di-n-butyl phthalate and di-(2-ethylhexyl) phthalate) derived from synthetic polymer biodegradation in both earthworm tissue and final composted substrate (vermicast) and soil. The physical decomposition will be determined by the median degradation time (DT50) of total raw plastic from initial load of residues by spectroscopic and microscopic techniques. Finally, a weight-of-evidence assessment of in situ plastic biodegradation will be determined. UNIPI will analyse plastic debris for morphology, chemical (FTIR) and thermal (TGA, DSC) properties.
Leader: UNIPI. Partners involved: OWS RF, IRIS, FEMTO.
Subtask 4.4.2. Inline monitoring disintegration and non-degraded plastic
Samples from task 4.1 will be supplied along the biodegradation tests and IRIS will use hyper spectral imaging (HSI) in VIS, NIR and SWIR spectral ranges to test the ability to quantify the size of the residual plastic fragments and of the distinction among different materials thanks to digital image processing. Different approaches for sample collecting and analysis will be studied to maximise the representativity of the results as it is likely that the analysis is limited to a few mm below the surface of the compost. The data, together with other analysis in the previous subtask, will support the selection of the most efficient treatment.
Leader: IRIS Partners involved: OWS RF, UNIPI
Subtask 4.4.3. Predictive modeling as support for biodegradation process optimisation.
IRIS will set up a database for the partners to centralise all relevant information for the modeling task by M3. It will be populated with the results from the previous tasks in terms of disintegration/biodegradation profiles both in and ex-situ. Next, the available data will be pre-analysed to select the most discriminant variable subsets and to define new factors using, for example, PCA (Principal Component Analysis) and Factorial Analysis. This will also rely on support and expert know-how of the partners. Pre-processing of data before data modeling will give optimum input sets to the machine learning algorithms. Then, the data will be modeled by benchmarking different machine learning algorithms using fitness functions as objective output to optimize different aspects of the known parameters influencing the biodegradation and disintegration such as composition (monitored in task 1.3), pretreatment, vermicompost composition and external conditions. The characteristic curves will be represented in a “parametric form”, including key values such as biodegradation time and biodegradation percentage achieved. The resulting predictive models will be used to select the most appropriate process and micro/macro-organisms mix for degrading a target type of plastic and/or reach a complete biodegradation in a given timing. As such, once the models are trained and validated with unitary testing, a Monte Carlo method will be developed to calibrate the system input parameters in order to obtain a closest match to a desired output (and/or fitness function). Once the calibration process is completed, a CBR historical case base and system will be defined, bootstrapped with initial cases (eg. from WP2), and as larger scale cases are processed in tasks 4.1, 4.2 and 4.3, they will be incorporated, based on a similarity matching algorithm, which will also be developed. Synthetic data can also be generated for the CBR to complement existing empirical results for data modeling needs, especially in early phases of the project. After modeling the single steps, DST will be used to support the optimal process in next task but also in the cascade process (defined in WP6).
Leader: IRIS, Partners involved: OWS RF, UNIPI, IDELUX, ENTO, NUTRINSECT

Task 4.5. Establishing protocol of action of the overall fortified system. Sequential study (M24-M36)

The sequential action of introducing the different fortified agents into the organic matrix, evaluating the optimal sequence, will be performed. It will comprise the three main ingredients developed in the project: fortified insects (FP-insect), fortified earthworms (FP-earthworms) and fortified microbial consortia (PMC). They will be inoculated all together and with the different sequential combinations to evaluate synergistic and potential antagonistic effects on the AWP biodegradation. The optimal combination will be decided as compromise between AWP biodegradation and the quality of the added-value products to be extracted from insects and worms (with feedback from WP5) such as chitosan and derivates from the chitin of insects and biofertilizers from the body rest after crushing, desiccating and chemical or enzymatic hydrolysis.
A mixed combination of AWP will be used as well as different pre-composting materials used as substrate to feed the fortified insects and compost, evaluating if it will be necessary for the FMC to be re-inoculated or if the initial feeding with the proposed substrate and the fortified effect is maintained through time.
Leader: UMH/NUTRINSEC. Partners involved: ASAJA, NUTRINSECT, UAL, UNIPI, UMH

Participation per Partner

Partner number and short name	WP4 effort
1 - UAL	
2 - UNIPI	
5 - UMH	
6 - ASU	
7 - NATURPLAS	
9 - NUTRINSECT	
10 - IRIS	
12 - IDELUX	
13 - FEMTO	
14 - ENTO	
17 - SAV	
18 - OWS RF	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D4.1	Report on operational conditions for enhance bioremediation process	18 - OWS RF	Report	Confidential, only for members of the consortium (including the Commission Services)	30
D4.2	Report on upscaled ex situ and in situ AWP biodegradation process	12 - IDELUX	Report	Confidential, only for members of the consortium (including the Commission Services)	36

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D4.3	Biodegradation monitoring and modelling	10 - IRIS	Other	Confidential, only for members of the consortium (including the Commission Services)	36

Description of deliverables

D4.1 : Report on operational conditions for enhance bioremediation process [30]

Written report linked to the task 4.1 including adjustment of process conditions for AWP degradation by PMC and insects to maximize the degradation rate as well as basic operational conditions for ex-situ and in-situ plastic treatment.

D4.2 : Report on upscaled ex situ and in situ AWP biodegradation process [36]

Written report linked to tasks 4.2, 4.3, 4.4 and 4.5 including results of upscaled trials on compost, vermicompost, insect rearing and soil as well as AWP biodegradation monitoring targeting percentage of plastic degradation >80% and protocol for the sequential process.

D4.3 : Biodegradation monitoring and modelling [36]

Written report linked to task 4.4 including the results of the different approaches used to monitor the biodegradation of AWP in the different EoL conditions followed as well an app based on a model to support the decision on optimal biodegradation scenario

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS9	Enhanced bioremediation process	18 - OWS RF	30	Results obtained in common conditions (temperature, moisture, aeration) adjusted to maximize the degradation rate and upgrade the operational boundaries. Process conditions for AWP degradation by PMC and insects to maximize the degradation rate as well as basic operational conditions for ex-situ and in-situ plastic treatment defined.
MS10	Defined up-scaled ex situ and in situ AWP biodegradation	12 - IDELUX	36	Defined standardized strategy to insects-earthworms-microorganisms consortia AWP degrading activity in two main scenarios: a) ex situ by composting, vermicomposting and insect rearing b) in situ by using

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
				vermicompost in soil reaching a plastic degradation >80%
MS11	Biodegradation Models	10 - IRIS	36	A decision support tool derived from developed models allow supporting the definition of the optimum biodegradation approach

Work package number [9]	WP5	Lead beneficiary [10]	2 - UNIPI
Work package title	Validation of value-added downstream products		
Start month	24	End month	48

Objectives

- To extract chitin and convert it to chitosan.
- To develop upgraded biofertilizers.
- To develop chitin and chitosan derived bioplastics for agriculture and food-packaging.
- To develop chitin and chitosan coated materials.

Description of work and role of partners

WP5 - Validation of value-added downstream products [Months: 24-48]
UNIPI, UAL, ASAJA, UMH, ASU, NATURPLAS, NUTRINSECT, IRIS, IDELUX, FEMTO, ENTO, MANOR, SAV, OWS RF
Task 5.1. Development of upgrade biofertilizers (M24-M36)
Subtask 5.1.1. Development of vermicompost derived from plastic-organic waste mixtures
Using the outputs of WP4 activities, selected earthworms with the most efficient plastic degradative capacities will be included into specific containers to produce vermicompost using at least 6 different mixtures of AWP with other substrates approaches. The device will be portable and modular design including stock and pre-treatment of plastic section and environmental control (humidity and mixing devices). The process will be optimized to produce adequate sizing and format of bio-fertilizers according to new European Fertilizer Regulation. The quality of produced biofertilizer will be tested and compared with commercial biofertilizers. UMH will test the fertilizing properties of the developed materials in greenhouse conditions using different application dose and comparing with equivalent NPK commercial fertilizers. The biopesticide and/or plant biostimulant activity of produced biofertilizers will be tested by UAL.
Leader: UMH. Partners involved: UAL, ENTO, IDELUX.

Subtask 5.1.2. Development of biofertilizers from insect manure and chitin extraction residues
Manure from insect breeding chambers (WP4), leftovers coming out from chitin extraction from insects (Subtask 5.3.1) will be identified by ENTO, UAL. Those considered potentially suitable for biofertiliser preparation will be analysed to determine their main nutrient and physicochemical properties (i.e. analysis of the N, P, K, aminoacids, total Carbon, total organic matter) by UMH. The formulation of the suitable mixtures will be investigated and post-treatment steps by chemical and enzymatic hydrolysis, dewatering, drying, composting, pelletizing will be proposed by ENTO in function of the final application.
UNIPI in cooperation with FEMTO will prepare biocomposites containing bio-fertilizer self-fertilizing agriculture items (sticks, yarns, etc) based on soil biodegradable polymeric matrices as a binder, such as polyhydroxyalkanoates or polybutylene succinate-co adipate to act as carrier for the bio-fertilizer.
The proposed biofertilizer development will be addressed attending to construct added-value products that would be improved by the incorporation of PMC, fortified worms and insects/larvae that will introduce important added-values such as biopesticide, biofertilization and biostimulation effects. The final composition of the organic fertilisers will be compared with market available fertilisers. The biopesticide and/or biostimulant activity of produced biofertilizers will be tested by UAL by a set of standardized procedures and compared with commercial biofertilizers. Added-value will be also determined by UMH studying yields, but also parameters related to soil and plant quality, such as reduction of chemical application of pesticides and fertilizers, antioxidant level, C vitamin in field experiments. SAV will also validate the fertilisers from the project on plant growth in rural fields and in city parks in comparison with conventional non-biobased fertilisers.
Leader: ENTO. Partners involved: UAL, UMH, ASAJA, UNIPI, NUTRINSECT, FEMTO, SAV.

Task 5.2. Extraction of chitin, and conversion to chitosan (M24-M48)
To circumvent the usage of alkaline solutions and acids for the production of chitosan from chitin, ASU in collaboration with NUTRINSECT and UNPI will extract chitin from the insects/worms coming from Task 4.1 using a combination of irradiation by electron beam (eBeam) treatment and chemical methods. Raw chitin material will be eBeam treated to introduce degradation of chitin and break up the tight chitin network. Initially, a wide range of doses (20-200 kGy) will be applied and the solubility of irradiated raw material in several (green) solvents will be tested at laboratory scale.

Additionally, it will be tested if irradiation allows the reduction of alkaline solutions and acids during conventional chemical extraction. Extracted chitin will be converted to chitosan by eBeam treatment. The extracted chitin and chitosan will be provided for Subtask 5.3.2, Subtask 5.3.3 and Subtask 5.3.4. Based on the results, the chitin extraction method will be further optimized regarding chitin and chitosan yields iteratively.
Leader: ASU Partners involved: UNIPI, ASAJA, NUTRINSECT, UMH.

Task 5.3 Processing of chitosan and chitin (M24-M48)
Subtask 5.3.1. Characterization and first trials of processing of chitin and chitosan
UNIPI will analyse the chitin and chitosan produced, by FTIR, NMR, and thermal analysis. Chitin and chitosan will be compared with commercial grade in terms of processability as additives in biobased and biodegradable polymeric blends and composites. The chitin and chitosan will be formulated with commercially available bio-based polymers and additives to allow thermoplastic processing. These are for instance plasticisers, stabilizers and processing aids. The different chitin and chitosan-based formulations with biodegradable polymers (polylactic acid, polyhydroxyalkanoates, polybutylene succinate), will then be evaluated regarding their process ability by extrusion and injection moulding by UNIPI in cooperation with FEMTO. Thereby an optimization of processing parameters (temperature, processing rate, etc.) will also be carried out. This activity will be performed in first instance, at laboratory scale using a Haake minilab extruder, with Haake mini jet at UNIPI to produce dog bone shaped test specimens. Specimen will be used to characterize the materials for thermal and mechanical properties relevant to evaluate properties and performances of the grades of chitin and chitosan as well as the different formulations obtained and thereof identify potential applications. Specimen produced by UNIPI will be evaluated by ASU concerning optical properties and surface energy in terms of contact angle measurements. Moreover, water vapor permeability & permeability against O2 will be monitored by ASU.
Based on the results of these characterisations, the formulations will be optimized in iterative loops. A maximum of five formulations will be selected for pilot scale trials conducted at UNIPI on pilot scale extruder (twin screw extruder, pilot injection moulding, calendering, films by blow moulding), and at FEMTO for industrial injection moulding. These tests will be conducted in order to further optimize formulations and processing parameters of polymeric formulation with chitin/chitosan, to verify lab results and to produce sufficient material for demonstrators for applications in agriculture (packaging, sticks, pots), and food packaging (trays, rigid containers, films).
Leader: UNIPI. Partners involved: ASU, NATURPLAS, FEMTO
Subtask 5.3.2. Chitosan and chitin for coatings.
Solubility of chitin and chitosan will be tested in view of applications as coating on plastic or cellulosic substrates. Up to 3 suitable solvents will be selected by using the Hansen solubility parameter approach. The chitosan and chitin solutions obtained will be further formulated by applying suitable bio-based additives such as plasticisers, antioxidants etc. These formulations will then be evaluated by applying them on different cellulose and plastic substrates. These coated materials will be characterized with respect to their mechanical and barrier properties as well as their antimicrobial activity by UAL. For further modification of the mechanical and barrier properties and to enhance the antimicrobial activity, the dried and wet coated materials produced by UNIPI on lab, scale will be irradiated via eBeam treatment by ASU to enhance barrier, mechanical and anti-microbial performances.
Based on the results, both formulations as well as coating parameters will be optimized in iterative loops. Up to two formulations for each substrate category will be selected and applied in pilot scale coating trials at ASU partner sites. These tests will be conducted in order to further optimize formulations and processing parameters, to verify lab results and to produce sufficient material for demonstrators (food packaging).
To evaluate the effect of chitosan coated films on perishable food, chitosan coated films will be applied as lids for food trays. These food trays are then used to analyse the shelf life of chicken in comparison to commercially available trays with support from IRIS.
OWS RF will perform biodegradability test on the developed formulations in different environments such as compost (ISO 14855), soil (ISO 17556) and water (freshwater ISO 14851 and under marine conditions ASTM D6691) according to international standards in order to define suitable end-of-life options and evaluate effect by accidental release in the environment.
Leader: UNIPI. Partners involved: IRIS, FEMTO, ASAJA, ASU, NUTRINSECT, UAL, OWS RF, MANOR
Subtask 5.3.3 Utilization of chitin/chitosan production by-products for optimized mulching films
Within this subtask chitin/chitosan by-products will be characterized and used to improve mulch materials for optimized conditions such as temperature and O2/CO2/H2O contents considering soil respiration. ASU in collaboration with UNIPI will use by-products of WP4 and Subtask 5.3.1 as well as the extracted chitosan to improve existing mulching polymer films for their light and heat transmission, absorption, reflection/emissivity (UV-VIS and IR-spectroscopy) and for UV stability and selective gas permeability (permselectivity). The chosen chitin extraction residues will be applied to existing mulching films by coating and/or by compounding to mulching film polymers closing the loop within the RECOVER approach. Mechanical & thermal properties include density, puncture resistance, specific heat capacity and permeability for gases including H2O, N2, O2, and CO2. The required mulch permeability will be determined in a

numerical simulation module based on existing models and the intrinsic permeation coefficients determined within this subtask. NATURPLAS will cooperate to films production, optical characterization and pre-testing to estimate films suitability to be used in mulching, the results will be fed back in iterative loops for further optimization. SAV will make the validation of the mulching films in real agricultural fields.
Leader: UNIPI, Partners involved: ASU, NATURPLAS, SAV

Participation per Partner

Partner number and short name	WP5 effort
1 - UAL	
2 - UNIPI	
3 - ASAJA	
5 - UMH	
6 - ASU	
7 - NATURPLAS	
9 - NUTRINSECT	
10 - IRIS	
12 - IDELUX	
13 - FEMTO	
14 - ENTO	
15 - MANOR	
17 - SAV	
18 - OWS RF	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D5.1	Report on up-graded biofertilizer products	14 - ENTO	Report	Confidential, only for members of the consortium (including the Commission Services)	26
D5.2	Report on eBeam-related optimizations	6 - ASU	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D5.3	Report on chitin-based bioplastics	2 - UNIPI	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	42

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D5.4	Optimized mulching films	2 - UNIPI	Report	Confidential, only for members of the consortium (including the Commission Services)	48

Description of deliverables

D5.1 : Report on up-graded biofertilizer products [26]

Written report linked to task 5.1 including the specifications of the processes to produce high-value fertilisers in comparison with similar and conventional products.

D5.2 : Report on eBeam-related optimizations [36]

Written report linked to Task 5.2 including protocol of novel method, also partially linked to 5.3 including properties improvement by eBeam.

D5.3 : Report on chitin-based bioplastics [42]

Written report linked to Task 5.3 including characterizations and prototypes production.

D5.4 : Optimized mulching films [48]

Written report on optimized mulching films for two selected soil respiration examples were successfully simulated and validated at lab scale

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS12	Up-graded biofertilizers formulation	14 - ENTO	36	Formulation matrix of several industrial mixtures of up-grade bio-fertilizer products compared with similar and conventional products.
MS13	Chitin-based extraction processes	2 - UNIPI	42	Verified sustainability of various extraction approaches of chitin (to chitosan conversion and uses).
MS14	Chitin/chitosan films and containers	2 - UNIPI	48	From mixture of polymeric matrices and chitin/chitosan from RECOVER production of prototypes of films for food packaging and/ or for mulching, and agriculture items such as pots, yarns, etc

Work package number [9]	WP6	Lead beneficiary [10]	3 - ASAJA
Work package title	Safety, environmental, social and economic sustainability assessments		
Start month	1	End month	48

Objectives

● To assess the environmental sustainability of the innovative biodegradation solutions, including the logistics system to handle RECOVER material, collection, sorting, selection and treatment by life cycle assessment (LCA).
● To assess the total economic costs and hot spots associated to project biodegradation pathways by life cycle costing (LCC) assessment.
● To evaluate the mass and energy balance of the innovative valorisation chain
● To evaluate the total social impacts associated to the biodegradation processes by social life cycle assessment (SLCA)
● To AWP biodegrading multi-step degradative cascade approach using sequences of fortified Insects and earthworms.

Description of work and role of partners

WP6 - Safety, environmental, social and economic sustainability assessments [Months: 1-48]
ASAJA, UAL, UNIPI, UMH, ASU, NATURPLAS, ASA, IRIS, OWS, IDELUX, FEMTO, ENTO, ENCO, SAV, OWS RF
Most of the work in this WP will be performed on data collected from operational demo site (WP4/5) representative of future industrial units. To facilitate the various economic, environmental and social, each innovative biodegradative process will be divided into functional units: receipt of samples, preparation, upgrading processes, end products and their final uses. By using LCA for environmental assessment and combining it with cost analysis and risk assessments/ social acceptance, stakeholders will be provided with high quality decision support.

Task 6.1 Mass, energy balance, techno-economic studies. and development of RECOVER cascade strategy (M24-M48)
Previous energy balances allow to demonstrate that all the concepts proposed into the project are reasonable. However, they are based on small scale experiences and some hypothesis on final yield products of the processes. Dynamic tool will be set up in the reactors used in WP4 for the continuous monitoring of the sub-products obtained inter-phases and energy balances of the different steps of the technological chains. It will allow identifying undesirable situations conducting to loss of efficiency/sustainability. For each step a single tool will be developed, integrated in a multistep system. ASAJA with the support of OWS RF will be responsible to create this dynamic tool and connect it to the Data Center. This tool will provide simplified reports for better understand the performance of the different steps and their contribution to the overall process. These data will also be useful for further techno-economic and Life Cycle Analysis (LCA). ASAJA will be in charge of techno-economic analysis. IDELUX will provide data from benchmark waste treatments for this and next task.
The most efficient integrated cascade concept will be discussed based on WP5WP4/5 results using subsequent treatment processes of specialised larvae-worm-insect plastic degrader capacities to produce a synergistic enhanced degradation; products of the piloting will be chitin/chitosan and vermicompost and a zero-waste scenario will be targeted. (theoretical exercise supporting future upscale). Mass balance, mass flow optimization and engineering procedures will be used to define the full scale strategy in both agricultural and municipal use scenarios.
In terms of the CO2 emitted during the performance of the biodegradation process, a concept of coupling of the reactor with a microalgae culture for CO2 capture will be designed with ALGAENERGY (BIC member in our AB).
Leader: ASAJA. Partners involved: OWS RF, UAL, UNIPI, UMH, IDELUX, UNIPI

Task 6.2 Life cycle analysis
The environmental impact of the developed Biotech processes will be screened, including the extraction of the different end-products, based on ISO 14040 and ISO 14044. Set of boundaries depends on the choice of products of previous WP's and will be defined during the project A detailed assessment of the different inputs and outputs will be performed in close cooperation with the partners of WP4 and WP5. (as well as the data from task 6.1). Impact assessment calculations will be performed using SimaPro LCA software in combination with e.g. the Ecoinvent and ELCD LCA databases. Major impacts in terms of human health, resources depletion and environmental protection will be determined. This way the 'hot spots' in the process will be determined and suggestions for improvement will be formulated. The results will be compared to the current environmental impact of pollution with plastics. Moreover a base scenario "the no implementation of proposed technologies" will be used, thus providing a simple and clear image of the advantages and disadvantages of implementing the proposed technologies.

Apart from the environmental impacts also the life cycle cost (LCC) and social life cycle aspects (sLCA) will be examined, resulting in a total life cycle sustainability assessment dealing with the 3 main pillars: people, planet and profit. The social risks will be calculated based on the Soca add-on to Ecoinvent in OpenLCA. The following social impact categories will be evaluated in this study: child labour, fair salary, forced labour, and safety measures. The economic costs will be calculated in terms of total production cost, total life cycle cost, and the main cost contributors.
Leader: OWS/OWS RF. Partners involved: ASAJA, NUTRINSECT, UAL, UNIPI, UMH.

Task 6.3 Risk assessment and social acceptance
As a key for the future uptake of the proposed AWP degrading process, the social sustainability will be investigated by risk assessments of the new value chains developed in the RECOVER project. ASAJA will be in charge to determine risks assessments by means of various danger categories identifications as i) Physical: noise, pressure, electrical hazard, temperature, machinery, etc; Biological: insects, larvae, worms, bacteria, viruses, fungi, etc; Chemicals: liquids, gases, fumes, dust, aerosols, heavy metals, organic contaminants, etc.; Ergonomic: posture, workstation, gestures, bad posture, etc; Psychological: pressure, workload, etc.
To assess the potential dangers and their associated risks, a survey will be made among the partners involved in the development of innovative biodegradation processes, especially for the technologies that will be developed at operative scale in the demo sites from WP4. Risks will then be classified according to their severity and likelihood of occurrence and preventive measures will be suggested. Complementarily, the social acceptance assessment will be done by ASAJA across their networks. Social impact assessment builds on local knowledge and relies on a participatory process to analyse the concerns of interested and affected parties. It includes study of public opinion, level of knowledge and/or concerns about communities of microorganisms, insects, inoculation technique, chitin-chitosan conversion, added-value new biomaterials, etc... Thus, opinions from society – i.e. the opinions of citizens, businessmen, representatives of ecological and social organizations, public authorities and politicians – will be taken into consideration. In this regard, at least 100 surveys (online, or during the project roadshow of events) will be carried out. Surveys will include closed and open questions.
Leader: ASAJA Partners involved: NUTRINSECT, OWS RF, UAL, UNIPI, UMH, ENCO, FEMTO, NATURPLAS

Task 6.4 Logistics approach for RECOVER large scale implementation
Following the preliminary approach developed in Task1.3, UNIPI in cooperation with IDELUX will design a general logistic approach to indicate the viability of the proposed solutions: a logistic feasibility evaluation, a comparison of costs, benefits and emissions coming from the Project Scenario relative to the Baseline Scenario will be assessed. In this analysis, will be conducted a logistic analysis of each future step of the future AWP new logistic chain, from the supply side, while it will be designed a Stated Preference Survey to be presented to agri-food stakeholders so to evaluate, by means of a Discrete Choice Model, the future demands for the innovative solution model able to understand the main features to be taken into consideration to increase the attractiveness of the proposed logistic solution. This, among other, will be discussed in front of the corresponding business model defined in WP7, whereby the option for cooperative AWP treatment is considered.
Leader: UNIPI, Partner involved: IDELUX, ASAJA, SAV

Participation per Partner

Partner number and short name	WP6 effort
1 - UAL	
2 - UNIPI	
3 - ASAJA	
5 - UMH	
6 - ASU	
7 - NATURPLAS	
8 - ASA	
10 - IRIS	
11 - OWS	
12 - IDELUX	

Partner number and short name	WP6 effort
13 - FEMTO	0
14 - ENTO	
16 - ENCO	
17 - SAV	
18 - OWS RF	
Total	0

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D6.1	Report of the mass, energy balance and techno-economic studies	3 - ASAJA	Report	Confidential, only for members of the consortium (including the Commission Services)	30
D6.2	Interim report of the carbon foot- print and LCA	18 - OWS RF	Report	Confidential, only for members of the consortium (including the Commission Services)	20
D6.3	Final report of the carbon foot- print and LCA	18 - OWS RF	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D6.4	Report on risk assessment including social acceptance assessments	3 - ASAJA	Report	Confidential, only for members of the consortium (including the Commission Services)	32
D6.5	RECOVER logistic approach	2 - UNIPI	Report	Confidential, only for members of the consortium (including the Commission Services)	48

Description of deliverables

D6.1 : Report of the mass, energy balance and techno-economic studies [30]

Written report linked to task 6.1on techno-economic studies including mass and energy balance

D6.2 : Interim report of the carbon foot- print and LCA [20]

Written interim report linked to task 6.2 reporting results and analysis of sustainability assessments based on RECOVER approaches.

D6.3 : Final report of the carbon foot- print and LCA [48]

Written report linked to task 6.2 reporting results and analysis of sustainability assessments based on RECOVER approaches.

D6.4 : Report on risk assessment including social acceptance assessments [32]

Written report linked to task 6.3 on health, safety and risks aspects as well strategy for RECOVER acceptance by social entities.

D6.5 : RECOVER logistic approach [48]

Written report linked to the logistic approach to indicate the viability of the solutions proposed in RECOVER.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS15	Social and environmental impacts, logistic strategies	3 - ASAJA	42	Environmental, logistic, social impacts and acceptance show viability of RECOVER implementation referencing 3 scenarios of rural development.
MS16	RECOVER Cascade strategy	3 - ASAJA	48	Recommendations and strategies for RECOVER cascade process upscaling towards future full-scale implementation available.

Work package number [9]	WP7	Lead beneficiary [10]	16 - ENCO
Work package title	Cross-cutting activities and Innovation impats		
Start month	1	End month	48

Objectives

● To lay down the basis for the establishment of RECOVER novel biotechnological solutions value chains and set up suitable business models to catch new market opportunities.
● To maximise the innovation impacts of the project for contributing to the uptake of the project results for growth and jobs by properly managing and protecting the RECOVER knowledge and results and establish a robust plan for the communication, dissemination and exploitation of RECOVER results.
● To connect in a strategic dialogue general public and policy makers to lobby for the new soil biodegradability standard at the EU level, as well as policy recommendations.

Description of work and role of partners

WP7 - Cross-cutting activities and Innovation impats [Months: 1-48]
ENCO, UAL, UNIPI, ASAJA, BRUNEL, UMH, ASU, NATURPLAS, ASA, NUTRINSECT, IRIS, IDELUX, FEMTO, ENTO, MANOR, SAV, OWS RF
Task 7.1 Innovation Management (M1-M48)
This task will ensure that an appropriate and systematic approach to Innovation Management is implemented during the project. The Innovation Committee will ensure that the innovation process (described in Section 3.2) is appropriately managed. During the meetings of the Innovation Committee, the Exploitation & IPR manager (ENCO) and the Communication & Dissemination manager (ASAJA) will co-chair the working session on Innovation Management activities. As a link between excellent technical results and market demand, the main outcome will consist in identifying real business opportunities, the related customer segments and which value propositions they require. The monitoring of the evolution of the results towards market readiness via a mapping of the TRL evolution for each KER will also be implemented (in link to milestones where applicable).
Leader: ENCO. Partners involved: all.

Task 7.2 Roadmap for the set up of RECOVER value chains and new cross-sector interconnections in the bio-based economy (M1-M24)
This task seeks to boost new bio-economy value chains based on RECOVER results, by strategically provide impetus to all the key actors along the value chain. The societal transition towards bioeconomy implies structural remodelling of the value chain players as discussed in section 2.1 in relevance to the expected impacts in the call topic. Thus, a stakeholder analysis (identifying and prioritizing who to mobilise and engage with) will be shaped approaching key actors across all stages of the value chains. The analysis will be based on a 3-step process: i) Mapping Stakeholders, ii) Recruiting Stakeholders (Acquisition) iii) Managing Stakeholders (Retention) considering their knowledge, interests, positions, alliances, relationships, key roles and responsibilities. To connect in a collaborative dialogue the identified players along the value chains appropriate measures will be adopted as 1) Workshops, roundtables, open days, action groups to bring together key players in new cross-sectorial value chains and maximise their pull effect. 2) Online consultations and pools, surveys and interviews in order to gather relevant information to feed into subsequent activities of this WP. Thereafter, a roadmap to create new cross-sector interconnections in bio-based economy will be designed to facilitate the establishment of new bonds among different sectors that may benefit from RECOVER innovation. It will be built on top of the analysis of the following settings: i) Economic & Market: insight of biotech & plastics value chains structure (quantity and quality), demand and supply, Users/Consumer perceptions and new potential business opportunities/ Market drivers/barriers. ii) Socio-cultural: Public awareness, Environmental awareness, Human health, Specialized knowledge/skills available; iii) Policy & Regulation: current political framework for plastics EoL, beneficial/detrimental legislation/initiatives, Government endorsed progressive programs, Prioritization on political agenda; iv) Finance: Mobilisation of capital, Favourable/ adverse environment, Policy incentives, Financial models and instruments. In addition, needs and expectations of consumers - which shall contribute to speeding up the time-to-market of the project solutions - will be investigated considering gender aspects.
Leader: ENCO,. Partners involved: ASAJA

Task 7.3 Knowledge and IPR management (M1-M48)

This task intends to capture and effectively manage the IP used in RECOVER project and generated by it. Measures and rules for knowledge management and protection will be set up to ensure an effective dissemination and a fair commercial/non-commercial exploitation of the project results. The IPR management will be carried out all along the project schedule to derive the greatest commercial value out of the intangible assets and intellectual capital. Patent watch will be performed periodically to confirm the novelty of the proposed R&I work and guarantee that the consortium partners are free to operate. The background information and knowledge held by the partners will be secured according to the IPR strategy (IPR ownership, Access rights, Non-Disclosure Agreement /Material Transfer Agreement if any 3rd party need to test the novel materials we aim at protecting, etc.). The results generated will be regularly screened in terms of novelty, patentability and protection to define the most appropriate IPR scheme to secure the project outputs. After assessing the commercial potential, all protectable results will be protected (patent, copyright, trademark, etc.) by the owner or joint owners. Business information or valuable know-how that cannot be protected by patents or other forms of registered protection will be protected via contractual mechanisms (NDA/MTA) or trade secret. A policy on publication of results will be implemented where a clearance procedure will be followed by all parties, to mitigate against the risk of publicly disclosing information before patent protection is in place and avoid that presentations/publications may hinder/jeopardise patent protection. The IP & Exploitation Manager will work closely with the knowledge owners to assist them in fulfilment of the IPR rules, as well as negotiating any additional agreements that might be necessary in line with developments as they unfold (Joint Ownership Agreement(s), etc.). The IP & Exploitation Manager will also support the inventors and knowledge owners in acquiring further insight on to how to deal with their new inventions.
Leader: ENCO. Partners will be involved: All.

Task 7.4 Exploitation and Dissemination Planning (M12-M48)
This task seeks to establish a Plan for the Exploitation and Dissemination of results (PEDR) to boost the impacts of the RECOVER action, paving the way for the broader post-project uptake of all tangible and intangible Key Exploitable Results (KERs) generated. The PEDR will detail how each of the KER will be used in commercial/non-commercial direct/indirect exploitation activities for business growth and competitiveness following project completion. Gender factors will be also contemplated in all stages of this activity.
Subtask 7.4.1 Business and exploitation planning.
All the possible potential exploitation routes for each KER will be mapped in support of the transition to the post-project phase (to advance the TRLs and reach the market). This plan will be drawn up as follows: i) KERs and their applications: development status, SWOT analysis, differences from competing products/services, main IPR issues to be clarified. ii) Market: from T7.2, including definition of strategies and actions to align the RECOVER value proposition to catch trends. iii) Technology watch: sight towards relevant technological developments and innovations from third parties with the aim to timely react to these trends and turn collected information into knowledge to feed the project. iv) Financial projections: investment needed for covering the costs, source of further funding to be identified focusing on applications where a return on investment is feasible, complementary or parallel projects. Pre-commercial procurement as a means to complete market-oriented exploitation processes by creating demand will be also monitored. The avenues for licensing the RECOVER solutions will be studied as viable option to support a wider adoption.
Leader: ENCO. Partners involved: All.
Subtask 7.4.2 Standardisation, certification, policy & regulatory framework
Relevant standardisation and certification landscape for RECOVER will be permanently assessed at international, EU and national level. The results from WP4/5 will be evaluated in terms of their compliance with applicable standards by their responsible partner. RECOVER findings will be analysed overall in the light of the different standard (e.g. regarding biodegradation, fertilisers, etc), certification (e.g. sustainable sourcing for bio-based materials) and labelling schemes. An in-depth exchange will be established with the relevant standardisation bodies, technical committees (CEN/TC 411-Bio-based products and CEN/TC 249 - Plastics), and with the most important independent certifiers in the field of bioplastics (Vinçotte and DIN CERTCO). In this task, ASAJA and OWS RF/IDELUX, will curate the provision of policy recommendations regarding waste management in different scenario (in/ex situ), that could be used by standardization bodies in the reflection about new/ improved standards. The outcome of the evaluation of the project results, especially in terms of end of life and LCA, will culminate in policy recommendations and position papers addressed to concerning relevant actors featured in D7.6.
Leader: ASAJA. Partners involved: OWS RF
Subtask 7.4.3 Dissemination of results.
All non-confidential information and knowledge generated within the project will be properly distributed. This will include: Publishing scientific articles and manuscripts in open access journals (OAJ), submitting papers in conferences (subject to the consortium's policy on publication of results consistent with IP protection rules. Organizing dedicated sessions & scientific workshops as a part of 3rd party large events with significant impact. Targeted transmission of articles in relevant trade and industry-focused media outlets. Establishment of synergies and cross-fertilisation with other projects, initiatives and networks in EU and beyond. This will guarantee knowledge interchange and reuse between

previous and future projects. Interaction with the other BBI projects will be particularly sought as well as interacting with project partners' respective regions/value chains. Policy briefs and case studies targeting policy makers.
Leader: ASAJA. Partners involved: All
Subtask 7.4.4 Knowledge transfer & Upskilling
A set of training modules will be prepared and delivered among partners (onsite training to the new technologies, documents sharing, transfer of knowledge within meetings) for ensuring that there is sufficient knowledge and competence transfer opportunity during the project. This also guarantees that the consortium is prepared for the future post-project use of the results. In addition, formal external training sessions will be delivered in web streaming and targeted at selected attendees such as scientists, end user industry and policy makers to ensure the RECOVER knowledge is exchanged, providing foresight at benefits gained from the new solutions. This educational effort will facilitate understanding, acceptance and future uptake and exploitation of the results of RECOVER. Feedbacks will be collected used to feed the project and improvements for post-project. During these sessions, partners will show illustrative video, studies and best practice materials. RTDs involved in educative actions will also include the project methodology in their lectures and involve PhD/Master students. All the materials developed will be freely available.
Leader: UNIPI. Partners involved: All

Task 7.5 Communication Campaign (M1-M48)
An effective communication strategy will be established to connect and engage with people around the project topics. This will be designed as a branding campaign where storytelling will be accurately designed on top of agreed key messages and conveyed through specific channels. In all cases, messages will be shaped on top of the benefits of RECOVER, educating the public about the end-of-life scenarios offered by our project. Appropriate measures (reported and Annexed to the PEDR) comprise: Graphics identity (M1). Cornerstone for a public communication, original and creative graphics will be developed to allow public to easy identify, and identify with, the project. The audiences' attention will be caught by using visual content, GIFs, icons imagery. Project website (M3). It will present the project brand and circulate relevant information about RECOVER, easy to understand infographics, links to other relevant websites, project news, educational materials, videoclips, etc. It will serve to follow up the development of the project and to spread the non-confidential science results leveraging emotion and passion of general public. It will also host the RECOVER Green Open Access Channel where project public reports, deliverables, publications will be made easy available to the audience. A SEO (Search Engine Optimization) study will be carried out in order to improve the project placement within the search engines as well as a detailed analysis of the web site visitors will be carried out. Project Roadshow of events, scheduling an agenda of EU and international events as trade fairs, exhibitions, round tables, etc., as well as the organisation of dedicated RECOVER events and competitions. Mainstream media (MSM) campaign (M1) pushing briefings and press releases to key media contacts (TV, radio and written press) to amplify the outreach of our core messages. The Cordis Wire service will be used to submit the project press releases facilitating the accessibility for journalists and other interested readers, as well as the communication channels of the Bio-Based Industries Consortium (BIC) and the National Contact Points (NCPs). Social media (M2). An extensive social media campaign (Pinterest, Twitter, and LinkedIn) will be implemented, capitalising on the extraordinary power of engaging audiences in a two dialogue. Digital and printed material (M2). Project flyers, posters, booklets, gadgets will be created and distributed at any events attended in order to support project communication and branding actions. E-newsletter. A 6-monthly newsletter will be transmitted to the project audiences. It will contain news and information on the project, upcoming events/conferences attended/organised by partners. RECOVER final event (M47) will be organised to showcase the project outputs. Particular efforts will be paid to approach and liaise with general public as well as policy makers, who have a strategic role in magnifying the impact of RECOVER communication results (also in line with previous task). Communication will be also streamlined through Horizon2020 and BBI communication channels.
Leader: ASAJA. Partners involved: ENCO for website and materials design, co-organisation of the final event. All partners for content, leveraging their communication department via their existing corporate newsletters, blogs, social media channels.

Participation per Partner

Partner number and short name	WP7 effort
1 - UAL	
2 - UNIPI	
3 - ASAJA	

Partner number and short name	WP7 effort
4 - BRUNEL	
5 - UMH	
6 - ASU	
7 - NATURPLAS	
8 - ASA	
9 - NUTRINSECT	
10 - IRIS	
12 - IDELUX	
13 - FEMTO	
14 - ENTO	
15 - MANOR	
16 - ENCO	
17 - SAV	
18 - OWS RF	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D7.1	Project website	16 - ENCO	Websites, patents filing, etc.	Public	4
D7.2	Communication Plan	3 - ASAJA	Report	Public	6
D7.3	Draft Plan for the Exploitation and Dissemination of the Results	16 - ENCO	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D7.4	Report on RECOVER value chain	16 - ENCO	Report	Public	12
D7.5	Final Plan for the Exploitation and Dissemination of the Results	16 - ENCO	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D7.6	Industry, stakeholders and policy guidelines regarding RECOVER waste management	3 - ASAJA	Report	Public	48

Description of deliverables

D7.1 : Project website [4]

Website giving access to the public to the project related news, goals, etc.

D7.2 : Communication Plan [6]

Written report including detailed communication intention and strategy.

D7.3 : Draft Plan for the Exploitation and Dissemination of the Results [18]

Strategy and measures to disclose and use the project results.

D7.4 : Report on RECOVER value chain [12]

Results of the survey of value chains where RECOVER results could apply.

D7.5 : Final Plan for the Exploitation and Dissemination of the Results [48]

Final updated version of D7.3, including the business plans, further Research Roadmap as well as report documenting the IPRs and patent status, communication & dissemination activities carried out, as well as policy recommendations, with Final Joint Ownership Agreements annexed. Including the report on the performed training activities annexing all materials used.

D7.6 : Industry, stakeholders and policy guidelines regarding RECOVER waste management [48]

Report on how to optimally manage RECOVER waste based on the analysis of the current waste management landscapes, results of EoL tests and LCA, annexing policy recommendations and position papers.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS17	Successful dissemination and exploitation	16 - ENCO	48	Final Joint Ownership Agreements, and business plan including a roadmap for commercialisation of project results.
MS18	Industry, stakeholders and policy guidelines	3 - ASAJA	48	Publication of "Guidelines for Industry, stakeholders and policy regarding RECOVER waste management" (optimal manage RECOVER waste).

Work package number [9]	WP8	Lead beneficiary [10]	1 - UAL
Work package title	Project Management		
Start month	1	End month	48

Objectives

● To carry out a comprehensive series of project management activities in order to manage and optimise the application of resources in this large, multicultural and multidisciplinary consortium by creating an appropriate management framework linking together all the project components and ensure that the contractual communication & reporting requirements are met.
● To coordinate the bodies involved in the project management structure
● To ensure the project management procedures are implemented
● To be committed to ethic requirements
● To coordinate with and report to the European Commission/ BBI-PPP

Description of work and role of partners

WP8 - Project Management [Months: 1-48]
UAL, UNIPI, ASAJA, BRUNEL, UMH, ASU, NATURPLAS, ASA, NUTRINSECT, IRIS, OWS, IDELUX, FEMTO, ENTO, MANOR, ENCO, SAV, OWS RF
The structure of the management system is discussed in detail in Section 3.2. It will involve the integration of effort between all partners. UAL will lead this WP and all partners will be involved to ensure the smooth running of the project. Note: this WP does not involve the scientific and technical management of individual WPs. This activity is performed as an integral part of each WP.

Task 8.1 Management of project resources (M1-M48)
The activities that will be involved in the management of the project resources include: i) Coordination and careful monitoring of the activities of the project (with the assistance of a project management tool); ii) Organising the dates, agenda and documenting the minutes of all meetings of the General Assembly (GA); iii) Overseeing the different managerial positions (TMC, Innovation Committee), iv) Ensuring that the Decision making procedures for the different management bodies of the management structure (as defined in Section 3.2) are respected at all times; v) Tracking and reporting progress; vi) Financial administration of the project and its funds; vii) Administrative management of the project; viii) Ensuring all partners adhere to the contractual obligations of the Grant Agreement; iix) Responsibility for maintenance of the Consortium Agreement. UAL will lead this task with support from ENCO and all partners will be involved to ensure the robust management of resources. Each partner must ensure effective use of their own resources through careful internal project management. Internal reports will frequently be prepared by WP leaders (at least quarterly) to fuel in the contractual reporting to the EC/BBI-JU in Task 8.3.
Leader: UAL. Partners involved: All

Task 8.2 Implementation of the management procedures (M1-M48)
The Project Coordinator will ensure that all partners are fully briefed on their role in its implementation. Different critical project management procedures will be managed in dedicated aspects:
Risk Management: Risk Management procedures will be in place, whereby the status of the Critical Risks identified in Table 3.2b will be continually reviewed, and additional risks and mitigation strategies will be devised as appropriate. Contingency plans will be brought into play in an early and timely fashion, in consultation with the TMC.
Data management: According to the data management plan set up in section 2.2 that will be further refined during the project implementation, all data, both for internal curation will be hosted in the project intranet and those benefiting from Open Access (publications, selected LCA and validation results among others possibly after anonymisation of the data regarding e.g. the formulations involved) for public distributions will be available from the project website as well as in OpenAire or Cordis among others respectively in case of publications and deliverables.
Responsible research and innovation (RRI) management: Overseeing ethical (in coordination with next task), societal issues (in link with WP6), regulatory aspects (WP7) and liaising with WP7 in terms of checking the proper outreach to policy makers, public and training. Overseeing the promotion of gender equality in the project. In addition, the proper implementation of the open Innovation Management activities will be checked, and in link with WP7, the Coordinator will assist the IPR & Exploitation Manager review all progress and proceedings with regard to IP protection, sharing, etc.

Management of the project AB: UAL with support from the exploitation manager ENCO will engage, keep updated and organise the collection of the required inputs from the AB as will be reported in the related deliverables.

Leader: UAL. Partners involved: All

Task 8.3 National Ethics Councils (NEC) requirements (M1-M48)
The coordinator will analyse the ethics requirements related to the activities of the project, monitor that appropriate procedures will be put in place by all partners to ensure compliance with these requirements and report on these procedures. Areas of focus will include:
• E1. Research involving work with humans beings
• E2. Research using, producing or collecting human cells or tissues.
• E3. Research involving animals.
• E4. Research that may adversely affect the environment, and the health and safety of researchers involved
• E5. Research which involves collecting or processing personal data
• E6: Research in non-EU countries, in particular regarding issues of import/export of materials (e.g. chemicals) and any other potential ethical issues related to research activities conducted by non-EU partners.
Ethics issues will be reviewed by an independent external expert.
Leader: UAL, Partners involved: All

Task 8.4. Ethic Position Questionnaire (EPQ) requirements (M1-M48)
The coordinator in cooperation with the partners will identify, monitor and report procedures put in place to ensure compliance with ethics requirement defined in 8.3. Detailed information on ethical aspects and procedure put in place by partners to deal with ethical requirements will be collected and analysed on a regular basis using specific questionnaires.
Leader: UAL Partners involved: All

Task 8.5: Coordination with and Reporting to the BBI JU (M1-M48)
The Project Coordinator is the sole point of contact between the EC/ BBI-JU and the project consortium. The Coordinator must ensure that all milestones are met and that all reports are prepared and delivered to the EC in accordance with contractual requirements. UAL will execute this task that will be greatly facilitated by the internal management plan and reporting described in the previous tasks. In addition, the PC will be in charge to collect partners' data and report on the yearly BBI questionnaire showing the evolution of the project and partners contribution to the KPIs set in the BBI SIRA.
Leader: UAL. Partners involved: All will assist in reporting

Participation per Partner

Partner number and short name	WP8 effort
1 - UAL	
2 - UNIPI	
3 - ASAJA	
4 - BRUNEL	
5 - UMH	
6 - ASU	
7 - NATURPLAS	
8 - ASA	
9 - NUTRINSECT	
10 - IRIS	
11 - OWS	
12 - IDELUX	
13 - FEMTO	
14 - ENTO	

Partner number and short name	WP8 effort
15 - MANOR	
16 - ENCO	
17 - SAV	
18 - OWS RF	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D8.1	Project Management Tool	16 - ENCO	Other	Confidential, only for members of the consortium (including the Commission Services)	1
D8.2	Project management Plan	1 - UAL	Report	Confidential, only for members of the consortium (including the Commission Services)	3
D8.3	Report on National Ethics Councils (NEC) requirements	1 - UAL	Report	Confidential, only for members of the consortium (including the Commission Services)	3
D8.4	Report on Ethic Position Questionnaire (EPQ) requirements	1 - UAL	Report	Confidential, only for members of the consortium (including the Commission Services)	3
D8.5	Data management plan	16 - ENCO	ORDP: Open Research Data Pilot	Confidential, only for members of the consortium (including the Commission Services)	6
D8.6	First Report from Advisory Board	16 - ENCO	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D8.7	Second report from Advisory Board	16 - ENCO	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D8.8	KPI & Impact Questionnaire 1	1 - UAL	Other	Confidential, only for members of the	16

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
				consortium (including the Commission Services)	
D8.9	KPI & Impact Questionnaire 2	1 - UAL	Other	Confidential, only for members of the consortium (including the Commission Services)	28
D8.10	KPI & Impact Questionnaire 3	1 - UAL	Other	Confidential, only for members of the consortium (including the Commission Services)	40
D8.11	KPI & Impact Questionnaire 4	1 - UAL	Other	Confidential, only for members of the consortium (including the Commission Services)	48

Description of deliverables

D8.1 : Project Management Tool [1]

Online repository restricted to the consortium for communication and document sharing

D8.2 : Project management Plan [3]

Defines the project monitoring, conflict, risk management, deliverable review procedures and templates, etc. to be used by the consortium for a smooth project implementation

D8.3 : Report on National Ethics Councils (NEC) requirements [3]

Report on National Ethics Councils (NEC) requirements

D8.4 : Report on Ethic Position Questionnaire (EPQ) requirements [3]

Report on Ethic Position Questionnaire (EPQ) requirements

D8.5 : Data management plan [6]

Defines how the project results will be curated, made accessible where applicable

D8.6 : First Report from Advisory Board [18]

Report on the feedback from the Advisory Board received along the project.

D8.7 : Second report from Advisory Board [36]

Report on the feedback from the Advisory Board received along the project.

D8.8 : KPI & Impact Questionnaire 1 [16]

The deliverable will consist of the questionnaire with both qualitative and quantitative information, indicating the expected results by the end of the project.

D8.9 : KPI & Impact Questionnaire 2 [28]

The deliverable will consist of the questionnaire with both qualitative and quantitative information, indicating the expected results by the end of the project.

D8.10 : KPI & Impact Questionnaire 3 [40]

The deliverable will consist of the questionnaire with both qualitative and quantitative information, indicating the expected results by the end of the project.

D8.11 : KPI & Impact Questionnaire 4 [48]

The deliverable will consist of the questionnaire with both qualitative and quantitative information, indicating the expected results by the end of the project.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

Work package number [9]	WP9	Lead beneficiary [10]	1 - UAL
Work package title	Ethics requirements		
Start month	1	End month	48

Objectives

The objective is to ensure compliance with the 'ethics requirements' set out in this work package.

Description of work and role of partners

WP9 - Ethics requirements [Months: 1-48]
UAL
This work package sets out the 'ethics requirements' that the project must comply with.

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D9.1	H - Requirement No. 1	1 - UAL	Ethics	Confidential, only for members of the consortium (including the Commission Services)	6
D9.2	POPD - Requirement No. 2	1 - UAL	Ethics	Confidential, only for members of the consortium (including the Commission Services)	6

Description of deliverables

The 'ethics requirements' that the project must comply with are included as deliverables in this work package.

D9.1 : H - Requirement No. 1 [6]

-The procedures and criteria that will be used to identify/recruit research participants must be submitted as a deliverable. -The informed consent procedures that will be implemented for the participation of humans must be submitted as a deliverable.

D9.2 : POPD - Requirement No. 2 [6]

• A description of the technical and organisational measures that will be implemented to safeguard the rights and freedoms of the data subjects/research participants must be submitted as a deliverable. • Detailed information on the informed consent procedures with regard to data processing must be kept on file. • Templates of the informed consent forms and information sheets (in language and terms intelligible to the participants) must be kept on file.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

1.3.4. WT4 List of milestones

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS1	Mapping of AWP and their characteristics	WP1	12 - IDELUX	6	AWP inventory and characterization ready as a basis for the selection of samples of 4-6 AWP types (including in each case virgin plastics and AWP counterparts picked individually from MSW or agricultural films) for later project activities.
MS2	Strategy for AWP collection, monitoring & sorting	WP1	2 - UNIPI	12	Matrix for the definition of the classification, characterization and procedure for obtaining AWP available (will feed in WP6).
MS3	Pre-treatment rules and procedures basing on composition of mixtures of AWP	WP1	10 - IRIS	12	Abiotic pre-treatment to be used depending on the mixture of AWP to biodegrade (fragmentation, micronization, thermal treatments, UV irradiation, oxidizing agents, pH and irradiation) defined and samples of 4-6 selected AWP after pretreatment available
MS4	Microorganisms and enzymes for AWP degradation	WP2	1 - UAL	12	Biodegradation capabilities of microbial consortia identified as well as a set of 3-4 microbial consortia and a preliminary selection of at least 2-3 enzymes for later project activities.
MS5	Novel synthetic enzymes	WP2	4 - BRUNEL	24	Novel enzymes samples available and new strategy for selected / directed evolution
MS6	Selection and pre-compatibility test of macro-meso-micro consortia	WP2	5 - UMH	18	Earthworm and insect selected and 2-3 microbial/ enzyme pool for each earthworm and insect according to pre-compatibility results (available for activities in WP3&WP4).
MS7	Formulations and production of stable plastic-degrading microbial/enzyme consortia (PMC)	WP3	8 - ASA	24	Set of at least 2-4 stable PMC formulations to be used as enhancers of AWP degradation consortia available for WP4

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS8	Insect and earthworm plus PMC with enhanced AWP degrading capability	WP3	5 - UMH	24	Protocol for fortified insect (FP-insect) and earthworm (FP-earthworm) formulations preparation validated and a set of at least 1-2 FP-earthworm and 2-5 FP-insects formulations for use in WP4.
MS9	Enhanced bioremediation process	WP4	18 - OWS RF	30	Results obtained in common conditions (temperature, moisture, aeration) adjusted to maximize the degradation rate and upgrade the operational boundaries. Process conditions for AWP degradation by PMC and insects to maximize the degradation rate as well as basic operational conditions for ex-situ and in-situ plastic treatment defined.
MS10	Defined up-scaled ex situ and in situ AWP biodegradation	WP4	12 - IDELUX	36	Defined standardized strategy to insects-earthworms-microorganisms consortia AWP degrading activity in two main scenarios: a) ex situ by composting, vermicomposting and insect rearing b) in situ by using vermicompost in soil reaching a plastic degradation >80%
MS11	Biodegradation Models	WP4	10 - IRIS	36	A decision support tool derived from developed models allow supporting the definition of the optimum biodegradation approach
MS12	Up-graded biofertilizers formulation	WP5	14 - ENTO	36	Formulation matrix of several industrial mixtures of up-grade bio-fertilizer products compared with similar and conventional products.
MS13	Chitin-based extraction processes	WP5	2 - UNIPI	42	Verified sustainability of various extraction approaches of chitin (to chitosan conversion and uses).
MS14	Chitin/chitosan films and containers	WP5	2 - UNIPI	48	From mixture of polymeric matrices and chitin/chitosan from RECOVER production of prototypes of films for food packaging and/ or for

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
					mulching, and agriculture items such as pots, yarns, etc
MS15	Social and environmental impacts, logistic strategies	WP6	3 - ASAJA	42	Environmental, logistic, social impacts and acceptance show viability of RECOVER implementation referencing 3 scenarios of rural development.
MS16	RECOVER Cascade strategy	WP6	3 - ASAJA	48	Recommendations and strategies for RECOVER cascade process upscaling towards future full-scale implementation available.
MS17	Successful dissemination and exploitation	WP7	16 - ENCO	48	Final Joint Ownership Agreements, and business plan including a roadmap for commercialisation of project results.
MS18	Industry, stakeholders and policy guidelines	WP7	3 - ASAJA	48	Publication of "Guidelines for Industry, stakeholders and policy regarding RECOVER waste management" (optimal manage RECOVER waste).

1.3.5. WT5 Critical Implementation risks and mitigation actions

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
1	Difficulties in the selection of APW. Probability: medium. Impact: high.	WP1	This risk is highly mitigated by the presence of farmers and food producers in the consortium. Nevertheless, as the expected plastic types used in agriculture and packaging are very broad. Priority will be given to AWP of interest from RECOVER consortium and of high impact on EoL and environment.
2	Difficulties in finding biotech tools for plastic degradation. Probability: low. Impact: high	WP2	This risk is mitigated by the results of previous researches and by the wide range of biotech tools that will be screened, including microorganisms, novel enzymes, insects and earthworms. In addition to the well-known biotech tools already available at BRUNEL and UAL, novel microorganisms and enzymes will be produced that will mitigate this risk. Also, RECOVER partners are very skilled in the isolation and selection of new microorganisms (UAL), development of novel enzymes (Brunel), earthworms (UMH) and insect (UAL)
3	Difficulties to obtain significant synergistic effects of PMC incorporated to earthworm/insects to plastic degradation. Probability: low. Impact: medium	WP3	This risk is mitigated by the preferential use of plastic-degrading microorganisms isolated from the earthworm/insect gut to enrich the gut natural microbiome. These will likely colonize the earthworm/insect gut much better than microorganisms coming from other environments (compost, soil) and will be used to replace them if gut colonization is not demonstrated in WP2. Anyway, synergy, colonization capability and compatibility are considered as key targets for the selection of PMC as part of WP2 tasks. The same applies to exo-colonizing microorganisms. Other factors that will be also considered for the success of the synergy effects PMC/ earthworms-insects in WP3 include the amount of inoculum, the requirement of reinoculation and the management conditions of the earthworms/ insects (by providing earthworm/insects-worms growth conditions that stimulate plastic-degrading native microbiome) needed to ensure that the symbiosis is settled and remains over time
4	Too low efficiency of plastic degradation. Probability: high. Impact: high	WP4	The operational parameters will be adjusted to increase the efficiency rate and select the best conditions.
5	Only specific types of plastics are degraded, while process should tackle broad range of plastics. Probability: medium. Impact: medium.	WP4	One of the biggest advantages of biodegradation-conversion of AWP by consortia of microorganisms-insects is the possibility of treating highly heterogeneous mixtures with organic traces and at room temperature. In that extreme case we could redefine the conditions of the plastic mixtures to biodegrade and analysing the components that have reduced

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
			the biodegradation capacity of the proposed process. (Almost certainly very atypical traces not common to most AWPs).
6	Loss of antimicrobial activity even after parameter adaption. Probability: medium. Impact: medium	WP5	Blending with chitosan of high molecular weight from other sources or additives.
7	Low production of chitin and chitosan, and of low quality Probability: medium. Impact: low	WP5	The production of chitin is directly proportional to the production of insects in the pupae phase, increasing the populations of insects and their feeding with AWP will directly increase the amount of chitin obtained and its possibility of obtaining the quantities required by the pre-industrial deacetylation processes necessary for conversion into chitosan.
8	Collection and transport of plastic waste complicated and costly. Probability: high. Impact: medium	WP6	Several scenarios of plastic collection and transport to degradation facilities will be considered from the very beginning of the project. Wide communication with farmers association (generating the waste), food producers and with waste-treatment plants will be kept in the project. Strategy for AWP pre-treatment (volume reduction, milling etc) before transport will be proposed. The possibility of making the withdrawal, pre-treatment and recovery of AWPs profitable for end users makes their involvement in the tasks indicated in the future very favourable
9	Confrontation among partners for IPR issues. Probability: low. Impact: high	WP7	This risk is considered low as it has been mitigated by establishing IPR rules at project stage as is documented in Section 2.2. In addition, before project start all the partners sign the Consortium Agreement that will address project IPR. The IPR & Exploitation manager with support from the Project Coordinator, with the assistance of consortium members, will control the proper exploitation of project results and associated IPR issues and when required a voting procedure is in place
10	Discrepancies in the technical approaches. Probability: low. Impact: high	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8, WP9	Regular communication among WP leaders to solve all technical issues. The Scientific &Technical Manager with support from the Project Coordinator, with the assistance of consortium members, will control this issue and when required a voting procedure is in place.
11	Problems in respecting the planned activities and resources allocation. Probability: medium. Impact: high.	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8, WP9	This risk will be mitigated by the implementation of the Technical Work plan Review and Update Procedures

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
12	Management issues in relation to the coordinated execution of the technical tasks due to the multi-disciplinary nature of the Project and its challenging scope. Prob.: medium. Impact: high	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8, WP9	Constitution of a Science and Technical Committee that will be chaired by the Scientific &Technical Manager, with relevant representatives of the most relevant disciplines in order to assure the required leadership.

communication (graphic identity, newsletters, press releases, promotional materials, gadgets, etc) and 2. 'Pull' communication and interactive (website, social media, liaison with other initiatives, joint communication, etc). The fact that our demonstrators are in part focused on consumers applications like packaging and in the broad part on plasticulture will allow maximising the awareness of the impact of bioeconomy, and indeed BBI programme, on their everyday life. In RECOVER, collaboration with the European Climate Adaptation Platform (Climate-ADAPT)[152] and agricultural European Innovation Partnership EIP-AGRI[153] will be particularly pursued to magnify the replicability of the project results, by sharing the project experience and lessons learned with stakeholders while simultaneously learning how these actors are managing and reacting to similar problems focused primarily on the problem of AWPs. In addition, to provide added value to the RECOVER end-users, EIP Multiactor Practice Abstracts will be prepared (in all partner languages) - 1.000 to 1.500 characters) highlighting main practical recommendation (how can practitioner make use of the RECOVER results).

Table 2.7: RECOVER Communication Toolbox and Approach

Approach	Measure	Timing	Impact	Key Metrics
Push Effect	Logo & Project Identity	M1	Recognizable project identity that will help increase awareness about the project	At least 1 logo, 2 templates, 1 set of imagery
	E-Newsletter	6-monthly	Create community sense & Give visibility to events and activities related to the project.	At least 150 enrolments per year
	Press Releases & Media presence (TV, Radio)	6 per year	Achieve maximum awareness of the project & share project vision	At least 5 press-releases, 10 articles and 5 interviews in media (Radio/TV)
	Digital/Printed Material	M1-M12-M24-M36-48	Create awareness of the project. Spread project results.	At least 3 flyers designed & 1500 printed; 3 posters; 3 infographics
	Videoclip	M30	Showcase benefits and results	At least 1000 visualizations
Pull Effect	Project Website	M1, regularly updated	Create awareness, disseminate information about the project.	At least 5k unique visitors
	Social Media	M2, content online	Reach & engage different target audiences	At least 1000 fans
	Interaction with ongoing initiatives/projects	Linking projects, platforms and database	Synergies with cross-cutting initiatives and cross-fertilization	At least 3 officially MoU signed for cross-fertilization
	Participation to events & conferences	At least quarterly from M3	Increase awareness of the project and its results	At least 2 events per partners
	Final Conference	M47	Engage public & stakeholders to amplify outreach of RECOVER	At least 100 attendees

3. Implementation

3.1 Work plan — Work packages, deliverables and milestones

Brief presentation of the overall structure of the work plan

A competitive work plan has been devised in order to deliver on the objectives of the RECOVER project providing tailored end of life solutions for a number of AWP using innovative biotech approaches. This will preserve our environment and resources but also meet the society and regulatory demands. The project objectives have been used as the starting point for planning the work that needs to be implemented within the project (Section 1.1). All the actions that are required to meet the objectives of this Research and Innovation project have been collectively analysed by the interdisciplinary consortium. It was agreed how best to distribute the workload among the various experts in keeping with their relevant fields of knowledge, capacity, infrastructure, resources and in the context of bioplastics supply, packaging textile industry value chains, as well as waste management companies. This bottom-up planning process has greatly facilitated the task of arriving at a clear, comprehensive and realistic work plan (see Gantt Chart and Pert Diagram below), whereby all partners their ownership.

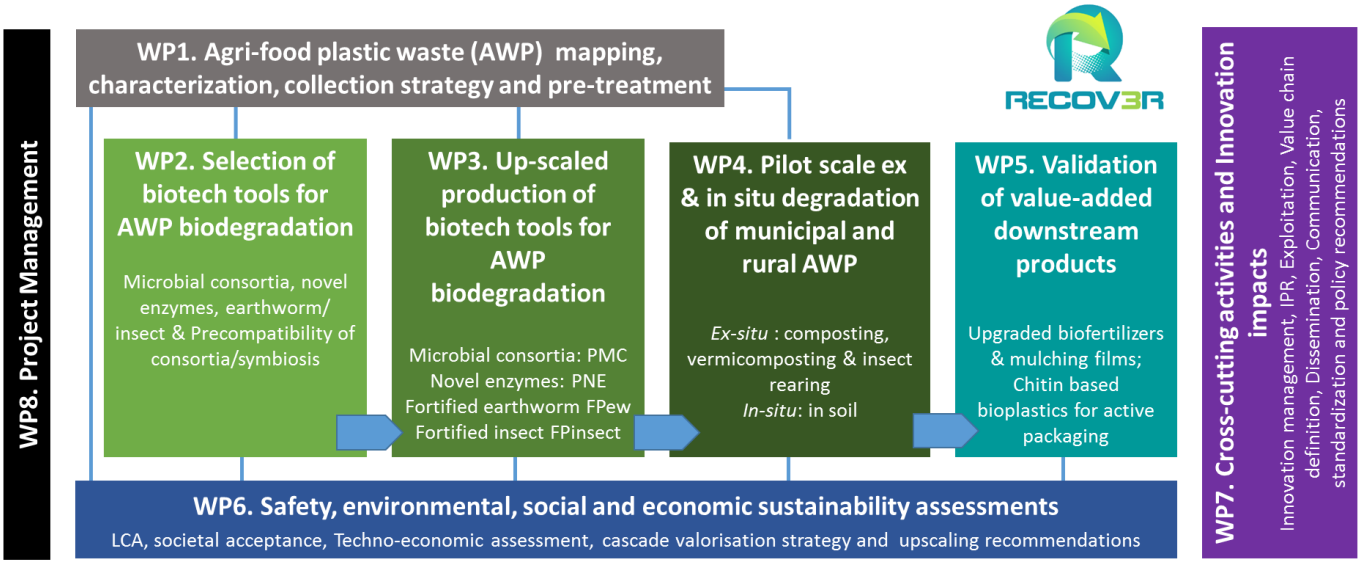
To fulfilling the project objectives, our work plan is split into 8 WPs with clear tasks that will enable the project's objectives to be fulfilled through a coherent structure that will also facilitate project. To ensure the clear focus towards the target application of this Research and Innovation Action building upon past successful research, most WPs are led by industrial partners. WP1 deals with industry-led specification in terms of AWP sourcing to match the market needs. WP2 deals with the lab scale development of the new biotech tools to degrade AWP and WP3 with the upscaling of their production and their combination. WP4 comprises field trials and large scale composting trials to validate the new process in an industrially preliminary scale.

152 https://climate-adapt.eea.europa.eu
153 https://ec.europa.eu/eip/agriculture/

WP5 considers the validation of the by-products of the process into high added values applications for different food and agriculture end uses. WP6 is a transversal stage that will be focus on the environmental and economic sustainability, and other assessment towards the safe and regulatory compliant uptake of the new process by the industry. To complement the first 6 WPs linked with scientific & technical (S&T) development and validation activities WP7 addresses the innovation, exploitation and dissemination activities to prepare the market uptake and maximise the project impact. The last WP, WP8, will address the full management of the project in order to make all the proposed developments in line with defined project and BBI JU objectives.

The work plan is clearly linked to the objectives set out in Section 1.1 and the consortium is confident that this work plan is robust to ensure that the project's S&T and innovation goals are fulfilled. The research and development work is encapsulated in a comprehensive risk assessment and management strategy, sound communication and training to heighten the impact of the knowledge generated, as well as knowledge management to pave the way for planning and implementing the project's exploitation and dissemination pathways, and robust project coordination and ISO certified project management procedures. The work packages and tasks form a complex inter-weaved matrix of actions, many interdependent, with very close interaction between the WP and task leaders, which is essential for the delivery of the full suite of outputs. During the work planning stage, consideration has been given to logistical matters, such as the time needed for ordering parts/components, the time needed for shipping of samples from development to test sites, installation times, the 'what if' scenarios of partial results not arriving, installation work throwing up challenges, delays in necessary inputs/results from feed in tasks or subtasks, etc. The link between WPs in terms of needed output material/sample quantity required by next activities were also accounted for. As this is a highly experienced development team, and in particular with many members of the consortium experienced in applied R&D under tight timeframes, as well as experience in integration work at industrial sites, the work planning has benefited from this 'on the ground' pragmatic experience, especially in terms of consideration of interdependencies and timings. The potential risks have been identified and contingency plans proposed, which are discussed in Section 3.1 under "Critical risks and mitigation measures". Importantly, the collective planning of the work plan has achieved the 'buy-in' of all the project developers and participants and an early understanding of the importance of a team effort, team planning and active communication among all WP and task leaders. Moreover, it is felt that the one-page Gantt itself (which also includes all the deliverables and milestones) will be a very effective tool for high level control and obtaining a "helicopter overview" as part of effective project management during the project.

Pert diagram of the components showing their interdependencies



Gantt chart showing the timing of the different WPs and their components outline

No	WPs and Tasks	Del. & MS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
1	Agri-food-plastic waste (AWP) mapping, characterization, collection strategy and pre-treatment							#						##																																				
1.1	AWP inventory and characterisation	D1.1-D1.3; MS1-MS3						■																																										
1.2	Selection of AWP to be used into the project.							■																																										
1.3	Strategy for AWP collection and sorting													■																																				
1.4	Pre-treatment of AWP.													■																																				
2	Selection of biotech tools for AWP biodegradation													#						#						#																								
2.1	Selection and building of microbial communities for AWP degradation	D2.1-D2.3; MS4-MS6																				■																												
2.2	Design and development of novel synthetic enzymes for AWP biodegradation																					■				■																								
2.3	Preselection of earthworm and insect larvae AWP-degraders																					■																												
2.4	Analysis of microbial community associated with earthworm and insect																					■																												
2.5	Pre-compatibility analysis of earthworm/insect-microorganisms																					■																												
3	Up-scaled production of biotech tools for AWP biodegradation																									##																								
3.1	Production of stable plastic-degrading microbial/enzyme consortia (PMC)	D3.1, D3.2; MS7, MS8																								■																								
3.2	Production of Insects/Earthworms fortified with PMC with enhanced AWP biodegrading capabilities																									■																								
3.3	Development of plastic-degrader formulations.																									■■																								
4	Pilot scale ex and in situ degradation of municipal and rural AWP																															#						##												
4.1	Development of biodegradation processes.	D4.1-D4.3; MS9-MS11																														■																		
4.2	Set up of pilot-scale ex-situ plastic degradation process																																					■												
4.3	Set up of in-situ bioremediation of AWP in soil.																																					■												
4.4	Monitoring of AWP degradation																																					■■												
4.5	Establishing protocol of action of the overall fortified system. Sequential study																																					■												
5	Validation of value-added downstream products																																					#						#						#
5.1	Development of upgrade biofertilizers	D5.1-D5.4; MS12-MS14																										■																						
5.2	Extraction of chitin, and conversion to chitosan																																					■												
5.3	Processing of chitosan and chitin.																																					■						■						■
6	Safety, environmental, social and economic sustainability assessments																																											#						#
6.1	Mass, energy balance and techno-economic studies and development of RECOVER cascade strategy	D6.1-D6.4; MS15, MS16																														■																		
6.2	Life cycle analysis																					■																												■
6.3	Risk assessment and social acceptance																																	■																
6.4	Logistics approach for RECOVER large scale implementation																																																	■
7	Cross-cutting activities and Innovation impacts																																																	##
7.1	Innovation Management	D7.1-D7.6; MS17, MS18																																																
7.2	Roadmap for the set up of RECOVER value chains and new cross-sector interconnections in the bio-based economy													■						■																														
7.3	Knowledge and IPR management																																																	
7.4	Exploitation and Dissemination Planning																																																	■
7.5	Communication Campaign					■		■																																										■
8	Project Management		●👤		¤			●👤 ¤			¤			●👤 ¤			¤			●👤 ¤			¤			●👤 ¤			¤		●👤 ¤			¤			●👤 ¤			¤			●👤 ¤			¤			●👤 ¤
8.1	Management of project resources	D8.1-D8.11	■		■															■																		■												
8.2	Implementation of the management procedures							■																																										
8.3	National Ethics Councils (NEC) requirements				■																																													
8.4	Ethic Position Questionnaire (EPQ) requirements				■																																													
8.5	Coordination with and Reporting to the BBI JU																	■												■												■								■

■ Deliverable / # Milestones / ¤ Technical Management Committee Meeting / 👤 Innovation Committee

4. Members of the consortium.

4.1. Participants (applicants)

4.1.1. Partner 1. University of Almeria (UAL)- Spain

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

The University of Almeria (UAL) is a research-education oriented public university, founded in 1993. It has currently 32 degrees available, 42 Master and 14 PhD programs available, counting with 854 professors and more than 13,000 students, including 600 doctorate students. A total of 14 departments and 11 research institutes are devoted to research and training. UAL has been beneficiary of more than 30 EU projects and it is a member of ceiA3, aimed at the advance and dissemination of agrifood-related activities that also offers a wide variety of technical services to companies and stakeholders. The Research Results Transfer Office (OTRI) links scientific activity with the business world. The European and International Research Projects Office (OPEI), in charge of administrative management of international projects, have extensive experience in managing European Projects and supports the researcher. The Unit called Outreach Unit (CCU- Unidad de Divulgación Científica) helps in dissemination activities. This unit is part of a network on Scientific Culture and Innovation (UCC + i) belonging to the Spanish Foundation for Science and Technology (Public Foundation of the Ministry of Economy Competitiveness). Central Research Services support R+D+i activities by providing instrumental analysis and scientific laboratories with advanced equipment and highly skilled technicians.

The UAL team involved in the project includes researchers from two groups of Dpt. Biology and Geology: BIO-175 group (Experts in Microbiology, Unit of Microbiology, "Development of microbiological techniques for improvement of soil of agricultural interest") and AGR-107 (Experts in insects, Unit Zoology, "Plant Protection in Greenhouse Crops).

BIO-175 is specialized in Microbial Biotechnology. Main research activity has been focused from the last 25 years on the biotechnological valorisation of organic wastes (mainly lignocellulosic wastes) by both composting and bioconversions to value-added microbial products and bioremediation/biodegradation of recalcitrant molecules. As one of the outcomes of this activity, the group has a large collection of microorganism isolated from diverse environments that are useful for recalcitrant molecules biodegradation, biological control and plant- growth promotion. These include microorganisms and consortia that produce ligninases, polyphenoloxidases, lipases as well as dye-decolorizing and other xenobiotics-degrading microorganisms. Biodegradation, compostability, eco-toxicity, lignocellulose biotransformation, enzyme production and analysis, bio-deterioration, bioremediation and microorganisms screening and identification are among the main tests developed so far by the team that could be relevant for the tasks attributed in the project. In particular, the use of polyurethane foam (PUF) either hard or soft was found to be an emerging technology for the production of ligninolytic enzymes by white rot fungi immobilized in this support. UAL in collaboration UNIPI developed this technology that give rise to EU patent Nr. EP2677030. Their activity has been financed by public and private funds through more than 30 national and international projects and led to publication of over 80 peer-reviewed papers, 5 books, 8 book-chapters, 3 patents and more than 130 congress national and international communications.

AGR-107 group has been working in the last 28 years in research, development, and innovation in the following lines: applied entomology, taxonomy, biology and ecology of insects, rearing insect and mite pests and their natural enemies (predators and parasitoids). Evaluation of entomo-pathogens (viruses, bacteria and nematodes). Arthropod pest control technology: Development of biological control programmes against arthropod pests. Mathematical models applied to arthropod pest ecology. The activity has been carried out through 119 research projects and 54 research agreements and contracts with companies and official organizations. The scientific output of the group has been covered by 450 published scientific papers, 37 books, 82 book chapters, and 20 patents.

Main role in the project *WPs envolvement and key task or activities to be done in each*

UAL will support the analysis and selection of biotech tools for APW biodegradation in WP2 where they will be involved in the microbial community development and strain selection (Microbiology experts) as well as insect selection and lab scale studies (Insect experts). In WP3, UAL will participate in the analysis and design of formulations containing AWP-degrading microorganisms and insect larvae, and provide feedback on their performance during pilot scale experiments in WP4 and further downstream processing in WP5. Finally UAL will contribute to providing data for environmental and socio economic sustainability analysis, dissemination activities and exploitation of their results. UAL will coordinate the project with support from ENCO to guarantee the Grant agreement technical and economical fulfilment.

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities*

Dr. JOAQUIN MORENO (*Male*) - Full Professor (1998) – He is the Leader of the BIO-175 group (1988). He received his PhD (1986) in Pharmacy. His main research activity is the study of microorganisms isolated from different environments, mainly composting, that are useful for industrial and environmental applications. He has been the PI of 12 national projects, and 1 international project. He published more than 80 scientific papers and 2 patents.

Dr. MARIA J. LOPEZ (*Female*) - Associate Professor (2002) - She received her PhD (1996) in Pharmacy. Her research is focused on the study of microorganisms that produce lignocellulose-degrading enzymes and the use of microorganisms for bioremediation. She has been the PI of 3 national projects, leader of UAL-team in 3 EU project and 1 USA project. She published 56 scientific papers and 3 patents.

Dr. FRANCISCA SUÁREZ-ESTRELLA (*Female*) - Associate Professor (2018) - She received her PhD (2001) in Biology. Her research is focused on the study of phythopathogen antagonists and Plant Growth Promoting Microorganisms-PGPMs for agronomic applications and the analysis of metabolites from microorganisms and analysis of microbial populations by respirometry. She has been the PI of 2 national projects, she published 42 scientific papers and 2 patents.

Dr. Ing. MACARENA M. JURADO RODRÍGUEZ (*Female*) - Assistant Professor (2019) - She received her PhD (2015) in Agricultural Engineering. Her research is focused on the study of composting, PGPMs for agronomic applications and identification of microorganisms. She published 12 scientific papers.

Dr. Ing. JUAN ANTONIO LÓPEZ GONZÁLEZ (*Male*) - Assistant Professor (2019) - He received his PhD (2015) in Agricultural Engineering. His research is focused on the study of microorganisms involved in microbiology of compost and culture collection maintenance and screening. He published 13 scientific papers.

Dr. Ing. TOMÁS CABELLO GARCÍA (*Male*) – Full Professor (1997) – He is the Leader of the AGR-107 group (1991). He received his PhD (1985) in Agricultural Engineering. His main research activity has been carried out in Agricultural Entomology: Pest species, taxonomy, biology, ecology, biological control. He has been PI or has participated in 45 national project and 2 international project, and also in 54 research contracts. He published more than 157 scientific papers. He has been director or co-director of 12 Doctoral's Thesis and 70 Grade and Master's Thesis.

Dr. PABLO BARRANCO VEGA (*Male*) - Associate Professor (2001) - he received his PhD (1997) in Biology. His research has been developed in several lines of Entomology: biology of crop pests and its control, biology and ecology of cave dwelling and subterranean fauna, and taxonomy in several orders of insect and arachnids. He has been the PI of 4 national projects. He published 128 scientific papers and 1 patent.

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

- AGRIMAX-G.A. 720719 (2016-2020). Agro and food waste valorisation co-ops based on flexible multi-feedstocks bio-refinery. H2020-BBI-PPP-2015-2-1, BBI.VC3.D5-2015.
- SABANA-G.A. 727874 (2016-2020). Sustainable Algae Bio-refinery for Agriculture and Aquaculture. H2020-BG-2016-1, BG-01-2016
- FORBIOPLAST- G.A. 212239 (2008-2012). Forest resource sustainability through bio-based composite development. FP7-KBBE-2007-1.
- Characterization of biological capabilities of the microbiota associated to composting" -AGL2012-36434 (2013-2015). Grant agency: Ministry of Economy and Competitiveness of Spain.
- Development of an environmentally friendly protection of sweet pepper and strawberry. QLK5-CT-2001-70484 (2002-2004). E.U. CRAFT.

Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Prederently 2013-2019 years*

- López-González, J. A., Suárez-Estrella, F., Vargas-García, M. C., López, M. J., Jurado, M. M., Moreno, J. (2015). Dynamics of bacterial microbiota during lignocellulosic waste composting: Studies upon its structure, functionality and biodiversity. Bioresource Technology, 175, 406-416.
- Jurado, M., López, M. J., Suárez-Estrella, F., Vargas-García, M. C., López-González, J. A., Moreno, J. (2014). Exploiting composting biodiversity: Study of the persistent and biotechnologically relevant microorganisms from lignocellulose-based composting. Bio-resource Technology, 162, 283-293.
- López, M. J., Suárez-Estrella, F., Vargas-García, M. C., López-González, J. A., Verstichel, S., Debeer, L., Wierinck, I., Moreno, J. (2013). Bio-delignification of agricultural and forest wastes: Effect on anaerobic digestion. Biomass and Bioenergy, 58, 343-349.
- Ricaño, J., Güerri-Agulló, B., Serna-Sarriás, M.J., Barranco, P., Lopez-Llorca, L.V. (2013). Evaluation of the pathogenicity of multiple isolates of *Beauveria bassiana* (Hypocreales: Clavicipitaceae) on *Rhynchophorus ferrugineus* (Coleoptera: Dryophthoridae) for the assessment of a solid formulation under simulated field conditions. Florida Entomologist, 96 (4), 1311-1324.
- Alarcon, F.J.; Martinez, T.; Barranco, P.; Cabello, T.; Diaz, M.; Moyano, F.J. (2002). Digestive proteases during development of *Rhynchophorus ferrugineus* (Olivier, 1790) (Col.: Curculionidae). Insect Biochemistry and Molecular Biology, 32, 265-274.

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

BIO-175 (Microbiology experts) has three research laboratories (microorganisms lab; molecular biology lab; sample

analysis lab), equipment for microorganism's culture, conservation, identification, enzyme production and analysis [culture chambers, -80ºC freezer, water baths, 2L Minifors bioreactor, microscopes, thermocycler (Biorad), RT-PCR (Biorad), UV transiluminators]; equipment for chemical and enzyme analysis/characterization [ultrafiltration units (Pellicon), spectrophotometers, Qbit 2.0 Fluorometer, microplate spectrophotometer and fluorimeter (EON-Biotek), luminometer (Thermofisher), electrophoresis chambers (Biorad), centrifuges; additional equipment for chemical analysis [COT Shimadzu analyser, Lignocellulose analyser (Atkom)]

AGR-107 (Insect experts) has equipment with two laboratories, one insect breeding chamber, 6 Climate Controlled Cabinets, and one Potter tower.

Other equipment available in UAL for the project at the Central Research Services are: LC/GC-MS (Hewlet Packard SERIES 1100), UPLC/GC-MS-Q-TOF (Waters), FTIR-RAMAN (Bruker Vertex 70), Micro-analyser CHNS (Elementar Vario Micro), Single Crystal Diffractometer (BRUKER AXS SMART APEX), X-ray Powder Diffractometer (D8 ADVANCE-DAVINCI), X-ray fluorescence spectrometer (BRUKER S4 PIONEER), TEM (JEOL JEM 2100), SEM (HITACHI S-3500N) microscopes, confocal fluorescence microscope, NMR (Avance III HD 300 and Avance III HD 500) and DNA/RNA analysis and sequencing equipment (Ion PGM (Personal Genome Machine) sequencer, with Ion OneTouch™ 2; System, from Life Technologies; Hitachi 3500 8 capillary genetic analyzer AB; MyGo Pro Real-Time PCR Instrument; GeneAmp PCR System 9700 Thermocycler)

4.1.2. Partner 2 University of Pisa (UNIPI) Italy

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

The Department of Civil and Industrial Engineering part of Pisa University (UNIPI) is located in Largo Lucio Lazzarino 1, in the center of Pisa. The Industrial Chemistry and Materials Science group, directly involved in RECOVER project activities, consists of about 30 researchers and 10 technicians. DICI was established in 2012, following a major reform of the Italian higher education system, as a merger of the former departments of Aerospace Engineering, Chemical Engineering, Industrial Chemistry and Materials Science, Mechanical, Nuclear and Production Engineering, and a part of the Department of Civil Engineering. The institute research covers a wide variety of skills in Materials Science: characterization and development of materials, polymer processing, design and modelling of mechanical properties. These research activities have a strong influence in areas of high social and economic impact, such as materials and environment, health, energy, industrial. The research activity is carried out in the DICI-UNIPI laboratories by employing advanced scientific instruments in collaboration with a variety of both national and international academic and private institutions.

UNIPI is very skilled in projecting polymers blends composition (additives selection, processing conditions) in order to tune properties of materials including biodegradability, morphology, thermal and mechanical properties as attested by papers, patent, and participation in European, National and regional projects but also by the direct cooperation with industries to bring, e.g., their patented results to the market. UNIPI staff has experience in test of compost, and biodegradability in soil and marine water, as well as reactive extrusion, copolymerization, selection of plasticizers, nucleating agents, and compatibilization of natural materials with polymeric matrices. UNIPI staff is also expert in studying and quantifying the crystalline and amorphous fractions of a material, whose ratio affects biodegradability many physical properties, as barrier, mechanical, thermal, ageing, adhesion, and, consequently, the final performance of a material.

Main role in the project *WPs envolvement and key task or activities to be done in each*

UNIPI will support WP1 where they will cooperate is s plastic characterization, in setting the logistic approach for plastic waste selection, collection and transport to the waste treatment facilities, WP2, WP3 and WP4, for analysis of plastic before and after degradation, WP5 for valorisation of chitin and chitosan and coordinating valorisation of other recover by-products, in WP6 for the logistic approach and for supporting for data collection, necessary for safety and environmental assessments, in WP7 with teaching, training, dissemination in conferences and papers, and as partner in WP8. UNIPI will also contribute to the project technical management.

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities*

Prof. Patrizia Cinelli (female) is associate professor of Pisa University for applied chemistry and Materials Technology at Master Degree student of Construction Engineering and Architecture at Pisa University, and master degree students of Materials and Nanotechnology. She was researcher at Inter University Consortium of Materials Science and Technology, and then researcher at National Research Council (CNR), Institute for the Physical and Chemical processes (IPCF) unit of Pisa. She graduated in Chemistry at the University of Florence, and then achieved a PhD in Chemistry at Pisa University partly performed at the United States Department of Agriculture (USDA), Peoria, IL, USA. 20 years of experience in materials science, polymer processing and characterization, biodegradation and life cycle assessment. She is co-author of over 70 papers in peer-reviewed journal, 9 book chapters, and 8 patents on innovative materials, with a scopus h-index of 25, and 2102 citations. She has worked in over 15 EC project from FP5, FP6, FP7 and Horizon 2020, directly following writing, scientific activity, and reporting. She is currently coordinator of the Horizon2020 project ECOAT GA 837863 BBI.2018.SO3.R9 started the 1st May 2019.

Prof. Maurizia Seggiani (female) is full professor of Synthesis of Chemical Processes and Chemical Industrial Instrumentation at Master Degree in Chemical Engineering, University of Pisa. She was graduated cum laude in Chemical Engineering at University of Pisa in 1993. She was junior researcher at the ENEL (Italian National Electricity Company) (1995-1997), in 2000 completed her PhD in Chemical Engineering at the University of Pisa. Assistant Professor (2001-2014) and from 2014 Associate Professor of Industrial and Technological Chemistry at University of Pisa. Her current research activities are focused on waste management (treatment/recovery/valorisation of industrial wastes), biomass gasification/pyrolysis, development of biodegradable/compostable composite materials and modelling/simulation of chemical processes. Over 70 papers in peer reviewed international journals and 1 patent. She coordinated Regional projects on gasification of industrial wastes and on the development of green biodegradable composites based on PHA and natural fibres for marine applications.

Prof. Antonio Pratelli (male) (DICI-UNIPI) is associate professor in Transport Planning at the Department of Civil and Industrial Engineering of the University of Pisa and professor of Distribution Logistic at the Leghorn Logistic Centre of the University of Pisa. He participated, with Dr. Petri, in many logistic projects like the Regional Rural Development project called Intraviva, regarding the nursery gardening logistics, the Linfa Project, in the Regional Research and Development Program, about drug logistics, the Go Smart Med and LIST-Port Project inside the Italy-France Maritime European Program, respectively about port freight logistic and noise emissions in the peri-portual areas and the project Furbot aims to develop a new vehicle to improve the urban city-logistic (European 7th Framework Programme).

Prof. Andrea Lazzeri (male) was graduated in Chemical Engineering at University of Pisa in 1982, in 1984 joined UNIPI as research associate. In 1991 completed his PhD at Cranfield University. He has been lecturer (1991-2002), Associate Professor (2002-2008) and from 1st January 2009 Full Professor of Materials Science and Engineering. He is specialized in preparation and use of micro-and nano-particles in polymeric composites, rubber toughening of plastics, deformation and fracture of materials and in ceramic matrix composites. He is author of 40 publications on international journals and 4 patents. Didactical activity in Materials Science and Engineering and Materials Science and Technology for Cultural Heritage. He coordinated two European projects on materials science in FP7 (FORBIOPLAST and HELM), and coordinated a project in Horizon 2020, NMP-21-2014-IA, Nanocathedral.

Dr. Massimiliano Petri (male) PhD in Territorial Spatial Analysis at University of Pisa and co-professor of a course for Logistic Distribution at Pisa University, actually he is consultant inside the Technical Group of the Italian Ministry of Infrastructure and Transport. He received a Doctoral Award from the Italian Association of Regional Science and a Research Award from the Italian Ministry of Youth for its PhD-thesis on urban mobility and logistic demand simulation (2001). In 2018 is scientific coordinator of the project SaveMyBike, admitted to the final of the European Regio Stars Awards. Petri has 15 years of experience in researches and studies regarding logistic and freight transport (he elaborated the Tuscany Region freight and passenger model, the Freight Demand Study for the European Rail Freight Corridor n.6 “Mediterranean” and actually is working to the national freight/passenger model called SIMPT). He published more than 30 articles regarding logistics/mobility and participated in many research projects with Prof. Antonio Pratelli at Regional, National and European level.

Prof. Maria-Beatrice Coltelli (female) (DICI-UNIPI) is associate professor in Materials Science at the Department of Civil and Industrial Engineering of the University of Pisa. She obtained the graduation in Chemistry on February 2001 and the PhD "Galileo Galilei" in Chemical Sciences on 2005. Her researches were focused onto polymer science, in particular recycling of polymer blends and composites. She is co-author of more than 30 publications and 2 patents. She followed the scientific activity of European projects on materials science, bio-based polymer processing and nano-composites. Currently she supports the didactical activity of courses for Chemical Engineers and Construction Engineers.

Dr. Francesca Signori (female), Researcher at Department of Civil and Industrial Engineering, Pisa University, PhD in Biomaterials at Trento University (Italy) in 2003. As PhD student, she performed part of the research (2001-2002) at the University of Twente, The Netherlands. In 2006 she received the European Biomaterials and Tissue Engineering Doctoral Award, by the European Society of Biomaterials. Her research topics focus on polymer chemistry and physics, mainly on the synthesis, the engineering and the structure-properties relationship assessment of biodegradable polymeric materials for biomedical applications. Moreover, she got interested in the use of biodegradable thermoplastic materials in the field of nanotechnology and for industrial applications. She is also active in the preparation and investigation of biodegradable based polymer composites and blends. Her scientific activity is testified by co-authorship of 35 papers in peer-reviewed international journals.

Dr. Nicoletta Barbani (female), is graduated technician in Biology, with elevated professionality (EP3) at the Department of Civil and Industrial Engineering (DICI) of Pisa University. Head of Infrared Spectroscopy Chemical Imaging Laboratory. Collaboration with important researchers from other research centers (Institute of Biomaterials, Department of Materials Science and Engineering, University of Erlangen-Nuremberg, Erlangen, Germany; Department of Mechanical and Aerospace Engineering, Polytechnic University of Turin Italy), responsible for numerous scientific research projects for important Companies working in the biomaterial field (Fidia Advanced, Chiron S.R.L, Novartis Vaccines). Co-author of scientific publication published both in national (9) and international journals (126), 125 communications at national and international conferences and 3 patents. Highly experienced in material science for application in biomedical and environmental fields. Expertise in physicochemical and mechanical characterization, by instrumental techniques (TGA, DSC, SEM, FTIR-ATR, UV, DMA); optimization of FTIR Chemical Imaging methods for the chemical characterization of materials.

Dr. Sara Filippi (female), PhD in Chemical Science at University of Pisa, has 20 years of experience on polymer science:

synthesis and characterization of LCP and energetic polymers, chemical modification of polymers, compatibilization of polymer blends, recycling of thermoplastic polymers, preparation and characterization of polyolefin organoclay nanocomposites, production and characterization of polymer modified asphalts with anti-ageing properties. She is co-author of more than 50 publications and she supports the didactical activity of courses for Civil and Construction Engineering.

Dr. Vito Gigante (male) is a Chemical Engineer graduated at University of Pisa on 27 February 2015 with a thesis on 'Properties of mixed compounds and base poly (lactic acid) (PLA) / polycarbonate (PC) made from recycled polymers'. He got a PhD in Chemical Engineering cum laude, at Pisa University. He studies biodegradability, life cycle assessment and the process-ability and mechanical properties of sustainable multiphase polymer systems for structural applications.

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

- NANO-CHITOPACK: Sustainable technologies for the production of biodegradable materials based on natural chitin-nano fibrils derived by waste of fish industry, to produce food grade packaging (GA no:315233, FP7-SME, completed)
- DIBBIOPACK: Development of injection and blow extrusion molded biodegradable and multifunctional packages by nanotechnology: improvement of structural and barrier properties, smart features and sustainability (GA no: 280676, FP7-NMP-2011-LARGE-5, completed)
- OLI-PHA: A novel and efficient method for the production of polyhydroxyalkanoate polymer-based packaging from olive oil waste water. (GA no: 280604-2, Completed)*
- AGRIMAX: Agri & food waste valorisation co-ops based on flexible multi-feedstocks bio-refinery processing technologies for new high added value applications (BBI GA no: 720720)
- POLYBIOSKIN: High performance functional bio-based polymers for skin-contact products in biomedical, cosmetic and sanitary industry "Horizon 2020 / BBI-JU, Call: H2020-BBI-JTI-2016, BBI PPP, GA no:745839)

Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Prederently 2013-2019 years*

- P. Cinelli, M. Schmid, E. Bugnicourt, J. Wildner, A. Bazzichi, I. Anguillesi, A. Lazzeri. "Whey protein layer applied on biodegradable packaging film to improve barrier properties while maintaining biodegradability", Polymer Degrad. and Stability, 2014, 108, 151-157.
- P. Cinelli, M. Schmid, El. Bugnicourt, M.B. Coltelli, A. Lazzeri, "Recyclability of PET/WPI/PE Multilayer Films by Removal of Whey Protein Isolate based Coatings with Enzymatic Detergents" Materials, 2016, Vol. 9(6), 473; doi:10.3390/ma9060473.
- P. Cinelli, M. Seggiani, N. Mallegni, V. Gigante, A. Lazzeri "Processability and Degradability of PHA-Based Composites in Terrestrial Environments". International journal of molecular sciences 2019," DOI:10.3390/ijms20020284,
- M.Seggiani,; P.Cinelli, E. Balestri; N. Mallegni, E. Stefanelli, Eleonora; A. Rossi, C. Lardicci, Claudio; A. Lazzeri, "Novel sustainable composites based on poly(hydroxybutyrate-co-hydroxyvalerate) and sea grass beach-CAST fibres: performance and degradability in marine environments", Materials, 11(5), 772 ,2018, doi: 10.3390/ma11050772.
- P. Cinelli, M.B. Coltelli, N. Mallegni, P. Morganti, A. Lazzeri, "Degradability and sustainability of nanocomposites based on polylactic acid and nanochitin" Chemical Engineering Transaction 60, 2017, ISBN 978-88-95608-50-1, ISNN 2283-9216
- A. Pratelli, M. Petri, C. Rindone, F. De Bari (2017) "A Feasibility Study for New Transport Connections between Italy and Algeria", Journal of Civil Engineering and Architecture 11 (2017) 691-700 doi: 10.17265/1934-7359/2017.07-006
- A. Pratelli, M. Petri, "Vehicle Routing Problem and Car-Pooling to solve home-to-work transport problem in mountain areas", International Conference on Innovation in Urban and Regional Planning nature, planning and ecosystem services, Cagliari (Italy), 24-26 June 2019

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

Fourier FT-IR Nicolet 380; Charpy Impact Tester CEAST 9050; Thermogravimetric analyser TG/DTA 7200 Seiko EXSTAR; Differential Scanning Calorimeter Perkin Elmer DSC4000 and DSC 8500 equipped with Intracooler III; Infrared spectrometer Spectrum Two Perkin Elmer equipped with diamond ATR; Size exclusion chromatography (SEC) instrument Agilent Technologies 1200 Series equipped with two PLgel 5 µm MiniMIX-D columns and a refraction index detector; Carver hot press; X'Pert PRO (PANalytical) powder diffractometer -Cu Kα radiation (1.541874 Å); Sonotrode Hielscher UP200S, Spectrophotometers Color quest Hunter Lab n°2, Universal testing machine Instron 4302, and Mod. 5966; MFI (melt flow index) CEAST automatic. Differential Scanning Calorimeter TA Q200DMTA; Gabo Eplexor® 100N; Microtome Leica Reichert Ultracut E; Scanning Electron Microscope, JEOL JSM-5600LV; Field Emission Scanning Electron Microscope, FEI.

4.1.3. Partner 3. Asociación de Jovenes Agricultores de España (ASAJA) - Spain

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

ASAJA (Asociación de Jovenes Agricultores de España) is the most important collective organization of Farmers in Spain (250.000 members from agriculture, farming, livestock and agrofood sectors) representing 100% of the Country' territory, 60% of the Agrofood turnover, and 34 % of the Spain farmers.

140 offices all over the Spanish Regions, more than 1600 employees, giving assistance to the thousands of associates (250.000) and representing the legitimate interest of the farmers versus the European and national Institutions and governmental and public Authorities.

- has a Young Entrepreneur Organization ANGA, an internal audit – REV -, a training agency - ENAPRA -, an editorial agency – SEPE – and a service Centre – CAAF.

- is a full member of COPA COGECA and CEJA in Bruxelles, of Farm Europe in Paris, and of the National Food Technology Platforms Food For Life.

Main role in the Project *WPs envolvement and key task or activities to be done in each*

ASAJA will support WP1, WP3, WP5, WP6, WP7:

WP1 - Mapping of APW and their characteristics: Samples and Written report linked to APW inventory and characterization task and 1.2 including the specifications of APWs and their availability as well as selecting of 4-6 types of most representative fossil plastics (including in each case virgin plastics and AWP counterparts picked individually from MSW or agricultural films).

WP1 -Report on AWP collection and monitoring strategies and the pre-treatment options to enhance biodegradation of AWP. Review in terms of state of the art for the research on the use of micro/macro organisms for their degradation. Written report on the outcome of tasks 1.3 and 1.4 and samples of 4-6 selected APW after pre-treatment.

WP1 -Strategy for APW collection and sorting. Obtaining the matrix for the definition of the classification, characterization and recommended biodegradation procedure. Manual of action in the points of generation of the biodegradable plastics mixtures following RECOVER.

WP1 -Pre-treatment rules and procedures basing on composition of mixtures of APW. Obtaining the matrix for the definition of each type of abiotic pre-treatment to be used depending on the mixture of APW to biodegrade (fragmentation, micronization, thermal treatments, UV irradiation, oxidizing agents, pH and irradiation) following RECOVER.

WP3 -Protocol for the production of insect and earthworm plus PMC with enhanced APW degrading capability. Formulations of plastic-degrader microorganisms, enzymes, microorganism-enzyme-fortified earthworms (EFW) and insects (EFI) tests results according their potential up-scaling formulation for production and commercialization.

WP5 -Up-graded bio fertilizers formulation. Determination of the formulation matrix to the obtaining of several industrial mixtures of up-grade bio-fertilizer products. Written report including the specifications of the processes to produce high-value fertilisers in comparison with similar and conventional products.

WP6 -Social impacts report. Including three surveys on social impacts (social impacts and social acceptance). Referencing 3 main scenarios of rural development.

WP7 -Industry, stakeholders and policy guidelines. Publication of the "Guidelines for Industry, stakeholders and policy regarding RECOVER waste management" (optimal manage RECOVER waste).

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities*

Pedro-Mauricio Barato Triguero (Male): ASAJA president - institutional representation and European Association Co-leader.

Ana-Leticia Plaza Simón (Female): journalist, communication manager of ASAJA. Wide experience in communication/dissemination, design/coordination and promotion/execution of corporative communication plans. Communication leader in BBI-JU projects.

Jose-Fernando Sánchez (Male): Industrial Engineer. Master in International Project Management in Univ. Polytechnic Madrid. 15 years' experience in high direction of innovation projects. Master in Standardization and Quality assessment.

Santiago Carretero-Martin (Male): Technical-admin. Support. European dissemination coordinator.

Manuel Lucena Marcos (Male): Technical Support. Grant-fulfilment manager. Dissemination support and managing.

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

- ECOTROFOOD - EIP-Eco-Innovation 2010 (ECO/10/256253).
- ENFFI - EUROPE INNOVA - EUROPEAN NETWORK FOOD INNOVATION FINANCING. FP6-2005-022491.
- FOOD WASTE TREATMENT - LIFE 11 ENV/ES/000601.
- Microbiome Support- "Towards coordinated microbiome R&I activities in the food system to support EU and international bio-economy goals (2018-2021; EU H2020; No 81811).
- GIST - EIP-Eco-Innovation 2010 (ECO/10/277217).

And more than 30 relevant projects related to the development of this project.

Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Prederently 2013-2019 years*

1. Markets legislation.
2. Food Hygiene and safety Regulations.
3. Local Initiatives on regional ground.
4. Insect role in agro-food sector including development of guidelines and mentoring to administration.
4. New food chains and farmers role in New business models and consumer behaviours biomasses, new macro and micro materials, food and no food.
5. New Business models and consumers behavior.

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

Infrastructure: ASAJA have a wide network of agricultural and farming and agro-food association. Specifically ASAJA will provide insect fluxes and expertise, including 4 insect producers in Spain to provide all the research in Europe in this topic. In addition all the field trials in Spain.

Equipment: testing agronomic field including controlled green housing and control equipment for in-situ validation tests.

Staff: 8 persons directly dedicated to the specific tasks/WP of RECOVER-project and 120 transference and innovation personnel in ASAJA.

4.1.4. Partner 4. Brunel University London, UK

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

Brunel University London in West London is a dynamic institution with over 12,500 students and over 1,300 academic staff operating in a vibrant culture of research excellence. With a turnover of more than £200 million, the University is a Higher Education and research establishment with considerable intellectual, financial, and social resources and makes major contributions to the economy and community of the West London region. It also plays a significant role in the higher education scene nationally and has numerous national and international links and partnerships with both academia and industry.

Brunel University London has climbed 14 places in the 2019 QS World University Rankings, and is now placed in the top third of universities worldwide, in the top 200 in engineering and technology, and in the top 50 for international outlook. Brunel prides itself on its multicultural environment and has gained countless benefits from having staff and students from over 113 different countries contributing to both its academic and social environment. The university holds a European Commission HR Excellence in Research award for its commitment to supporting its researchers.

Brunel has a long history of successful bidding for European funding and of managing EU projects, Brunel was involved in 120 projects within FP7 with a cumulative value of over 37 Million Euros, and has been successful with over 60 Horizon 2020 proposals to the value of 30 million Euros, of which it is leading and coordinating 9 projects.

Since the launch of the university in 1966, research has been at the very heart of Brunel's academic activities; consistent with its establishment mission as a 'Wilson University' to apply science to the practical work through translational and applied research and education. Today, we have three Research Institute's covering most of these areas. The research groups within them are world leading, with highly cited papers, substantial grant income, established reputations for particular methodological approaches, and a community of PhD graduates pursuing successful careers in industry and academia.

The Institute of Energy Futures has a holistic approach to the energy costs of food distribution, bringing together researchers from a range of disciplines, as well as mainstream engineering research. We have particular strengths in refrigeration, heating and cooling and in environmental design.

The Institute main themes include Advanced Powertrains and Fuels, Energy Efficient and sustainable technologies, Smart Power Networks and Resource Efficient Future Cities.

Institute of Environment, Health and Societies aims to achieve internationally acknowledged excellence in all forms of research relating to the quality of our environment, and to our health and wellbeing. From assisting governments with environmental and health policies, to offering lung disease sufferers drug-free, medically proven treatments, to helping consumers with food allergies make better choices, our practical research has helped millions of people worldwide lead healthier, happier lives.

The Institute main themes include Environment & Health, Ageing Studies, Health Economics, Synthetic Biology, Biomedical Engineering & Welfare Health and Wellbeing.

Institute of Materials and Manufacturing research aims to improve the performance of materials and structures, including their

design, manufacturing, integrity and use, through a combined theoretical approach. Its vision is to become the leading recognised international provider of materials and manufacturing research, addressing current major societal questions.

The Institute main themes include Structural Integrity, Liquid Metal Engineering, Micro-nano Manufacturing, Materials Characterisation and Processing and Design for Sustainable Manufacturing.

The expertise relevant to the current project lie within Synthetic Biology, one of the 15 University Strategic Research Themes, and will also have access to the newly established expertise in enzymes and biotechnology in the newly established Department of Chemical Engineering. It will be hosted within the dedicated synthetic biology research labs which provide an integrated wet-dry, bioinformatics-laboratory research environment; and led by the head of the Synthetic Biology lab and Research Theme.

Main role in the project *WPs envolvement and key task or activities to be done in each*

UBRUN will primarily contribute to Work Package 2, through the development and application of new methods for the generation of 'better than nature' enzymes based upon new strategies exploiting comparative genomics and synthetic DNA-based methodologies. They will also contribute to the integration of these enzymes with other parallel and later Work Packages. In addition, they will provide expert genomics support for other Work Packages studying microbe content and changes within compost and insects. Finally, they will contribute to the project dissemination and strategies for the further development and translation of enzymes.

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities*

Dr Ronan McCarthy BSc (Hons), PhD, FHEA. (male) Senior Lecturer in the Biosciences Division in the College of Health, Medicine and Life Sciences since 2018. Formerly: Higher Scientific Officer at the Animal and Plant Health Agency- Weybridge UK (2017-2018), Visiting Academic at the MRC Centre for Molecular Bacteriology and Infection at Imperial College London (2017-2019), Postdoctoral Research Associate Visiting at the MRC Centre for Molecular Bacteriology and Infection at Imperial College London (2014-2017).

Since joining Brunel University London Dr McCarthy has established himself as one of the Departments foremost researchers. This is reflected in him being selected as the Universities nominee for the Blavatnik Award for Young Scientists in Life Sciences 2019. Dr McCarthy has established an interdisciplinary international research collaborations with key leaders in his field and has secured funding to travel to China, Sweden and Kazakhstan to present his research and cultivate collaborations (Newton Fund, Global Challenges Research Fund, Young European Research Universities Travel Award). Dr McCarthy has extensive bioinformatic expertise particularly in bacterial genomics and in the development of novel strategies to modulate enzyme activity (Atomwise Artificial Molecular Screening Award x2). He has secured funding from leading funding bodies including the British Society for Antimicrobial Chemotherapy (2019), Biotechnology and Biological Sciences Research Council (2018) and the Wellcome Trust (2018). In 2019, he was also awarded an Innovate UK Smart Grant to work with synthetic biology company CC Bio Ltd. This grant is highly relevant to this proposal as it uses synthetic biology approaches such as modular assembly to develop novel enzymes capable of tackling microbial dysbiosis. In 2021, Dr McCarthy was awarded a highly prestigious BBSRC New Investigator award along with an Academy of Medical Sciences Award.

Dr Ashley Houlden (Male) is an experienced molecular microbial ecologist having studied the impact of disease and injury in animal models on the GI microbiome. Some key research findings include how the intestinal parasite *Trichuris muris* manipulates its host microbiome to protect itself and has its own specific internal microbiome, published in Science Advances. This experience, in both biological and bioinformatic microbial ecology analysis in a range of disease situations would be a key part to fit the research team for this proposal. Dr Houlden was part of the team that secured the BBSRC funded FoodBioSystems Doctoral Training Partnership and the UKRI supported Partnership for Sustainable Food Future – Centre for Doctoral Training at BUL. Dr Houlden also has experience with analysing the microbial communities capable of degrading plastic waste.

Dr Sophie Howard (Female) is an experienced molecular microbiologist who has extensive in analysing bacterial regulatory pathways and recombinant protein production and purification. She completed her PhD in Imperial College London in 2020. During her PhD she published a number of high impact studies in publications such as PNAS, Cell Reports, mBIO and eLife.

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

- Valor Plus (FP7). Project valorisation project for C5 and lignocellulosic products. NS work package lead (based in UK, Spain, Austria, and Germany) for microbial aspects of the project, in addition to contributions to 2 work packages: microbial strain optimization for C5-metabobolism, furfural tolerances, and butanol / product tolerance; and conversion of waste glycerol as a feedstock. Plus, support role for Technical University of Munich for optimization of strains for C5-rich hemicellulose-derived sugars (2013-2017).
- SupraBio (FP7). Work package (added for last 2 years) to address issues of strain safety and optimization for the production on of 2,3-BDO in Gram negative bacteria. (2012-2014)
- SynbiSTRAIN. A project supported 75% by SynbiCITE (the UK Synthetic Biology Accelerator) and 25% by Brunel, to establish core strain collection and genomics tools for the (10) priority species nominated by the academic and commercial partners of SynbiCITE to support synthetic biology translational research in the UK and beyond. (2015-17, and ongoing)

Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Prederently 2013-2019 years*

Our relevant work in these areas of biotechnology are largely subject to commercial and partner agreements that do not lead to publishing in the academic literature, or are part of ongoing development towards commercialization through Sygenious, a company listed through Brunel University as a vehicle for development of resources and project development work in several areas.

- The SynbiSTRAIN collections and supporting genomics are available to academic and commercial collaborators and co-developers, directly, or through SynbiCITE.

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

There is an integrated synthetic biology research lab, with both bioinformatics and wet-lab facilities. The wet lab has three key parts, for strain storage, microbiology and culture, and a clean lab for molecular biology. There are dedicated within-lab facilities for QPCR, bioanalyzer, nanodrop, qubit, FastPrep, small-scale fermenters, and 96-well spectrophotometer for strain growth profiling and comparisons (for sequencing and strain / enzyme performance analysis), in addition to standard molecular and microbiology laboratory equipment.

In addition to standard workstations, there is a dedicated central server for bacterial bioinformatics (64 cores, 1.5TB RAM), and as a host on the particle physics grid we have access to 4-10,000 cores if / when necessary (for which we are an established / registered non-physics user group).

There are a number of bespoke and unique bioinformatics resources and pipelines specifically developed for strain identification, typing, population genetics, and comparative and evolutionary genomics established on these resources.

4.1.5. Partner 5. Miguel Hernandez University (UMH) - Spain

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

UMH of Elche is a public university located in southeast Spain, an area of high economic and business potential (about 2000 employees, 18000 students). 28 bachelors, 51 Masters and 13 Doctorates are included in our teaching scheme. A total of 27 departments and 5 research institutes are devoted to research and training. UMH has been beneficiary of more than 40 EU projects. The Research Results Transfer Office (OTRI) links scientific activity with the business world and specifically the European and International Research Projects Unit is in charge of administrative management of international projects, has extensive experience in managing European Projects and supports the researcher. Pre-doctoral expertise in formation on waste treatment, management and valorization, courses on recycling and advanced instrumental techniques for waste characterization is relevant to this proposal.

The **Applied Research Group on Agro-chemistry and Environment (GIAAMA) team,** involved in this proposal belong to the Dept. of Agro-chemistry and Environment. GIAAMA has extensive research work about waste valorization throughout the last 15 years including:

a) waste fluxes characterization using classical and new techniques (vg NIRS),
b) co-composting strategies for several organic waste fluxes, specially manures and digestato,
c) bio-fertilizer design and development mainly based on compost and vermicomposting
d) Enzymatic profiling of soil-plant systems and organic matter degradation in forced and natural environments.
d) Agronomic field validation of bio-fertilizers and new bio-products in commercial conditions using of wastes coming from the agriculture, the agro-food industry, and from urban origin (bio-solids and urban solid wastes).

Their activity has been financed by public and private funds through more than 25 national and international projects and led to publication of over 120 peer-reviewed papers, 20 books, 24 book-chapters, 8 patents and more than 200 congress national and international communications

Main role in the project *WPs envolvement and key task or activities to be done in each*

UMH will support in the pre-treatments to enhance AWP biodegradation in WP1. In WP2, UMH will provide earthworm samples for microbiomes determination to other partners and develop lab scale assays with earthworms to monitor key activities related to plastic degradation. In addition, pre-compatibility tests will done between earthworms and 2-3 microbial/enzyme pools, including survival and key enzymes. In WP3 (leaded by UMH) we will participate in the production of microbial/enzyme consortia (PMC) and earthworm fortified with PMC. UMH will help in WP4 to ex-situ and in-situ pilot-scale plastic degradation processes using fortified the earthworm collection from WP3. In WP5 UMH will help in the development of vermicompost products and residues of the AWP degradation processes including greenhouse tests. Finally, UMH will also contribute to providing data for environmental and socio economic sustainability analysis, dissemination activities, exploitation of their results and project management.

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, who will be*

primarily responsible for carrying out the proposed research and/or innovation activities

RAUL MORAL HERRERO (male)**.** BSc, PhD in Agro-chemistry, Professor at UMH (H-index 35) since 1994 and full professor since 2010. Extensive experience with research project management (15 projects), supervision (13 PhD students, 22 MSc theses), teaching (3 MSc). Extensive expertise in development of commercial organic fertilizers (7 within official Spanish fertilizer list).

JUAN C. SANCHEZ-HERNANDEZ (male, UCLM member). PhD in Environmental Science (Univ. Siena, It), postdoc Marie-Curie Fellowship (Univ. Lancaster, UK). Associate professor at UCLM since 2002 (H-index 25, Scopus author ID: 7004278012). Expertise in terrestrial ecotoxicology and bioremediation of pesticide-polluted soils using earthworm and biochar. 70 SCI research papers (85.7% in Q1), leadership of 10 national/regional projects, 9 book chapters, 1 edited CRC/Press book on bioremediation of agricultural soils, and 1 national patent on biological procedures for removal pesticides in soil (ref. P2013001180, 29/02/2016).

CONCEPCIÓN PAREDES GIL. (Female) BSc, PhD in Chemical Sciences. Professor at UMH from 2001 to the present. She has supervised 7 PhD students and 23 MSc theses. Her scientific activity is focused on the characterisation of agro-industrial waste, their composting and agricultural valorisation. She has participated in different research projects and in advisory services with companies and administrations and she is author of 74 papers in SCI journals (h-index: 31; Scopus Author ID: http://www.scopus.com/authid/detail.url?authorId=7006458594).

MARIA ANGELES BUSTAMANTE MUÑOZ. (Female) PhD in Chemistry (UMH), postdoc Juan de la Cierva (CEBAS-CSIC). Assistant professor at UMH since 2014 (H-index 21). 83 research papers published, 70 in SCI journals (56% in Q1), 3 books (1 of them as co-editor) and 51 book chapters. Participant in 24 competitive research projects, 7 of them financed by the European Union. Supervisor of 5 PhD theses, 11MSc theses and 21 BSc theses. Expertise in organic waste management, obtaining of organic added-value materials by composting and use of organic materials in different agricultural sectors (organic and intensive farming and soilless cultivation).

MARIA DOLORES PEREZ MURCIA. (Female) BSc, PhD in Chemical Sciences. Professor at UMH from 1994 to the present. She has supervised 5 PhD students and 27 MSc theses. Her scientific activity is focused on the analytical chemistry related to agronomy and environment including chemical and instrumental determinations. She has participated in different research projects and in advisory services with companies and administrations and she is author of 58 papers in SCI journals.

XAVIER BARBER VALLÉS (Male) PhD in Statistics, Associate Professor at UMH (H-index-13). Extensive experience with supervision (Msc theses) and teaching (3 MSc). Extensive expertise in designing experiments and Bayesian methods applied to ecology and agronomy.

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

- REGROW-Reclamation of olive waste abandoned lagoons using bio-recovering strategies in a circular economy scenario. Ref: LIFE16 ENV/ES/000331, 2017-2020 (partner).
- Earthworms and biochar: testing synergistic interactions to reduce pesticide impact in soil. CTM2014-53915-R. Ministerio de Economía y Competitividad (Madrid), 2017-2018.
- Recovery and Use of Nutrients, Energy and Organic Matter from Animal Waste–ReUseWaste. Ref: 289887. Program PEOPLE, Initial Training Network. 2012–2015 (assoc. partner).
- Enzymatic bioremediation of pesticide-polluted soils. PEII-2014-001-P. Consejería de Educación, Gobierno de Castilla-La Mancha (Toledo, Castilla-La Mancha), 2014-2017.
- Contribution of earthworm activity to enzymatic bioremediation of pesticide-contaminated soils. CTM2011-25788. Ministerio de Economía y Competitividad (Madrid), 2012-2014.

Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Prederently 2013-2019 years*

- Sanchez-Hernandez, J.C. 2018. Biochar activation with exoenzymes induced by earthworms: A novel functional strategy for soil quality promotion. Journal of Hazardous Materials, 350, pp. 136-143.
- Vico, A., Pérez-Murcia, M.D., Bustamante, M.A., (...), Pérez-Espinosa, A., Moral, R. 2018. Valorization of date palm (Phoenix dactylifera L.) pruning biomass by co-composting with urban and agri-food sludge. J. Environ. Manage., 226: 408-415;
- Sanchez-Hernandez, J.C., Ríos, J.M., Attademo, A.M. 2018. Response of digestive enzymes and esterases of ecotoxicological concern in earthworms exposed to chlorpyrifos-treated soils. Ecotoxicology, 27 (7), pp. 890-899.
- Jouni, F., Sanchez-Hernandez, J.C., Mazzia, C., Jobin, M., Capowiez, Y., Rault, M., 2018. Interspecific differences in biochemical and behavioral biomarkers in endogeic earthworms exposed to ethyl-parathion. Chemosphere, 202, pp. 85-93.
- Sanchez-Hernandez, J.C., Notario del Pino, J., Capowiez, Y., Mazzia, C., Rault, M. 2018. Soil enzyme dynamics in chlorpyrifos-treated soils under the influence of earthworms. Science of the Total Environment, 612, pp. 1407-1416.

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

Infrastructure: GIAAMA group facilities in EPSO campus (Orihuela, Spain), incl. technical labs, pilot scale composting facility, computer-controlled tunnels for compost optimization, open-air composting area for commercial scaling. Full-

equipped biochemistry lab at the UCLM.

Equipment: Culture chambers for earthworms, liquid chromatograph Agilent 1200 Series equipped with diode-array and fluorescence detectors, microplate-scale spectrophotometers, refrigerated centrifuges, MicroResp equipment, tissue homogenizers, vertical electrophoresis units for proteins.

Staff: 7 Ph.D. permanent researchers, 2 post-docs, 3 PhD students, 1 technician.

4.1.6. Partner 6. Albstadt-Sigmaringen University (ASU) - Germany

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

Hochschule Albstadt-Sigmaringen
Albstadt-Sigmaringen University

Food technology & nutrition is the core part of the education programme and the applied research in the faculty of Life Sciences of Albstadt-Sigmaringen University. Its activities belong to food product & process innovation (e.g. research projects in optimisation of food processing, functional & nutritional properties of food and food packaging) and production of cosmetic products with botanical extracts. Business innovation modelling is also a central part in the faculty of Business Science & Management of ASU. Advanced materials are the core research and teaching field for the faculty of Engineering. ASU is a project partner in the field of safeguarding food quality & sustainable food packaging and in the field of developing new business models. It aims to expose students & scientific staff to real world problems in an EU-scale, thus enabling them to interact with other skilled actors in an international environment.

Main role in the project *WPs envolvement and key task or activities to be done in each*

ASU will support in WP1 where they will irradiate AWP via electron beam (eBeam) as abiotic APW degradation pre-treatment to initiate chain scission. ASU will also support in WP5, where they will establish a more sustainable method to extract chitin from insects. This method will be based on eBeam irradiation. Within the same WP, they will evaluate coated specimen concerning optical properties and surface energy in terms of contact angle measurements as well as water steam permeability and permeability against O_2. Also in WP5, ASU will modify chitosan coated materials to enhance barrier, mechanical and anti-microbial performances for food packaging materials. Furthermore, ASU will characterize chitin/chitosan by-products and will use them to improve mulch materials for optimized conditions such as temperature and O_2/CO_2/H_2O contents considering soil respiration. The chosen chitin extraction residues will be applied to existing mulching films by coating and/or by compounding to mulching film polymers closing the loop within the RECOVER approach.

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities*

Prof. Dr. Markus Schmid (male) is a graduate in business administration. He holds a diploma degree in Food Technology and a master degree in Food Processing from the University of Applied Sciences Fulda as well as a Doctor´s degree from the Technical University of Munich, Germany. As Professor at Albstadt-Sigmaringen University his major research and teaching areas are in the field of Process Technology and Process Design in Life Sciences with emphasis on food preservation and sustainable food packaging technology. He is actively involved in preparing project proposals and works as project manager in food packaging related public founded and industrial projects. He is author and co-author of more than 50 peer-review publications in journals of high repute and is regularly invited to speak at workshops and conferences. He was involved in several National and European projects dealing with the use of residue derived fractions in food packaging. (e.g. ThermoWhey: EraNet-ManuNet 02PN2020; Bio4MAP: FP7- 606144, BioBoard: FP7- 315313, ProgRESS: CorNet).

Prof. Dr. Astrid Klingshirn (female) holds a diploma degree in Home Economics and Nutrition from the Technical University of Munich as well as a Doctor´s degree from the University of Bonn, Germany. As Professor at Albstadt-Sigmaringen University her major research and teaching areas are in the field of Physical Food Analysis, Sensory Analysis, Food Product Development and Innovation Management, Appliance and Food Service Technology, with a special focus on the impact of food preparation and storage technologies on physical and sensory food quality parameters. She is currently actively involved in research projects driving the standardization of home appliances and professional catering equipment related to performance parameters (e.g.: IEC SC 59 M/WG 4, CENELEC/TC 59X WG18).

Kristina Eissenberger (female) holds a bachelor and a master degree in Biology from the Ludwig-Maximilian University Munich. As PhD student of University of Hohenheim she has a strong background in food microbiology and bacteria-plant-interactions. As researcher at Albstadt-Sigmaringen University she is actively involved in studies on the effect of eBeam irradiation on the properties of bio-based polymer films.

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

- Biontop (H2020 BBI, in grant preparation): Novel packaging films and textiles with tailored end of life and performance based on bio-based copolymers and coatings, 2019-2023
- Humidity Loss of Fresh Produce in Refrigerator Storage Compartments: Evaluation of test method options with high repetitious accuracy on behalf of the European Committee of Domestic Equipment Manufacturers (APPLiA), 2016.
- Scientific evaluation of test methods for Refrigerated Food Storage Quality Performance on behalf of the European

Committee of Domestic Equipment Manufacturers (APPLiA), 2016.

- Coolfresh: Development of internationally applicable test standards to assess the essential food preservation characteristics of household refrigerating appliances (WIPANO-Project, 03TNG001B), running
- AlpBioEco - Valorisation of innovative bio-economical potentials along bio-based food & botanical extract value chains in the Alpine Space. WP T1 (walnut value chain - food/cosmetic applications) & WP T2 (Business models in regard to the walnut value chain), running project.

Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Prederently 2013-2019 years*

- Mueller, K., Zollfrank, C., Schmid, M. (2019):"Natural Polymers from Biomass Resources as Feedstocks for Thermoplastic Materials" Macromol. Mater. Eng.,
- Kaiser, K., M. Schmid and M. Schlummer (2018). "Recycling of Polymer-Based Multilayer Packaging: A Review." Recycling 3(1): 1.Schumann, B., & Schmid, M. (2018). Packaging concepts for fresh and processed meat–Recent progresses. Innovative Food Science & Emerging Technologies. Sängerlaub, S., Schmid, M., & Müller, K. (2018). Comparison of water vapour transmission rates of monolayer films determined by water vapor sorption and permeation experiments. Food Packaging and Shelf Life, 17, 80-84.
- Cinelli, P., Schmid, M., Bugnicourt, E., Wildner, J., Bazzichi, A., Anguillesi, I., Lazzeri, A. (2014):"Whey protein layer applied on biodegradable packaging film to improve barrier properties while maintaining biodegradability". Polymer Degradation and Stability, 108: 151-157

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

Access to laboratory equipped with latest state-of-the-art extraction units and eBeam technology (Comet). At ASU the faculty of Life Sciences scientific staff is very experienced in the development and characterization of food packaging. This includes the biopolymer extraction and processing, surface modification of bio-based films as well as the incorporation of active packaging constituents and the full characterisation of packaging relevant properties. Additionally, the faculty has the analytical equipment for characterizing the mechanical as well as barrier and optical properties of packaging films.

4.1.7. Partner 7. NATURPLÁS PLÁSTICOS AGRICOLAS, S.L. (NATURPLAS)- Spain

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

NATURPLÁS is a SME company founded in 2012, producer of plastics for a wide variety of agricultural applications. We offer products adapted to the agriculture needs; plastics for greenhouse covering, small tunnels, double chambers, padding, disinfection / solarisation, silage and geomembranes.

Naturplas was set up in 2012 in order to meet the farmers' needs and since then we are growing with them. Our business history of more than 30 years in the sector confirm us as a quality and confidence benchmark. Our factories are in Portugal, because we are part of the Grupo Polivouga, one of the main plastics manufacturers in the Iberian Peninsula and among the top ten in Europe. We have two warehouses and centres for format transformation in Andalucía, located in Almería and Huelva. Logistically, they allow us to supply, within 24 hours, any area in the Iberian Peninsula.

Naturplas products reach all over Spain and also Latin American and Maghreb markets, areas where there is an important growth in the consumption of agricultural plastics.

As an European Community company, we are submitted to the European Union standards and protocols, which ensure safety and quality of our products.

Naturplas is ISO 9001:2015 certified.

Naturplas differentiates from competitors for the customization level of its products, which are always custom-made products, in order to meet our customers' requirements, according to the type of cultivation and production area.

Main role in the project *WPs envolvement and key task or activities to be done in each*

NATURPLAS will support in understanding the market, logistics and requirements from agricultural films (WP1), will provide samples of agricultural films for the test of degradability (WP4). and for the development of new active mulching films enriched with fertilizers from by products from recover cascade to test (WP5). NATURPLAS will also contribute by providing data for environmental and socio economic sustainability analysis, dissemination activities, exploitation of their results and project management.

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities*

Jose María Navarro. *(male) -BSc in Chemistry. 1 year of experience as plastics producer in Naturplás Industrial company.*
Rocío Castelló. *(female)- BSc in Chemical Engineering + MSB Environmental, industrial and food biotechnology. Over 6 year of experience in the agri-business.*

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

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Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Prederently 2013-2019 years*

Among NATURPLAS leading products are:
- Agricultural films for greenhouse cover.
- Mulch films
- Films for low tunnels covering a variety of crops.

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

1 cast-film extrusion production lines, up to 5 layers.

4.1.8. Partner 8. ASA Special Enzymes GmbH - Germany

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

The business activities of ASA Spezialenzyme GmbH are the development and production of enzymes, microbial mixed cultures and other biotechnological products.

The product range covers the fields of application food and feed processing, bioenergy (bioethanol, biogas), organic synthesis, analytics, biosensors, textile industry and paper industry.

Since the foundation in the year 1991 several research projects were carried out in the area of enzymatic degradation of renewable raw materials resulting in new cellulase and hemicellulase products for the utilization in the production of biodiesel, bioethanol and biogas. Furthermore ASA was engaged in several r&d-projects dealing with the conversion of organic waste materials (e.g. lignocellulytic material as grain stillage), whereby ASA developed several new cellulases, hemicellulases, pectinases etc. Moreover within the scope of a r&d-project ASA developed hydrolases degradation of bioplastic material, which might be useful for the RECOVER project.

Main role in the project *WPs envolvement and key task or activities to be done in each*

Involvement in WP2 and 3 for the selection and upscaling of enzymes for AWP degradation as well as for exploitation planning of the same in WP7.

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities*

- Dr. Arno Cordes (Male), Project leader; genetic engineering; management, LCA
- Petra Hoferichter (Female) Dipl.-Ing, Scientist; Development and scale-up of fermentation processes
- M. Sc. Marc Schreiber (Male), Project engineer; downstream processing of enzymes; scale-up of downstream processing; production of enzymes

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

- **The BERBION - Project**: Efficient material & energy recovery from Organic waste and waste water in the Hamburg district Bergedorf (BMBF, 2009-2013)
- **Valor plus:** Development of processes for the production of polyhydroxyalcanoates (EC-FB7; 2013 – 2017)
- **Biotechnological recycling of electronic waste:** Development of processes for the production of polyhydroxyalcanoates (EC-FB7; 2013 – 2017)
- **ARBOCAR:** Enzyme design and enzymatic modification of lignin for the refining of thermoplastic lignin compounds for car interiors (BMBF, 2008 – 2011)

Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Prederently 2013-2019 years*

- Products: Special enzymes, cellulases, hemicellulases, amylases, proteases, lipases
- Services: contract manufactoring, contract research
- Cordes, A., Kunze, G., Dornig, D.: Development of an enzymatic process for the improvement of biogas yield and quality from plant and animal biomass. Research report, Fachagentur Nachwachsende Rohstoffee.V., FKZ 22011504

(2006)

- Bischoff, Felix, et al. "Three new cutinases from the yeast Arxula adeninivorans that are suitable for biotechnological applications." Applied and environmental microbiology 81.16 (2015): 5497-5510.
- 5. Tonin, Fabio, et al. "Comparison of different microbial laccases as tools for industrial uses." New biotechnology 33.3 (2016): 387-398.

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

Fermentation: incubation chambers for flask shake cultures, pilot to industrial scale automated stainless steel bioreactors (10 L, 20 L, 100 L, 300 L, 1.000 L)

Downstream processing: ultrasonic cell disruptor, glas bead mill, ultrafiltration systems from laboratory to technical scale, chromatography columns up to pilot scale, freeze dryer

Gene technoology: PCR, electrophoresis systems, electroporator

QC and analytics: electrophoresis systems, UV/Vis scanning spectrophotometers, 2 analytical HPLCs, 1 preparative HPLC

4.1.9. Partner 9. NUTRINSECT SRL (NUTRINSECT) Italy

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

Nutrinsect is a biotech company with headquarters both in Italy and Spain specialized in alternative protein and nutrients production by breeding and industrial insect transformation. Nutrinsect provides a sustainable solution to address the organic waste increasing production and nutrient deficiency. We use insects as the food waste bio-transformers. Through organic waste disposal and treatment, we obtain natural fertilizer rich in nutrients ideals to biological culture and insects' protein used such as nutrient source for animal feed.

As innovative aspects we highlight: 1. Vertical breeding, maximizing space / land 2-Feeding by organic waste 3-High conversion rate, which allows to obtain large productions in short periods of time (between 14 days and 2 months) 4-Reduced environmental impact, (livestock Traditional is responsible for 16% of greenhouse gases, compared to 3% of insects) 5-Comprehensive use, 100% of the insect is edible (no other livestock is used 100%) and greater digestibility 6-Circular Economy, enhancing zero waste. One of the main costs in obtaining current proteins (raising livestock and fish for animal protein, or cultivating land for vegetable protein), is the feeding of livestock and the use of natural resources such as water. Insects feed on organic waste and their growth is exponential (unlike livestock and crops), which favours the reduction of costs and obtaining the product in less time. Faced with the instability of the prices of vegetable proteins (usually commodities) and the problem of supply (70% of soybeans are imported) we offer agreed price / deliveries because we do not depend on weather conditions or crop losses. The main suppliers for the operation of our activity are producers of vegetable waste, as they are the food of insects, and whose feeding will influence the quality of our final product. We have to take into account that the waste must be clean of possible pesticides, with what the ideal supplier are food industries that have done some type of treatment such as canning, freezing, and waste treatment plants, since they have washed, scalded etc, that have contributed to eliminate any type of insecticides applied at source. In addition, some of these suppliers can become customers of both our main product (protein meal) and other products obtained in the transformation process, such as organic fertilizer.

Main role in the project *WPs involvement and key task or activities to be done in each*

To follow the circular economy policy Nutrinsect will support the project for the agricultural plastics biodegradation by insect breeding (mainly T. molitor) on an industrial scale. Finally they will contribute to the project dissemination and plan exploitation of their results.

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities*

- Tommaso Pinto, Tommaso Pinto, doctor in Marine biology, R&D manager and operative breeding phase responsible, 100%
- Silvio Piccinini, doctor in Agriculture, operational manager of the breeding and production phases, 100%.

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

- *Assistance Regional program the " I + D Empresarial del Gobierno de Navarra"; PLEAMAR 2018 program, European Maritime and Fisheries Fund of the European Union;*
- *ALTERNFEED project, Substitution of fish meals and oils by alternative products; Industrial Technological Development Center of the Government of Spain;*
- *NEOTEC financing line, where we developed 2 investigations: 1-Evaluation of alternative methods (probiotic-postbiotic) as prophylactic tools in rainbow trout (Oncorhynchus mykiss) " , Ichthyopathology Laboratory of the University of Zaragoza;*
- *H2020-EIC-SMEInst-2018-2020:Determination of the positive effects of insect-derived proteins as a protective and*

immunoenteroendocrine barrier of the gastrointestinal tract, MoBioFood research group of the Rovira Virgili University (Tarragona).

- *AFRODITA H2020-MSCA-ITN-2019: Innovative training Networks*

Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Prederently 2013-2019 years*

- First IPIFF International conference (21-22 /11/2017) in Brussels : "Insect for food and feed: opportunities for tackling societal challenges"

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

Nutrinsect is based on the breeding of different insect species (*Tenebrio molitor, A.domesticus*) and is composed of a single body of 500 square meters. The Nutrinsect facilities includes breeding and housing room, each room is autonomously heated with a 12000 btu heat pump, and also an epa filter and forced air recirculation; UV-C lamp, 100 ° hot water jet washer, substrate (food for crickets) given daily; Abatement (cold room -20°); Microbial decontamination in air conditioning system.

4.1.10. Partner 10. IRIS Technology Solutions S.L. (IRIS)- Spain

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

IRIS Technology Solutions S.L. is part of IRIS Technology Group (http://www.iristechnologygroup.com/) is an advanced engineering Group that was established in Barcelona in 2007. It specialises in the manufacture and integration of Process Analytical Technology (PAT)-based real-time quality monitoring solutions for the process industries and IoT systems for smart manufacturing, predominantly targeted at the food, pharmaceutical and chemical industries. In particular, our product line comprises proprietary state-of-the-art IR analysers for at-line and in-line process monitoring.

On the one hand, our solutions are based on the latest developments in photonics in the fields of MEMs, Near Infrared Spectroscopy, Hyperspectral Imaging, and Raman. On the other hand, we leverage ICTs, whereby we apply Machine learning, Artificial Intelligence and Data Mining techniques for predictive and active control, as well as for decision support systems. In addition to photonics-based analysers, we also work with IoT sensors for enabling equipment and machinery for smart manufacturing and transitioning industry to the Industry4.0 paradigm. Our solutions are cloud-based for mobility, remote monitoring and multi-site applications.

Our Innovation Division which is governed by an Innovation Committee comprising Senior Management, and an Innovation Unit for managing and executing the companies innovation roadmap, has 3 core strategic lines, for shaping the Factory of the Future:

1. New optical photonics-based technologies and PAT systems for production process optimisation
2. ICTs (Artificial intelligence / machine learning, IOT & cloud platforms)
3. Circular and Bio-based Economy: digitalization and waste valorization

In addition to running an internal R&D programme, as well as carrying our various joint private projects with industry partners, our Innovation Division is highly experienced in working in collaborative R&D projects in H2020 (both as coordinators and partners), principally in the SPIRE, FoF, and BBI programmes. The company was established in 2007 and currently employs 60+ staff with a turnover of € 5 M. It is equipped with engineering workshops (3), food testing and wet chemistry laboratory, optics labs (2), and an electronics/telecommunications lab. IRIS is ISO 9001:2008 certified and also is certified under UNE-EN-ISO 166002: Management Systems R&D&i. We are also certified by the local (Catalonian) government as a provider of technological R&D services (Tecnio Certification). IRIS is a member of the following sectorial networks: SPIRE, SPIE and SECPHO, in addition to regional industrial collectives such as Smart Space (association of companies and organisations that specialise in the 'smartification' of industry).

IRIS has a matrix of multidisciplinary knowledge groups in the company with strong expertise in engineering and science. With its team and resources, including electronics, mechatronics, mechanical, optical, industrial design engineers, as well as specialists in system engineering, application programming and web development etc. the group can carry out full system design and engineering, including electronic design and assembly, firmware and software development, electrical harnesses and interconnections, mechanical design for enclosure and prototype integration, as well as installation, validation and industrialisation of prototypes and equipment in production plants. As such they have recently delivered advanced plastic sorting devices. In addition to this broad cross-disciplinary engineering team, IRIS' research team consists of data scientists, chemiometricians, chemists and physicists, among others, who have the expertise to validate the technologies developed.

Main role in the project *WPs envolvement and key task or activities to be done in each*

IRIS will support WP1 where they will test different photonic-based systems for the end of life material identification to ensure they can appropriately be handled with the new process but also in WP4 for the monitoring of the disintegration under the different tested conditions and the development of predictive models that can support the optimisation and upscaling of the new biodegradation approach. Finally they will contribute to the project dissemination and plan an outmost exploitation of their results.

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities*

Dr. Ing. ELODIE BUGNICOURT. (female) Project manager and Leader of the innovation unit in IRIS since 2009. Masters in Material Science & Engineering (2002), PhD in Polymers and Composite Materials (2005), both from National Institute of Applied Science (INSA), Lyon, France. Coordinated several European projects in the fields of bio-based packaging materials development, process monitoring, surface treatment. Previous industry experience: Arkema; Wacker Chemie, Ashland.

Dr. LAURA RODRÍGUEZ TURIENZO. *(female)* B.S in Food Science & Technology (2008) Masters in Innovation in Safety & Food Technologies (2010), PhD in Biological Sciences (2011), all from University of Santiago de Compostela (USC), Spain. Leads the Science and Engineering department at IRIS. Has been involved in a number of EU projects.

ALEJANDRO ROSALES *(male)* has a M. Sc. in Physics from the Universidad de La Habana (1986). His background comprises theoretical and technical skills in several scientific and technological areas: optics, lasers, high vacuum, colorimetry, UV-VIS-NIR-MIR spectroscopy, chemometrics and design of measurement instruments. He has extensive practical experience in programming, electronics and mechanical CAD design as well as R&D methodology and management. Currently, he is the Science & Technology Manager in IRIS.

Dr. CUAUHTEMOC ARAUJO (Male) Chemometrician. Degree on Physics, master degree in Material Sciences, and doctorate degree in Applied Sciences. Conducted research in the fields of materials sciences, optical spectroscopies and chemometrics. More than 20 years of experience working on a wide range of biological and non-biological applications of vibrational spectroscopies (i.e. Raman, MIR and NIR). Also worked closely with other characterization techniques such as: UV/Vis and fluorescence spectroscopies.

Dr. LAURENT PHILIPPET (Male) Scientist. Bachelor's degree in science, speciality applied physics (Université Paris Sud). Engineer's degree in Optical Science (Institut d'Optique Graduate School). PhD in Optical Science (University Paris XIII). Experimental physicist specialised in optics and photonics with specific skills in laser physics, spectroscopy techniques, microscopy, interferometry and optical design.

Dr. DAVID NETTLETON. (male) Data Mining Analyst. Has a Ph.D in Artificial Intelligence from the Universitat Politécnica de Catalunya, Barcelona, Spain, M.Sc. In Computer Software and Systems Design from the University of Newcastle upon Tyne, U.K., and B.Sc. (Hons) Computer Science from the University of Wolverhampton, U.K. Has over 20 years experience of applying data mining algorithms and statistical techniques in a diversity of data analysis and modeling projects. As well as commercial projects, he has conducted research and lectured at two universities.

Dr. JOSE ANTONIO IBARRA FUSTE (male). Scrum Master. PhD. in Signal Theory and Communications, Universitat Politècnica de Catalunya, Spain (2014), Thesis: "Design Rules and Optimization of Electro-Optics Modulators Based on Coplanar Waveguides". Master Degree in Telecommunication Systems from Universidad de Oriente, Cuba (2005). Telecommunication Engineer Degree from Universidad de Oriente, Cuba (2003). Experienced in optical communication softwares, in electromagnetic structures simulators and in many programming languages.

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

- Hypera, Affordable hyperspectral Imaging System for on-line foreign body detection and chemical composition analysis, H2020- SMEInst, 2016-2017
- MULTICYCLE (H2020 SPIRE) Advanced & sustainable recycling processes and value chains for plastic-based multi-materials, 2018-2021
- Biontop (H2020 BBI, in grant preparation): Novel packaging films and textiles with tailored end of life and performance based on bio-based copolymers and coatings, 2019-2023
- COMPO-BALL; Project full title: Novel on-line composting monitoring system; Grant agreement no.: 243625; Feb 2010- Jan 2013. Funding programme: FP7
- B2B contracts in the field of multilayer plastic films sorting.

Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Prederently 2013-2019 years*

- Relevant product lines: Monitoring devices for real-time quality control:
  - VisumPALM: NIR technology for incoming material quality control and detection of adulteration.
  - VisumNIR: In-Line NIR monitoring for product standardization and increase of production efficiency.
  - Hypera: Hyperspectral imaging solutions for the industry
  - Services: Chemiometrics model development and training
- A. A. Gowen, Y. feng, E. Gaston, V. Valdramidis (2015). Recent applications of hyperspectral imaging in

microbiology. Talanta 137, p. 43-54.

- Whey protein layer applied on biodegradable packaging film to improve barrier properties while maintaining biodegradability, Patrizia Cinelli, Markus Schmid, Elodie Bugnicourt, Jessica Wildner, Agostino Bazzichi, Irene Anuillesi, Andrea Lazzeri, Polymer Degradation and Stability (Impact Factor: 2.63). 07/2014; DOI: 10.1016/j.polymdegradstab.2014.07.007
- A. Rosales, O. Mc Nerney. Sensing Technology. Chapter 6. Resource Efficiency of Processing Plants. Monitoring and Improvement. Edited by Stefan Krämer and Sebastian Engell. Wiley-Vch. Print ISBN: 978-3-527-34074-3
- "Wireless Sensor Network for Smart Composting Monitoring and Control". Oscar Casas, Marga López, Marcos Quílez, Xavier Martinez-Farre, Gemma Hornero, Carlos Rovira, Mirta R Pinilla, Pedro M Ramos, Beatriz Borges, Hugo Marques, Pedro Silva Girão Measurement 01/2014; 47:483 – 495

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

Spectroscopic Laboratory equipped with several instruments, namely, a FTIR-ATR (Interspectrum), 2 UV-VIS (PerkinElmer and Ocean Optics), an Extended-SWIR (StellarNet) and a NIR (Mightex) spectrophotometers as well as a SWIR Hyperspectral imager (Specim-Xenics). The associated staff are properly trained and skilled in chemometrics modelling by using a commercial suite (SOLO+MIA from Eigenvector) as well as ad hoc non-commercial chemometric tools (Matlab and Python toolboxes).

Electronics, mechanical and optical workshops and laboratories equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others.

Our optical engineers work with ZEEMAX software for optimally designing of optical systems and to setup the designs by means of advanced optical components and facilities (optical tables, windows, lenses, lasers and light sources and related "optical ware").

Food and wet chemistry lab equipped with a DSC, a Surface Energy analyser, an HPLC, a Cobb tester, microbiological equipment, microscope and microtome for structural analysis, migration cells, colorimeter (for colour and opacity measurement), Ford cup viscometers, water vapour permeation cells, turbid meter, climatic chambers, manual and automatic coaters, electrospray, different drying technologies, plasma surface activation, etc.

4.1.11. Partner 11 Organic Waste Systems (OWS)- Belgium (Terminated)

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

Organic Waste Systems (OWS) is a stock company under Belgian law, constituted in 1988. The company is mainly active in following areas: the area of engineering, construction and operation of composting and biogasification plants, the area of biodegradation and composting laboratory testing, the area of waste management consulting services and environmental sustainability assessment. OWS developed the anaerobic Dranco-process to treat several waste streams. The OWS laboratory is fully equipped to perform the necessary tests and analyses to simulate and manage a digestion process. Moreover OWS has a longstanding experience in the field of compostability testing and biodegradation in all kind of environments (compost, soil, anaerobic, freshwater and marine conditions). OWS has tested more than 3000 products in this field for more than 800 different clients. The lab is recognized as testing lab on compostability and biodegradation by all certification institutes worldwide (TÜV AUSTRIA Belgium, Din Certco, BPI, ABA,…). OWS was, and still is, actively involved in the development of test procedures (and international standards) to evaluate the biodegradation of materials under different environments (compost, soil, anaerobic, water, marine). OWS is involved in various standardisation committees and standardisation projects: ISO TC61/SC5/WG22 - Biodegradable Plastics (Off. Belgian delegate); CEN TC261/SC4/WG2 : Packaging-Organic Recovery (Off. Belgian delegate); CEN TC249/WG9 : Plastics – Degradability (Off. Belgian delegate); ASTM D 20.96 Degradable Plastics; E 47.06 Chemical Fate; D 10 Packaging; D 34 Waste Management. Moreover OWS has an experienced staff performing life cycle assessment with technical knowledge of production chains in the chemical, plastic, food and energy industry, in the transport sector and in the field of waste treatment.

Main role in the project *WPs envolvement and key task or activities to be done in each*

OWS is WP leader of WP4 and involved in the development of the biodegradation processes and upscaling. OWS will examine the effectiveness of the process on the biodegradation of plastic waste. Moreover OWS will perform the LCA and check biodegradation of chitosan and chitin formulations.

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, Who will*

be primarily responsible for carrying out the proposed research and/or innovation activities

Mr. BRUNO DE WILDE (male), lab manager, Master in bioscience engineering with a specialization in environmental technology at the State University of Gent and started working in OWS at 1989. He is the author or co-author of many scientific articles on biodegradation and compostability. For many years he has been actively involved in several standardization committees such as ISO, CEN and DIN in the field of biodegradable and compostable materials. He is also a member of the Certification Committee Biodegradable Materials of DIN CERTCO (Germany).

Mr. STEVEN VERSTICHEL (male), lab project manager, Master of Science in bioscience engineering with a specialization in environmental technology at the State University of Gent. He is the author or co-author of several scientific articles on biodegradation and compostability. He started working at OWS at 1999 and is head of BCE (Biodegradation, Compostability and Eco-toxicity) department. Within OWS he is responsible for organisation of EU projects.

Ms. NIKE MORTIER (female), lab project manager, Master of Science in bioscience engineering with a specialization in environmental technology at the State University of Gent. She started working in 2008 and is project manager for compostability and eco-toxicity testing and was responsible for EU-projects as FP7 KBBPS and FP7 OPEN-BIO. She was and is active in several standardization committees on biodegradation. She is also representative of ECOS (European Environmental Citizens Organisation for Standardisation) in standardization groups on biodegradation.

Ms. LYNN SERBRUYNS (female) is Master of Science in Biology at the State Univer. of Gent. She started working in 2011 and is project manager for biodegradation testing within OWS.

Mr. STEVEN DE METS (male) is an engineer with a specialization in biochemistry. He is a project engineer of the LCA department.

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

- FP7 FORBIOPLAST 'Forest resource sustainability through biobased composite development', tasks OWS: biodegradability and compostability (organic recycling) testing, LCA
- FP7 SYNPOL 'Biopolymers from syngas fermentation'; tasks OWS: biodegradability and compostability (organic recycling) testing, ecotoxicity, LCA
- FP7 BIOBOTTLE 'Biodegradable solutions for packaging of liquid dairy products'; tasks OWS: biodegradability and compostability testing (organic recycling), ecotoxicity, LCA
- FP7 BIOCLEAN 'New BIOtechnologiCaL approaches for biodegrading and promoting the environmEntal biotrAnsformation of syNthetic polymeric materials'. Investigation of bioremediation and bioaugmentation processes, LCA
- H2020 OPTINANOPRO 'Processing and control of novel nanomaterials in packaging, automotive and solar panel processing lines'; task OWS: organic recycling (composting), LCA.

Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Prederently 2013-2019 years*

- De Wilde, B., Mortier, N., Verstichel, S., Briassoulis, D., Babou, M., Mistriotis, A. & Hiskakis, M. (2013). Report on current relevant biodegradation and ecotoxicity standards. (http://www.biobasedeconomy.eu/media/downloads/2013/01/130226%20KBBPPS%20Deliverable%206_1.pdf)
- Phuong T.V., Verstichel S., Cinelli P., Anguillesi I., Coltelli M.-B., Lazerri A. (2014). Cellulose Acetate Blends – Effect of Plasticizers on Properties and Biodegradability. Journal of renewable materials 02/2014; 2(1).
- Seggiani M., Cinelli P., Verstichel S., Puccini M., Vitolo S., Anguillesi I., Lazzeria A. (2015). Development of Fibres-Reinforced Biodegradable Composites. Chemical Engineering Transactions Vol. 43, 1813 – 1818.
- Mortier N., Velghe F. and Verstichel S. (2016). Organic Recycling of Agricultural Waste Today: Composting and Anaerobic Digestion. Book Elsevier: Biotransformation of Agricultural Waste and By-Products. The Food, Feed, Fiber, Fuel (4F) Economy, Chapter 4, Pag 69-122.
- Biodegradable Plastic Blends Create New Possibilities for End-of-Life Management of Plastics but They Are Not a Panacea for Plastic Pollution. Tanja Narancic, Steven Verstichel, Srinivasa Reddy Chaganti, Laura Morales-Gamez, Shane T. Kenny, Bruno De Wilde, Ramesh Babu Padamati, and Kevin E. O'Connor. Environ. Sci. Technol., 2018, 52 (18), pp 10441–10452.

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

Incubation rooms for composting vessels (200 liter vessels) and reactors (2 liter, 4 liter and 6 liter reactors). Incubation at different temperatures possible.

Equipment for biodegradation tests in all kind of environments: compost, anaerobic, soil, water, marine…

Analytical equipment for compost analyses (drying oven, muffle furnace, titrators, pH-meters, centrifuge, manual and automatic sieves, gas chromatographs, spectrophotometer, AAS, flow injection analyser, grinding equipment, TOC analyser…).UVA weather meter

LCA software: SimaPro LCA software in combination with the Ecoinvent and ELCD LCA databases.

4.1.12. Partner 12. IDELUX Environnement - Belgium

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

The group IDELUX is made of five inter-municipals, respectively named "IDELUX Développement, IDELUX Environnement, IDELUX Eau, IDELUX Finances and IDELUX Projets publics". Their common mission is to improve the well-being of the populations of their territories of action. The Group exercises two big jobs related to the prospect of the sustainable development: the economic development and the management of the environment. As a major concern, the assistance to the municipalities is also a big part of the work performed.

IDELUX Environnement is responsible for the waste treatment from their collection to their valuation. IDELUX Environnement is managing household waste within the Luxembourg province of Belgium (4400 km² and about 340.000 inhabitants). In total, 55 municipalities entrust to IDELUX Environnement the management of their waste. This mission of public service is essential for the municipalities, but also for the citizens. In addition to the collection door to door or in the civil amenities, IDELUX Environnement operates three waste treatment centres including an anaerobic digester, a substitution fuel preparation unit, different paving stones of composting, two halls of sorting for recycling waste, a recycling of inert waste unit, two wastewater treatment plants, a sludge drying unit, a platform of sorting the deck for agricultural covers and finally when no other options remain, the waste landfill facilities. IDELUX Environnement has experience with several types of waste streams.

IDELUX Environnement knows very well the problematic of plastics ending in the compost made of municipal waste due to sorting mistakes made by the citizen. As an industrial partner specialized in the waste treatment, the new technology that will be developed through the RECOVER project is of great interest. This will help the actual mechanical sorting and reduce the plastics ending in the environment despite all our efforts. Our public company makes a perfect showcase of the plastics end-of-life issue and will considerably contribute to the life cycle assessment or the business case analysis. Therefore, IDELUX Environnement will be helpful for organic waste, digestate, sludge or other biomass supplies. IDELUX Environnement will share its knowledge and properties in the biowaste flow as well as the operation of the composting facility.

Main role in the project *WPs envolvement and key task or activities to be done in each*

IDELUX Environnement will lead the WP1 for the characterization, selection and collection of the most representative agri-food plastic waste. IDELUX Environnement will continuously provide biowaste samples to all the other partners but more importantly, IDELUX Environnement will participate in WP4 to the elaboration of some pilot tests of the new technology. IDELUX Environnement will help to improve the new vermicompost reactor development. For the WP6, IDELUX Environnement will contribute to analyse the RECOVER developed technology for both agricultural and municipal use scenarios and will indicate the logistic viability of the proposed solutions. Furthermore, IDELUX Environnement will provide plant operating data to perform the LCA (WP6). IDELUX Environnement will also contribute for communication and dissemination tasks (WP7). The idea is to stimulate a direct public engagement from the citizens and the local policy makers about plastic sorting mistakes.

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities*

Marie-Aline PIERRARD (Female) has a Master of Biological Science with a specialization in ecology of fresh water, and a PhD in Biological Sciences, Environmental Toxicology, both at Namur University. She holds a post-Doc from the Luxembourg Institute of Health. She is a project manager in IDELUX Environnement and is responsible for the EU-project H2020-Volatile as well as internal research projects.

Hugues HANOSSET (Male) is an Electromechanical Engineer from ECAM, Bruxelles Graduate School. He is a Waste Processing Plants Project Leader in IDELUX Environnement.

Marc LUC (Male) is the Operating Manager of the waste treatment plant of Tenneville.

Christophe ARNOULD (Male) graduated as an Electromechanical Engineer at ECAM, Bruxelles Graduate School. He is the Waste Processing Plants Director

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

- H2020-VOLATILE – Biowaste derived volatile fatty acid platform for biopolymers, bioactive compounds and chemical building blocks (Grant Agreement no: 720777).

Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Prederently 2013-2019 years*

- 35000 tons/year of municipal biowaste treatment
- *25000* tons/year of green waste treatment
- *25000* tons/year of digestate are mixed with the volume of green waste in the composting facility
- *20000* tons/year of compost made of municipal biowaste are certified and sold.

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

Industrial household biowaste treatment facilities (35 000 tons biowaste/year).
Logistic of various plastic waste streams.
Composting facilities.

4.1.13. Partner 13. Femto Engineering (FEMTO) / Italy

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

FEMTO is a mechanical engineering and manufacturing company of a scope and size able to cater to product design, prototyping, engineering, production equipment design and construction and techno-polymer component production requirements.

FEMTO is an SME at the moment but is in process of acquisition by a Large group (LCI) and will therefore adopt the financial conditions related to large companies in BBI-RIA projects upfront by providing their full budget as in kind contribution.

FEMTO furnishes a global engineering service, working alongside customers and providing support from the initial development phases of a new product up to and including complete industrialisation, with a complete suite of skills and instruments covering the entire development cycle, from the initial design phase to the finished product.

With FEMTO, companies requiring technical and technological operating support can study a new product from the initial aesthetic/functional design phase, simulate its manufacturing processes, construct prototypes, production equipment and produce this in complete continuous and univocal partnership, without fragmentation and with fully coherent methods of operation, all to the advantage of development times and the substantial quality of the product.

Since it was set up in 1995 as an engineering company oriented towards design using state-of-the-art Cad/Cam/Cae instruments, FEMTO has gradually expanded through the setting up of a suite of operating structures able to provide customers with increasingly complete support.

Today, FEMTO is able to follow up the entire development process comprising design, engineering, calculation, model shop, rapid prototyping, design of equipment, construction, tooling and production, in a wide range of application sectors.

FEMTO cooperates with a number of leading manufacturing companies operating in market sectors such as: automotive, transport, household appliances, electro-tools, biomedical, electro-medical, marine, telecommunications, sports equipment, and consumer goods in general.

Investing more than 10% of annual turnover in R&D, FEMTO furnishes not only technical expertise but also innovative machinery and technologies using software and hardware systems able to machine and manage all types of materials with up to concurrent 5-axis controls.

FEMTO's team has specific expertise in the plastic materials, metalwork construction and composite materials sectors, always based on experience acquired through studies of strictly technical, precision parts and projects in forefront technical environments.

FEMTO operates also with our laboratory of electronic microscopy, where we have a Field Emission Scanning Electron Microscope (Zeiss Merlin) able to operate structural, morphologic and chemical characterization of solid materials in a big range of applications.

We are, therefore, able to offer a qualified high-value added service based on substantial completeness of the cooperation furnished.

Main role in the project *WPs envolvement and key task or activities to be done in each*

FEMTO will perform plastic conversion, tests with chitosan for packaging, provides specifications and samples of packaging to recover.

FEMTO have a good expertise at this level, and in particular in the context of FP7 where it participated with a similar role. FEMTO have realized numerous applications based on the use of bio-polymers, such us more than 30.000 pieces of ECO PHONE, a cordless device produced for Telecom Italy, pots based on bran and potato fibres with biodegradable polymeric matrices. They will have very important roles in WP1 providing support in setting logistic approaches for plastic collection and handling, based on their experience in plastic production and processing, in WP5, for producing pots, tips and other packaging or agriculture products valorised by chitin/chitosan presence, and in WP6 to refine logistic of RECOVER and to support in data for safety, risk and sustainability analysis,.

FEMTO is able to cover all the stages of development of a new product from conception to realization with his expertise on materials, design and technology applied to production process. With FEMTO is possible to define a new object, a new

device and after provide with a complete development, indeed FEMTO has a design department, a mechanical workshop, a prototyping department and a production department, in addition to a microscopy and measurement laboratory certificated.

With his multifunctional team can ensure an integrated management of the entire process reducing overlap between functional steps and the uniqueness information.

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities*

- Dr. Ing. Giuseppe Amato (male) Co-founder and CEO of FEMTO srl. 1995–2004 Founder and only administrator of Femto srl, an engineering company working for projection and production with computer and machines assistance, by offering services of products development and of support for Italian and foreign manufacturers.1989-1995 Freelancer. Consultant senior for CAD/CAM/CAE for various manufacturers strongly collaborating with national and international producers of software for projection and production with computers and machines assistance (Parametric Technolgy Corp., Open Mind, Auton, Think3..). 1987-1989 Manufacturing Manager for Officine Meccaniche DP srl.
- Dott. Andrea Cavallo (male) Master Degree in Scienze Geologiche Roma University. Final rating 107/110.Scientific research and technological applications of Scanning Electron Microscopy (SEM) and the Electronic Microprobe (EMPA) in the field of mineralogy, petrology, experimental petrology and geology of the microstructure and the material sciences.

Responsible for the use and maintenance of the following equipment installed in the laboratory of micro-analysis of the National Institute of Geophysics and Volcanology (High Pressure and High Temperature Laboratory of Experimental Geophysics and Volcanology (HP-HT): Scanning Electron Microscope Emission field (FESEM model JEOL 6500F) and the Electronic Microprobe (EMPA JXA model JEOL 8200). (Service Order INGV, Roma 1 Section 2/2011 and subsequent Internal Register no. 0000383 of 03/02/2015)

Support activities for users outside the laboratory (INGV, other research organizations, universities and third-account)

Activities of maintenance and development of high-tech FESEM and EMPA.

Preparation of natural samples and experimental, electronic microanalysis for high vacuum (EMPA) .and for high resolution imaging (FESEM)

Activity of design and realization in Labview development environment of a SW control and data logging that integrates data and controls the pressure and temperature of a press in solid medium supplied to the laboratory HP-HT of the section of Roma1.

- Ing. Tommaso D'Amico (male) Bachelor Degree in Industrial Engineering (2005), from Florence Engineering University. Senior Production Engineer and CAD/CAM Manager in Femto Engineering. He has participated in a number of R&D regionally, nationally and internationally funded projects aimed at developing and validating new bio-based material and new production technology systems.
- Dott. Ing. Alessandro Velenosi (male) Master Degree in Mechanical Engineering, ex Department of di Mechanics and Industrial Technologies, thesis: "Sviluppo di un Modello di Taglio del Legno in funzione dell'Orientazione della Fibratura". Finale rating 110/110.

Assistant Researcher for the activity: "Sviluppo di strategie per l'ottimizzazione di un processo di asportazione di truciolo nell'ambito del progetto finanziato ATENE". The research is directly applied to the analysis of geometrical errors and unbalance sources of 3D Full Milled Impellers for Centrifugal Compressor in Nuovo Pignone GE Oil & Gas, focused on: Tool Wear and effects on cutting conditions to evaluate vane surface errors variability, tool life, chatter occurrence Machine Tool Kinematic Accuracy for a full optimization of Impeller geometrical errors and an increase of process capabilityVirtual Reconstruction Method given by the integration of CAM/CAD software with developed numerical tools for data processing and representation

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

- 2010-2104 - Customer Brandani Gift Groups production of trays and vessel in PLA for domestic tools and accessories
- 2008-2014 – Design and construction of moulds compatible with bio-materials for the production of table accessories
- Agri and food waste valorisation co-ops based on flexible multi-feedstocks bio-refinery processing technologies for new high added value applications AgriMax H2020 BBI GA 720719
- Development of Personalised Food using Rapid Manufacturing for the Nutrition of elderly Consumers PERFORMANCE FP7 GA 312092
- Integrated cascades of PROcesses for the extraction and valorisation of proteins and bioactive molecules from Legumes, Fungi and Coffee agro-industrial side streams Prolific H2020 BBI GA 790157

Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Prederently 2013-2019 years*

- 2005-2012: Project MIUR-FIRB "Developing new technologies for the protection and defense of the territory from

natural hazards." Within the project I supervised the launch of the electronic laboratory microanalysis (FESEM and EMPA), paying particular attention to the chemical composition of the products of effusive and explosive volcanic districts Italians, in the setup and calibration of electronic microanalysis.

- 2014-2015: Project "EPOS". As part of the Preparatory Phase of the project I have taken care of the task called "LabService" Work Package "WP6".
- 2007-2009: Project INGV-DPC V1 UNREST: Realization of an integrated method for the definition of the unrest phases at Campi Flegrei.

INGV-DPC V4 FLANK (2007-09). UR V4 / 08 and UR V4 / 09. Hazard connected to the flank of Etna dynamics INGV-DPC (2004-2006) V3_2. RH 9 Campi Flegrei

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

Microanalisys laboratory equipped with several instruments, among which the FE-SEM high resolution microscope Zeiss Merlin.Mechanical department equipped with different machines : Milling system Sigma MISSION 5M, Milling system Mikron VCP 600, Milling system Hermle C600V, Milling system Fagima SPINFAST 220, EDM system Agie AGIETRON IMPACT 3, Wire EDM Sodick A 535, Grinding machine Delta SYNTH 800x500 COMPACT PLUS, Radial Drill 40x1250 416.02, Circular saw Shark 280SX Jet, Late CU 400/1000, Column drill IM115 and 35ME

Production department equipped with different machines : Injection machine 9HP300/1300 SANDRETTO, Injection machine 9HP75/285 SANDRETTO, Injection machine 9HP120/430 SANDRETTO, Injection machine BMbiraghi mod. SINTESI 100, Injection machine BMbiraghi mod. SINTESI 35, Inkection machine BMbireghi mod. SINTESI S580, Injection machine Negri Bossi serie CANBIO mod. VS250…

4.1.14. Partner 14 (Entogreen – Ingredient Odyssey, Lda)- Portugal

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

The agri-food sector faces two major challenges, desertification of soils, with the growing need for nutrient input, and reliance on imported nutritional sources for animal feed. However, on the other hand, society faces another major challenge, food waste, resulting in high inefficiencies in natural resource management and nutrient loss along the value chain.

EntoGreen is the brand for the company Ingredient Odyssey, Lda and is dedicated to the valorization of agroindustrial by-products, with the mission of reducing the nutritional waste that occurs in the value chain, returning the environmental sustainability to the agrifood sector and contributing to the sustainable management and efficient use of natural resources.

The biological system we have developed is based on the principles of circular economics and returns the otherwise lost nutrients to the value chain, giving them a second life in the nutrition of both animals and plants. The link that allows nutrient circularization is the Black Soldier Fly (*Hermetia illucens*) larvae, an insect that has a large capacity for conversion of organic matter.

Thus, by-products that would otherwise be lost on landfill or composting serve as feed for these larvae, which after about 15 days reach a suitable larval phase to be separated and processed. This process allows the production of three differentiated end products, insect protein concentrate and insect oil (used in animal feed) and organic fertilizer (for plant nutrition). In this way, we reduce food waste by acting at the agricultural producer and processor level (Figure 1).

For their part, these new sources of animal nutrition contribute significantly to the sustainability of animal production in Portugal and in Europe, reducing the importation of other protein sources on which our continent is highly dependent, such as soy and fishmeal. In addition, these new nutritional sources contribute to the sustainability of aquaculture by reducing this sector's reliance on wild fish catches, and poultry and pig production by reducing the area needed to produce their feed. On the other hand, the use of organic fertilizers in agricultural soils reduces the dependence on chemical and mineral fertilizers, which contribute to groundwater contamination.

We hope to build a facility capable of converting significant quantities of by-products with a significant regional impact, reducing waste and offering more sustainable, locally produced, alternative nutritional solutions, making local farming nutritionally more independent.

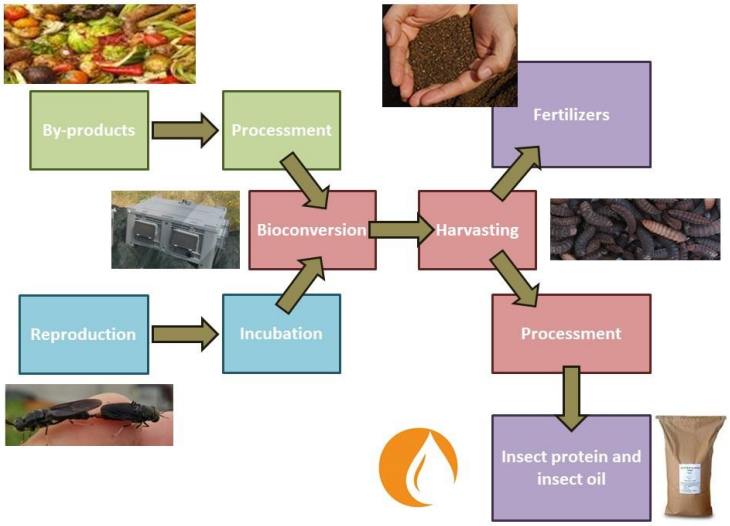

Figure 1 - EntoGreen production process. Green - Waste, Blue - Production of inoculated larvae, Red - Production, Purple - End products.

Model

EntoGreen is headquartered in Santarém, having a pilot unit (Figure 2) built at the National Zootechnical Station, an R&D center belonging to INIAV, an institution with which we have several projects submitted and approved. Among the approved projects we have a strategic one which is EntoValor, a project under co-promotion under COMPETE2020 and led by us and which serves as proof of concept for our process and business plan.

Figure 2 - Processes developed at EntoGreen pilot unit in Vale de Santarém

EntoGreen's business model is based on nutrient circularization, a process in which we receive plant by-products from agro-industrial partners which are used as a nutritional source for our insect larvae. These insect larvae, produced at EntoGreen's premises, are inoculated into the by-products at the right doses and at the most suitable larval phase, ensuring by-product bioconversion into three main products, insect protein and oil and organic fertilizer, thus creating a by-product solution. which do not always find a suitable destination (Figure 3).

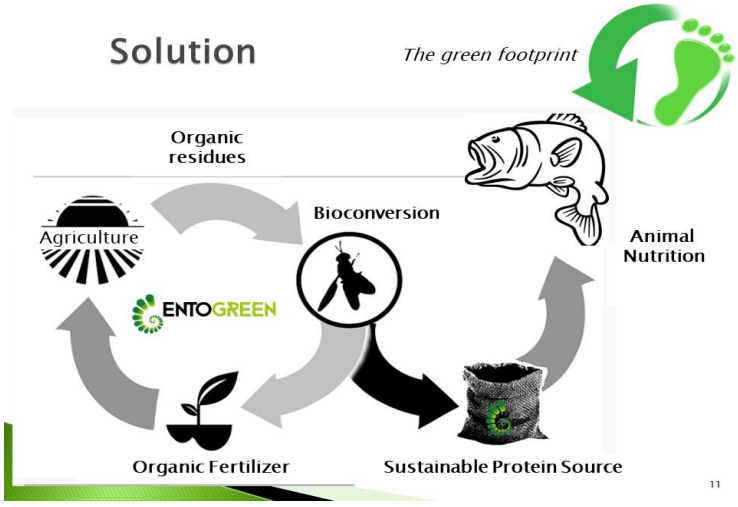

Figure 3 - The nutrient cycle in EntoGreen's production process

EntoGreen's plans are to build a unit capable of converting 3000T of plant by-products per month, being able to produce about 750T of organic fertilizer, 214T of insect protein and 42T of insect oil. In addition, this unit will be able to employ over 40 people and will be ideally located in the Santarém area (Portugal).

The dedication to this area and the quest to make this sector a reality in Portugal recently led us to create the Portuguese Insect Producers and Transformers Association - Portugal Insect – which is presently chaired by EntoGreen.

Over the years, the excellence and innovation of Entogreen's work has been recognized, not only through the approval of national and international projects, but also through the awarding of several awards.

Recently the company received 4 awards that should be noted:

- The Entrepreneurship and Innovation Award of Banco Crédito Agrícola;
- The Born from Knowledge Award from the Portuguese National Agency of Innovation;
- The Agri-Innovation Summit Award;
- Also the European Union INvertebrateIT Project Award, recognizing Entogreen as one of the three most promising startups in the sustainable food field.

Main role in the project *WPs envolvement and key task or activities to be done in each*

Entogreen will support WP 1 - Agri-plastic waste (AWP) mapping, characterization, collection strategy and pre-treatment, contributing with its expertise on insect. For the WP2 we will cooperate in testing the ability of earthworm, and insects to digest plastic. In WP3, we will support the up-scale production in terms of knowhow on insects insects, including maintenance/depletion of plastic-degrading capacities in time. For the WP5 we will work on bio-fertilizers, and will transfer to scale up in WP6. We will collaborate in WP7 and WP8.

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities*

Dr. Daniel Murta (male): Project manager and Leader of the innovation unit in Entogreen. PhD in Veterinary , PhD in Veterinary, Researcher at CIISA(University of Lisbon) and Lecturer at Lusofona University

Dr. Rui Nunes *(male):* BsC Industrial Chemistry with MBA from Coimbra University, experience in Food and Drug Industry in the past an at the moment fully dedicated to development of Entogreen.

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

- The Entrepreneurship and Innovation Award of Banco Crédito Agrícola;
- The Born from Knowledge Award from the Portuguese National Agency of Innovation;
- The Agri-Innovation Summit Award;
- Also the European Union INvertebrateIT Project Award, recognizing Entogreen as one of the three most promising startups in the sustainable food field.
- 2Colaborative Labs:
    - INOV FEED with INIAV (National Institute for veterinary and agriculture investigation)
    - B2E – Blue Bioeconomy

Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Prederently 2013-2019 years*

Dezembro 2015 Os insectos na suinicultura em Portugal

- *Murta, D., Nunes R., Moreira, O. (2015). Suinicultura, 110.*

Julho 2016 Insetos: uma solução nutricional e ambiental

- *Murta, D., Nunes R., Moreira, O. (2016). Alimentação Animal, 96.*

Abril 2017 Fórum Politécnico, Valorização da Agroindústria: Redes de experimentação agrária e internacionalização, Ovibeja, Beja, Portugal.

- *Murta D. - Economia Circular: Produção de insetos para a alimentação animal.*

Março 2017 Green Business Week, Lisboa, Portugal.

- *Murta D. – Insetos – Uma oportunidade na valorização de resíduos.*

Março 2017 Seminário – Potenciar Sinergias na Atividade Pecuária, Santarém, Portugal.

- *Murta D.– Insetos como uma solução na atividade pecuária – Uma resposta a dois desafios.*

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

EntoGreen's plans are to build a unit capable of converting 3000T of plant by-products per month, being able to produce about 750T of organic fertilizer, 214T of insect protein and 42T of insect oil. In addition, this unit will be able to employ over 40 people and will be ideally located in the Santarém area (Portugal). At the moment, Entogreen is installed in INIAV (National Institute for veterinary and agriculture investigation) and have capacity to produce 1 ton/ moth of live BSF. We work with INIAV laboratories to evolve our technology developing new ways to produce Protein and Fat from BSF. We need to invest in new equipment to test and manage insects digesting plastics.

EntoGreen is working with a team of investors and aims to lay the foundation stone for the construction of a factory by the end of 2019. This unit, to be built in the Santarém District, will employ over 50 people and will be able to produce 9000T of fertilizer per year (plant nutrition) and about 2500T of protein and 500T of insect oil per year (animal nutrition). We will therefore use about 3000T of plant by-products per month, giving them new life and returning them to the value chain. New Partnerships EntoGreen seeks to engage the agro-industry sector both upstream and downstream of its process, thereby seeking to partner with both farmers and industrialists and compound feed producers. These partnerships have allowed us not only to give new destination to the generated by-products but also to create new end product lines for animal and plant nutrition. We anticipate that these partnerships will not only be developmental but possibly also investment in our process. The use of insects in animal feed has been a hot topic in recent months, and EntoGreen has invested heavily in research and development to make this opportunity a reality. An example of this is our involvement not only in R&D projects but also in the creation of master theses with students from various institutions, from ISA, FMV-ULisboa and UTAD. These 3 theses were all developed for feeding birds such as chickens and laying hens using larvae produced by us. We are preparing three more lines of research, but with the use of insects to feed swine as early as September 2019, receiving three students from FMV-ULHT.

We need to invest in infrastructures to manage new developments. We will work with actual equipment, boxes, labs (INIAV), productions of new eggs rooms and developing new BSFL (black soldier fly larvae) forms of processing increasing the efficiency of the process. Working with plastics is important to test the ability to produce BSFL with plastics and test in the end how protein and lipids will be affected by the plastic digestions.

4.1.15. Partner 15. Carton Bros -T/A Manor Farm (MANOR) - Ireland

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

Carton Brothers - an Irish Family Tradition since 1775

Carton Brothers is the name of the company that produces Manor Farm chicken. It is one of the oldest family companies in Ireland and dates back to 1775. Manor Farm is the largest chicken processor in the Republic of Ireland currently producing approximately 900,000 birds on a weekly basis. The company, trading under its own flagship “Manor Farm” brand, distributes to all major retail outlets in the Republic of Ireland and many of the catering institutions. The company also produces “Own Label” product to all the major retailers in Ireland. The company is highly active in the export market where its products may be seen in the U.K., France, Spain, Eastern Europe, South Africa and the Far East. The company is vertically integrated in that it has its own hatchery, feed mill and processing plant. The farms where the birds are reared are under contract to the company and must follow all guidelines and procedures as laid down by the company. This allows complete traceability for the products. Processing moved to Shercock in Co. Cavan in 1970 becoming the first plant in Britain and Ireland to be EEC approved. The company whilst already at the forefront of the industry in terms of size has spent over €14m in 2005 increasing the size and capacity of its processing plant at Shercock. Further investment is currently underway with an additional €25million planned over the next 5 years. Consistent within Manor Farm at all times is dedication to Quality and the production of unique and very tasty chicken products from the best chicken that we ourselves produce. Manor Farm believes innovation is critical to their success and support new product development to ensure that we are at the forefront of new & emerging technologies. Manor Farm was first to market in Ireland with cook in the bag packaging technology & continues to invest resources in new packaging developments.

Main role in the project *WPs envolvement and key task or activities to be done in each*

Carton Bros (Manor) will support WP1 with information on plastic used in food packaging, and in WP5 where they will validate the performance of the new material in an industrial, commercial environment from both migration & shelf life perspectives in comparison to existing PET & PE materials. We distribute through all fresh food channels throughout the island of Ireland & are therefore in a unique position to test new materials. The project also offers interesting new complementary opportunities for our farmers chicken suppliers.

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities*

Mr JUSTIN CARTON (male) Fellow of the Association of Chartered Management Accountants, MSC MIS, BSC Marketing and Management**.** Director of Information Technology at Carton Bros Ltd, Ireland's largest poultry industry business. Has worked in the poultry industry since 1984 on all aspects of integrated poultry production. He is responsible for ensuring that the business maintains its leading-edge position in the poultry industry, by directing its research and development programme, with particular emphasis on the application of "smart systems" within the industry.

Mr Declan Kennedy (male) Manor Farm processing director. Qualification of BSC in Food Business from University College Cork. Prior to joining Manor Farm, Declan has over 20 years' experience in food production in senior management positions with Dawn Farm Foods, Irish Country Meats and Green Isle.

Mr Declan Prunty (male) Manor Farm Project Manager. Leads all major projects for the Manor Farm site to including new & emerging technologies. Qualifications include:

2013 Lean Six Sigma Green Belt / 5S
2009 L5 in AutoCAD Design & Computer Aided Draughting
2008 C&G Certificate in Handling Refrigerants (Ammonia)
2006 Project Management
1992 C&G Advanced Certificate Refrigeration & Air Conditioning (Part 3)
1990 National Craft Certificate Refrigeration (Part 1&2)

Mr Liam Hilliard (male) Manor Farm Food Safety & Quality Assurance Manager. Liam has held Quality Management positions since 2003 working with large organisations including Kerry Foods, Irish Country Meats and Dawn Farm Foods before joining Manor Farm. Qualifications include:

DIT Cathal Brugha Street, MSc. Food Safety Management 2011-2013

DIT Kevin Street, Kevin Street, Dublin 8, Graduate Diploma (FETAC Level 8) in Food Science and Technology – 1995-1996

Carlow RTC, Kilkenny Road, Carlow, Diploma in Food Quality Management – 1992-1995

Dr Zandra Reed (Montgomery) (female) Manor Farm NPD Manager. NPD Manager at Manor Farm since 2007, previously working as senior food technologist in Manor Farm since Oct 2005. Previously, NPD Manager at Dawn Farm Foods, The Mauldins, Naas, Co. Kildare (2003-2005), NPD Project Manager, Dawn Farm Foods, Oct 2001 – Nov 2003. 1998-2001, completed full time PhD in University of Ulster, thesis title "Sensory Quality & Consumer Acceptance of Chilled Ready Meals", September 2001. Graduated from University of Ulster with 1st Class Hons Degree in Consumer Studies with DIS in 1998.

Ms. Rebecca O'Brien (female) (Manor Farm NPD Project Manager). Rebecca joined Manor Farm in 2016, having held positions in quality, NPD, and project management in a number of flagship Irish food companies such as Kerry Foods, Carroll Cuisine, and Queally Food Group. Her remit in Manor Farm includes Export NPD, packaging and process development.

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

- RefuCoat (H2020 BBI): Full recyclable food package with enhanced gas barrier properties and new functionalities by the use of high performance coatings, 2017-2020

Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Prederently 2013-2019 years*

- Haughton, P.N., Lyng, J., Fanning, S., Whyte, P. (2013). Potential of a commercially available water acidification product for reducing *Campylobacter* in broilers prior to slaughter. British Poultry Science, 54, 319-324.
- Haughton, P.N., Lyng, J., Cronin, C., Fanning, S., Whyte, P. (2012). Effect of crust freezing applied alone and in combination with ultraviolet light on the survival of *Campylobacter* on raw chicken. Food Microbiology, 32, 147-151.
- Haughton, P.N., Gomez Grau E., Lyng, J., Cronin, D., Fanning, S., Whyte, P. (2012). Susceptibility of *Campylobacter* to high intensity near ultraviolet/visible 395±5nm light and its effectiveness for the decontamination of raw chicken and contact surfaces. International Journal of Food Microbiology, 159, 267-273.
- Haughton, P.N., Lyng, J.G., Cronin, D.A., Morgan, D.J., Fanning, S., Whyte, P. (2012). Efficacy of pulsed electric fields for the inactivation of indicator microorganisms and foodborne pathogens in liquids and raw chicken. Food Control, 25, 131-135.
- Haughton, P.N., Lyng, J.G., Morgan, D.J., Cronin, D.A., Noci, F., Fanning, S., Whyte, P. (2012). An evaluation of the potential of high intensity ultrasound for improving the microbial safety of poultry. Food and Bioprocess Technology, 5, 992-998. (doi 10.1007/s11947-010-372-y).

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

- Manor Farm Innovation Centre
- Contracted Laboratories for Migration & Shelf life Testing

4.1.16. Partner 16. ENCO SRL (ENCO)- Italy

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

ENCO is an innovation and research-consulting firm based in Naples, active since 1987. Originally specialized in business consulting and industrial engineering, it has expanded its range of activities becoming an experienced advisor for both private business and public authorities involved in territorial development and opened two new offices in Rio de Janeiro and Brussels. The company participates in national and international R&D&I projects as a facilitator for business, product, technology and process innovation in different industrial areas, including energy, environmental management, the agro-food sector and logistics. The company has a long track record participating in EU-funded projects as WP leader, task leader and project coordinator. The company is specialized, inter alia, in two main fields:

• Development of market analysis, business plan and exploitation plans: studies (elaborated at European and international scale) for innovative products, technical solutions and systems are focused on market trends, market profitability and distribution channels. ENCO conducts market surveys with potential customers, Desk analysis, SWOT analysis, identification of economic, legislative and behavioral barriers, customer segments analysis and competitor analysis, market feasibility studies and Risk analysis. Moreover, ENCO develops innovation and business strategies, quantitative assessment of pricing and cost-benefit analysis, identifying Capital and Operational Costs.

• Communication and dissemination strategies: developing action plans for dissemination and communication activities; identifying target stakeholders, tools and actions to be implemented; developing tailor made campaign to raise audience's awareness on the project rationale, activities and results; identifying and selecting events, workshops and meetings; organizing information and dissemination events; producing publications, newsletters, brochures, leaflets, promotional and marketing materials; setting up and managing project websites, and participating in relevant events and exploiting institutional communication channels; building a broader community around the project topics. https://www.enco-consulting.it/

Main role in the project *WPs envolvement and key task or activities to be done in each*

ENCO will lead WP7 Cross-cutting activities & Innovation Impacts to support the full deployment of the project outputs and results, planning and implementing dissemination and communication activities targeted to the project key audience groups. To achieve this, a set of activities has been outlined already in Section 2.2. An extensive professional communication campaign to reach key stakeholders will be designed by ENCO and implemented in a spirit of full cooperation by all the project partners.

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities*

Dr. Ing. MARCO DE LA FELD *(male). Senior Project Manager*, since 2008. He graduated in Mechanical Engineering in 2003 and obtained a Research Doctorate in "Engineering of Materials and Production" in 2006. Mr. de le Feld is the Project Coordinator of SHEALTHY (H2020- GA 817936) and as been the Project Coordinator of SMART GROUND (H2020-GA 641988), and the IP & Exploitation Manager of SaltGae (H2020-GA 689785) and has successfully managed more than 8 European research projects including the FOODIE project (CIP-ICT-PSP GA 621074), ALGATEC II (FP7-SME GA 315469) and OPTISOLV (ETB-2012-26). He is very skilled in delivering business scenarios and plans for internationalization and fast replication of innovative products and services. In addition to his vast experience as a project partner/manager and as a research advisor in numerous European projects, Mr. de la Feld has several experiences as EU Project trainer and has led several capacity building workshops. *(in-house consultant).*

Ms. SIMONA MINCIONE (*female*). *Project Manager,* since 2014. She holds a Master degree in International Relations and European Studies. She got the certificate in EU Funding and EU Project Design & Management in 2012 and WIPO certificate for IPR Management. She as excellent skills in the management, planning and execution of exploitation activities, besides managing activities of EU-funded project as FOODIE (ICT-PSP-GA 621074), Waste2Fuels (H2020-GA 654623. She is currently assisting the SHEALTHY (H2020- GA 817936) project coordinator. *(employee)*

Ms. ANTONIETTA PIZZA *(female). Senior Project Manager*, graduated in International Relations, she specialized in European Funding and Project Design in 2014 and got a 2nd level Master in Open Innovation and IPR in 2017. In her position at ENCO, she is responsible for developing project proposals and carrying out project activities, having experience in EU project management and operational activities of various funding programmes (H2020, CIP Eco-Innovation; FP7, EuroTrans-Bio), in particular she assisted the Project Coordinator in the management of SMART GROUND project (H2020-GA 641988) and in the planning and execution of exploitation activities. In 2016, Ms. Pizza obtained the ISIPM Certificate in Project Management. (employee).

Mr. Ing. GIAMPIERO DE LA FELD (*male*). *Managing Director of ENCO* since 1986, thanks to his more than 30 years of experience and his engineering background, Mr. de la Feld has been involved in his company's projects and has managed and coordinated diverse national and international projects with strong focus on their technical and financial aspects.

Furthermore, Mr. de la Feld was president of Confindustria Campania in 2000/2002, which is a non-profit organization uniting different business associations from the Italian region of Campania and representing their interests, needs and demands vis-à-vis political and administrative institutions, trade unions and other social forces. He has thus gained further insights into the European business world, its opportunities and difficulties, as well as experience with administrative and political institutions/ procedures involved in business development and innovation. Moreover, Mr. de la Feld has sound experience in supporting start-ups in developing their business as well as planning all the actions to undertake to facilitate market penetration of innovative systems, developed under EU-funded projects. (SME owner)

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

- BIOnTOP "Novel packaging films and textiles with tailored end of life and performance based on bio-based copolymers and coatings" (H2020-BBI GA 837761). ENCO leads the Market strategy and Innovation impacts.
- SHEALTHY "Non-Thermal physical technologies to preserve fresh and minimally processed fruit and vegetables" (H2020-SFS GA 817936). ENCO acts as Coordinator, leads the Exploitation Management and Dissemination activities.
- REMADYL "Removal of Legacy Substances from polyvinylchloride (PVC) via a continuous and sustainable extrusion process" (H2020-SC5 GA 821136). ENCO leads the Dissemination and Exploitation activities.
- FOODIE "Open and interoperable agricultural platform hub on the cloud for the management of spatial and non-spatial data relevant for farming" (CIP-ICT-PSP GA 621074).ENCO leads dissemination activities and responsible for the IPR Management.
- WASTE2FUELS "Sustainable production of next generation biofuels from waste streams". (H2020-LCE GA 654623). ENCO is responsible for conducting in-depths market research and communication activities.

Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Prederently 2013-2019 years*

- Rossetti, P., Dino, G. A., de la Feld, M., Pizza A., Coulon, F., Wagland, S., Gomes, D., and the SMART GROUND Team (2016). SMART GROUND Project: SMART data collection and integration platform to enhance availability and accessibility of data and information in the EU territory on SecoNDary Raw Materials, Geophysical Research Abstracts Vol. 18, EGU2016-5881-1, 2016 EGU General Assembly 2016
- Rossetti, P., Dino, G. A., Biglia, G., De La Feld, M., Pizza, A. (2017) Characterization of extractive waste facilities for SRM recovery from mining sites: case studies from the SMART GROUND Project, Associazione Mineraria Sarda, IV SIMPOSIO Attività Minerarie nel Bacino del Mediterraneo 2017.
- Dino, G. A., Rossetti, P., Biglia, G., Sapino, M. L., Di Mauro, F., Heikki, S., Coulon, F., Gomes, D., Parejo Bravo, L., Zapata Aranda, P., Lorenzo Lopez, A., Lopez, J., Garamvölgyi, E., Stojanovic, S., Pizza, A., De La Feld, M., SMART GROUND Project: A new approach to data accessibility and collection for Raw Materials and Secondary Raw Materials in Europe (2017), Environmental Engineering and Management Journal 2017.
- Charvát, K., Kepka, M., de la Feld, M., Mincione, S., Šplícha,l M., Šrédl M. (2015) POMODORE – Smart Agri Food Project, ISAF 2015, 15 September 2015
- de la Feld, M., Lorenzo A., Pizza, A., (2014). ALGATEC II: ottimizzazione delle soluzioni di riciclaggio dell'acqua di lavaggio delle olive, Acta Italus Hortus, Bari 2014

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

As a leading business consultant with extensive experience in business planning and development, communication, dissemination and exploitation of the project results, ENCO possesses the suitable human resource and operational capacity for the tasks to be fulfilled within the project.

4.1.17. Partner 17. S.A. Agricultores de la Vega de Valencia (SAV)- Spain

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

SAV is a large company founded in 1900, which conducts its activity in the environmental sector, from Gardening, agriculture, preservation of natural parks, logistic, street cleaning, and integral waste management to water treatment plant maintenance. Nowadays SAV has decided to diversify its activities by expanding into bioproducts markets.

SAV has been collecting urban waste for more than 100 years. Its experience is more than a guarantee regarding the

innovation in waste life cycle. The company provides environmental services in more than 80 municipalities across the country of Spain in strict collaboration with the local government.

SAV has operated since 1900 in "l'horta de València" and has several owned agricultural fields such as vegetables, vineyards and olive trees. Also SAV belongs to the cooperative agricultural of Montserrat with the partner number 575 and is in charge of the collection of meat and vegetable remains in Valencia Central market and in numerous supermarkets. SAV regularly collaborates on activities which aim to promote local food and reduce food waste. SAV has different lands and crops to use as AgroLabs (dry farming and irrigation) located in the provinces of Alicante and Valencia (Spain).

Another of SAV activities is the management, protection and maintenance of natural parks.

SAV has 30 years of experience guarantees the work of landscape services, work such as the design, execution and maintenance of green areas, parks and gardens, all under the strict criteria of quality and environmental sustainability. Being the first European organization to have their activities in gardening certified in the Water Footprint.

Our company has the most modern plant of sorting household in Valencia. This plant currently processes 264Tm daily. From the waste processed, only around a third goes to the landfill. The rest is separated between organic and byproducts (around 15%) from this almost half of it is plastic. This number is subjected to change since the plant is going to receive waste from other municipalities.

In the field of water treatment, SAV manages the purification and treatment of wastewater in more than 200 wastewater treatment plants. The population served is more than 8 million equivalent inhabitants and the volume of treated water reaches 640 cubic hectometers per year. SAV is a member of the European Water Platform.

Regarding technological solutions, SAV has a software development team and has always tried to be an early adopter of innovative technology. In this contest, for example within the field of the logistics, CO2 emissions have been reduced by 12% through the use of sensors in the fleet of vehicles that helped to operate with efficient driving (national project winner of the 2019 sustainable city awards) and is developing an optimal route system that will be use in the cities of Utrecht (Netherlands) and Alba iulia (Romania) within a European project. Besides a smart City platform has been developed by SAV as integral parts of the H2020 project Plasticircle (No 730292) and is currently working on an massive Internet of Things (IOT) platform for use in different activities.

SAV has expertise in project coordination in this way at the present is the coordinator of the proposal WaysTUP! Value chains for disruptive transformation of urban biowaste into biobased products in the city context (818308).

The department of R&D and European projects is formed by a multidisciplinary team of doctors in science and engineering as well as technical experts in ICT, circular economy, agriculture, food, water purification and many other disciplines. Our team works actively managing environmental projects and developing project proposals through the H2020 instrument, and is certified in the project management methodology Prince2® and Agile - Scrum.

Main role in the project *WPs envolvement and key task or activities to be done in each*

SAV will support with information on plastic use in agriculture (WP1) and in soil bioremediation experiments (WP4), will make the validation of the mulching films and the fertilisers from the project (WP5) not only in agricultural production in rural fields but also in city parks. SAV is responsible for the maintenance of urban gardens in several cities of the Valencian region.

Finally, SAV will contribute with data for safety, risk and sustainability analysis as well as with the project dissemination and management.

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities*

Dr Jerónimo Franco PhD. *(male)* has a Ph. D. mechanical engineering department and construction doctoral program in industrial technologies and materials; Design of a methodology for the obtaining of solutions to the collection of OFMR in Smart Cities environments by the Jaume I University . He is also Industrial Chemical Engineer by the Technical school of Industrial Engineers of Valencia (UPV) with the specialty of sewage treatment. He has a master in environmental sustainability and participated in the UPV, in the department of machinery and thermal engines in research groups working with physical and chemical modelling of internal combustion processes. He participated also in the program of thermofluid processes of alternative internal combustion engines. For seven years he has been responsible for the cleaning and waste collection services of the city of Valencia managed by SAV. He is certified by Prince2(PRojects IN Controlled Environments) Project management method (04086683-01-RCK4)

Ms María Pilar Uceda. *(female)* Agricultural Technical Engineering. Specialty: Farming operations by the University of Castilla la Mancha. She also has a course of Recognition and Management of Plagues and Diseases in Urban Gardening (30 h) - (Nov-11). polytechnic university of Valencia, course of Recognition and Management of Plagues and Diseases in Gardening (30 h) - (Jan-09). polytechnic university of Valencia and course of Recognition and Management of Plagues and Diseases in Gardening (30 h) - (Jan-09). polytechnic university of Valencia. She has been Technical Engineer of the Department of Gardening and Landscape (since Oct.97) in SAV.

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

- PlastiCircle (H2020). (GA 730292)The project will rethink the different phases involved in transforming waste into valuable products: from the development of smart containers for separate waste collection to the improvement of transport routes and sorting technologies. The overall aim is to achieve recovery within the same value chain. (2017-2021).
- WaysTUP (H2020) (Coordinator) (GA 818308) The project will demonstrate the establishment of new value chains for urban biowaste utilisation to produce higher value purpose products (i.e. biobased products, included food and feed ingredients,biochar for application in agriculture) though a multi stakeholder approach in line circular economy

Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Prederently 2013-2019 years*

- F. Ample, J.M. Ricart, A. Clotet, D. Curulla Ferré and J.W. Niemantsverdriet (2004), "Competitive CN and N2 formation on Rh(111) : a case of entropic stabilization" , Chemical Physics Letters, 385, 52.
- Franco Gonzalez, Jeronimo; Gallardo Izquierdo, Antonio; Carlos Alberola (2019), Alternativas en la gestión de la recogida separada de la fracción orgánica biodegradable de los residuos sólidos orgánicos, scientific poster, VIII Simposio Iberoamericano De Ingeniería De Residuos, Asunción Paraguay.
- J.Franco (2017), Design of a methodology to obtain solutions for the collection of the urban waste organic fraction in Smart Cities environments, Department of Mechanical and Engineering Construction, Jaume I university of Castellón.
- J. Franco (2019), Waste4think & Plasticircle two different methodologies to improve the collection of waste in containers, scientific poster, Word congress ISWA 2019

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

SAV manages the purification and treatment of wastewater in more than 200 wastewater treatment plants. The population served is more than 8 million equivalent inhabitants and the volume of treated water reaches 640 cubic hectometers per year.

SAV has the most modern plant of sorting household in Valencia. This plant currently processes 264Tm daily. From the waste processed, only around a third goes to the landfill. The rest is separated between organic and byproducts (around 15%) from this almost half of it is plastic. This number is subjected to change since the plant is going to receive waste from other municipalities.

SAV has a large fleet of garbage trucks. We have been collecting bottles and cans for more than 30 years.

In order to reduce our emissions and to provide a cleaner city, we are currently developing an Internet of Things (IOT) platform for the use in environmental and activity sensors for the use of a smart city project.

SAV is a member of the agricultural Montserrat cooperative in Spain.

SAV has different lands and crops to use as AgroLabs.

Number of employees 920.

SAV turnover is more than 100 million euros per year.

4.1.18. Partner 18. OWS Research Foundation PS (OWS RF) - Belgium

Organization description *a description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal*

The OWS Research Foundation is a private foundation and as organization type: non for profit. It's goal is to contribute to the scientifical know-how and optimization of technologies in the field of biogas, biodegradibility, compostability and ecotoxicity, alone or in cooperation with third parties, in Belgium or outside Belgium.

OWS Research Foundation may therefore develop the following activities:

- the development and execution of laboratory analyses and research, among others for the determination of the biological biodegradibility, compost quality, environmental sustainability, compostability, ecotoxicity and other environmental and biological aspects of products and packaging;
- the delivery of recommendations, consultancy, laboratory analyses, feasibility studies, process monitoring and engineering studies, among others in the field of municipal solid waste treatment and waste treatment, composting, aerobic and anaerobic fermentation, gas fermentation, fatty acids fermentation and other fermentation, biological degradability, renewable energy and the environment in general;
- contribute to the development and optimization of special installations, more specifically systems and plants for the treatment of municipal solid waste, organic fractions of municipal solid waste or equivalent organic waste streams, energy crops or organic residues, systems and plants for the production of energy, compost, chemical and organic products, products for soil amelioration and substrates from agricultural crops, systems and plants that contribute to the remediation of the environment or the production of renewable energy, installations for the conversion of gases via fermentation to components with higher value;
- contribute to the development and optimization of the collection, the transport, the storage, the treatment and the recycling of raw materials, semi-finished products, the treatment and recycling of raw materials, agricultural crops and residues and waste;
- the delivery and maintenance of certificates related to compostability, biodegradibility, environmental sustainability, recyclability, sustainability.
- participation to international, national and local scientifical or commercial research, innovation and development projects.
- the application for, receiving and utilization of international, national or local governmental or private subsidies, financing, loans, incentives or other economic benefits.

Main role in the project *WPs involvement and key task or activities to be done in each*

OWS RF is WP leader of WP4 and involved in the development of the biodegradation processes and upscaling. OWS will examine the effectiveness of the process on the biodegradation of plastic waste. Moreover OWS will perform the LCA and check biodegradation of chitosan and chitin formulations.

CV and key personnel *a curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities*

- **Bernard Moerman (Male),** Permanent Representative of Euander BV and Authorized Representative of OWS Research Foundation PS. He studied at the State University of Gent and New York University School of Law. He is legal advisor.
- **Steven Verstichel (Male),** lab project manager, Master of Science in bioscience engineering with a specialization in environmental technology at the State University of Gent. He is the author or co-author of several scientific articles on biodegradation, compostability and anaerobic digestion. He is responsible for organization of EU projects within the company. He was active in projects as TRANSBIO, VOLATILE, AGRIMAX and FUNGUSCHAIN.

Relevant projects. *A list of up to 5 relevant previous projects or activities, connected to the subject of this proposal*

- Not applicable

Relevant publications, products and services. *A list of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content. Preferently 2013-2019 years*

- Not applicable

Relevant infrastructure and equipment a description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work. *Outline the key facilities and infrastructure available and demonstrate that each team has sufficient capacity*

- Not applicable

4.2. Third parties involved in the project (including use of third party resources)

The partners below will use 3rd party resources.

Partner 1. Universtity of Almeria (UAL) - Spain	
Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	**YES**
UAL will select a sub-contractor as external independent Ethics Advisor.in order to comply with the Ethics requirements, to ensure the best Ethical practice and to address Ethical issues in all their shades *The best value for money will be looked for a strong expertise in ethical issues* *The subcontract budget is 17K.*	
Does the participant envisage that part of its work is performed by linked third parties[1]	**NO**
If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party	
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	**NO**
If yes, please describe the third party and their contributions	
Does the participant envisage that part of the work is performed by International Partners [2] (Article 14a of the General Model Grant Agreement)?	**NO**
If yes, please describe the International Partner(s) and their contributions	

Partner 5. Miguel Hernandez University (UMH) - Spain	
Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	**YES**
UMH will select a sub-contractor in order to test earthworm pre-compatibility with microorganism and enzyme pools in WP2 (task 2.5) and for up-scaling production of earthworm fortified with PMC consortium in WP3 (task 3.2). *The best value for money will be looked for a strong expertise in Lumbicus terrestris earthworms and capabilities in environmental restoration, terrestrial ecotoxicity of pesticides, enzymatic bioremediation and biochar uses.* *The subcontract budget is 50K.*	
Does the participant envisage that part of its work is performed by linked third parties[1]	**NO**
If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party	
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	**NO**
If yes, please describe the third party and their contributions	
Does the participant envisage that part of the work is performed by International Partners [2] (Article 14a of the General Model Grant Agreement)?	**NO**
If yes, please describe the International Partner(s) and their contributions	

Partner 17. S.A. Agricultores de la Vega de Valencia (SAV) - Spain.	
Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	**YES**
SAV will select a sub-contractor to support services in field sampling techniques, specific agricultural techniques and fertilizer application *The best value for money will be looked for the subcontract* *The subcontract budget is 12K.*	
Does the participant envisage that part of its work is performed by linked third parties[1]	**NO**
If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party	
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	**NO**
If yes, please describe the third party and their contributions	
Does the participant envisage that part of the work is performed by International Partners [2] (Article 14a of the General Model Grant Agreement)?	**NO**
If yes, please describe the International Partner(s) and their contributions	

5. Ethics and Security

5.1. Ethics Issues

ANIMALS:

An Ethics assessment was performed with the completion of the Ethical Issues Table, and this project resulted to involve Ethics issues. Specifically, the Ethics issues identified are summarized below

MOTIVATION:

[156]The discovery of the presence of PE-degrading microorganisms in wax worm guts has provided a promising direction for further study of the microbial degradation of both PE and other plastics, along with new insights into the life cycle of synthetic plastics in nature. In addition, the isolation and characterization of more PE-degrading microorganisms from this source and a better understanding of the enzymatic system involved in PE degradation could be helpful in the development of remediation approaches for plastic wastes, which could eliminate plastic pollution concerns.

Information to be provided:

1) Details of the species and rationales for their use, numbers of animals to be used, and nature of the experiments, procedures and techniques to be used.

Species.

For the development of the RECOVER project the inclusion of several species of insects and earthworms has been determined, which are listed below (indicating number of animals to be used) approx:

- *Tenebrio molitor (150,000 uds)*
- *Acheta domestica (30,000 uds)*
- *Plodia interpunctella (30,000 uds)*
- *Galleria mellonella. (120,000 uds)*
- *Ephestia kuehniella, (30,000 uds)*
- *Sitotroga cerealella, (30,000 uds)*
- *Tribolium castaneum, (30,000 uds)*
- *Sitophilus granarius, (30,000 uds)*
- *Eisenia andrei (5,000 uds)*
- *Lumbricus terrestris (2,000 uds)*

Reasons for its use.

- The uses contemplated for these commercial species of insects include the work of research, extraction, characterization and isolation of microorganisms endogenous to them.
- The capacity of biodegradation of plastics that these specific communities of insects and associated microorganisms have is evaluated.
- Insects are a means vector of endogenous and exogenous species of fungi and specific bacteria that allow them to chemically attack plastics and cause their enzymatic degradation.
- Insects are a useful means of colonizing plastic surfaces and activate their accelerated degradation.
- The by-products of accelerated biodegradation can be used by insects in their conversion to protein for use in various industries (fertilizers, chemistry, pharmaceuticals, biomaterials, nanotechnology, etc.).

Nature of the experiments, procedures and techniques to be used

- Insects will only be inoculated with several potentiated microorganisms.
- Inoculation techniques are the ones determined by development protocols and techniques published and reported as ethically and environmentally safe.

2) Justification of animal use (including the kind of animals to be used) and why alternatives cannot be used.

The use of the indicated insects is necessary because it is the key to the associated symbiotic biodegradation solution. This

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[156] **156 Evidence of Polyethylene Biodegradation by Bacterial Strains from the Guts of Plastic-Eating Wax worms**
Jun Yang,*,†,‖ Yu Yang,†,‖ Wei-Min Wu,‡ Jiao Zhao,§ and Lei Jiang†
†Key Laboratory of Bio-Inspired Smart Interfacial Science and Technology of Ministry of Education, School of Chemistry and Environment, Beihang University, Beijing 100191, P. R. China ‡Department of Civil and Environmental Engineering, William & Cloy Codiga Resource Recovery Research Center, Center for Sustainable Development & Global Competitiveness, Stanford University, Stanford, California 94305-4020, United States §Shenzhen Key Laboratory of Bioenergy, BGI-Shenzhen, Shenzhen, Guangdong 518083, P. R. China

solution will allow us to perform complex plastic degradation processes under common environmental conditions without the need for activation of complex processes. Insects with their symbiont communities of microorganisms can have a great technical and even economic repercussion in the management of plastic waste.
In this case, insects are the best biotechnological alternative to carry and deposit specific communities of microorganisms that generate films on plastics, thus initiating their biodegradation through contact and reaction.
Insects by nature generate in their own bodies and adhere to the eggs all the microbiome necessary for their use in the following morphological phase components of their usual life cycles. Insects are the best-prepared beings to use in symbiosis both endogenous and acquired microorganisms. That is why we have chosen them without the possibility of an alternative.

In order to ensure the best Ethical practice according to the European regulations, an Ethics **WP6** was included in the project work plan **WP6. Safety, Environmental, social and economic sustainability.** It is aimed at taking into account the abovementioned Ethical issues in all their shades thanks to the expert guidance of an external independent Ethics Advisor.

5.2 Security[157]

Please indicate if your project will involve:

- Activities or results raising security issues: (YES/NO): **NO**.
- 'EU-classified information' as background or results: (YES/NO): **NO**.

[157] See article 37 of the Model Grant Agreement. . For more information on the classification of Information, please refer to the Horizon 2020 guidance: https://ec.europa.eu/research/participants/data/ref/h2020/other/hi/secur/h2020-hi-guide-classif_en.pdf.

Annex: Letters of support from RECOVER advisory board

Summary of initially appointed AB members and respective roles:

Name (Short name)	Type	Country	Role
Remedia	Association	Spain	Advisoring in environmental impact related to GHG mitigation and carbon sink in agriculture of RECOVER solutions
Biocampo	SME	Spain	Provide requirement for organic fertilizers. Act as potential adopter of the solution.
Compostingreen (CinG)	SME	Spain	Complement the consortium with additional emphasis on biofertilisers, compost and vermicompost.
Carbiolice ***(affiliate of Carbios)***	SME	France	Provide insights on existing solutions for biodegrading different types of plastics they have recently brought to the market. Create synergies for exploitation, e.g. through common advocating for new standards supporting the EoL routes pursued in the project.
SOTRAFA S.A.	Large	Spain	Provide mulching films to valorize in the project and test the application of chitosan on active mulching films
METGEN	SME, BIC member	Finland	Provide complementary enzymes samples vs. those available at ASA. Be part of the future biotech value chain.
Repsol	Large	Spain	Provide requirements and plastic samples as large petrochemical company.
AlgaeEnergy	Large	Spain	Study the ability of coupling the biodegradation of APW with microalgae culture to reach a carbon neutral cascade process.
Fomento de Costrucciones y Contratas (FCC)	Large	Spain	Interest in learning more on the Recover process to implement it is the largest MSW management sites in Spain.
Centro Lombricoltura Toscano (CLT)	Association	Italy	Interest in providing earthworms knowledge and exploitation in vermicompost
PackagingCluster (PackCluster)	Association	Spain	Interest in providing insights to heighten the sustainability of the plastic packaging sector which is at stake in the public opinion and in joint dissemination
International Solid Waste Association (ISWA)	Association	Global (based in NL)	Provide requirements on feasible waste treatments approaches on a global basis, as well as access to many potential licensees of the RECOVER process
COMITE DES ORGANISATIONS PROFESSIONNELLES AGRICOLE DE L UNION EUROPEENNE COPA ASSOCIATION DE FAIT – COPA-COGECA	Association	EU	To disseminate the project results to the widest network of farmers in EU

As discussed in section 1, these AB members will be consulted along the project depending on the needs. Further members could be appointed especially in order to extend the RECOVER network throughout Europe.

remedia
RED CIENTÍFICA DE MITIGACIÓN DE EMISIONES DE GASES DE EFECTO INVERNADERO EN EL SECTOR AGROFORESTAL

To the attention of the Coordinator:
ASAJA, Jose Sanchez
Date: September 4th, 2019

Re: Letter of Support for "Recover: Developing innovative biotic symbiosis for plastic biodegradation and synthesis to solve their end of life challenges in the agriculture and food industries."
Call: BBI-2019-S02-R3

Letter of Support

There are huge challenges with the currently available end of life treatment of the plastics used in the agriculture and food industries. Recover will help providing novel biotechnological solutions applying microorganisms, enzymes and insects to degrade conventional plastic packaging and agricultural films waste streams but also result in new feedstock for the bio-based industries. The process will be optimized and performed either ex-situ in composting reactors (preferred route for 11011-recyclable plastics from municipal solid waste or for agricultural films that can be collected easily) or in-situ in the case of bioremediating soil pollution, by e.g. mulching films, further to contributing to the long-term removal of non-biodegradable polymers from the environment. In the former case, the project will also deliver chitosan extracted from the vectorized insects after their use, which can be turned as a new source of bioplastics used in our target sectors, as well as in new fertilizers valorizing in cascade any by-products of the process.

Red REMEDIA herein confirms its interest in joining the Advisory Board that will be established under the Recover project. Based on our knowledge and market experience, we can contribute to the project and we are interested to follow the relevant project results (remotely and joining selected project meetings) in conditions that will be agreed in a Non Disclosure Agreement.

We strongly believe in the positive impact of this project and we are also willing to inform our network about the existence and outcomes of the project and support other dissemination activities.

Signed at Valencia, September 4th, 2019,
Name: Salvador Calvet Sanz

biocampo

BIO CAMPO S.L.U.
C/SAN LUIS, 46
30730. SAN JAVIER. MURCIA

To the attention of the Coordinator:
ASAJA, Jose Sanchez

Re: Letter of Support for **"Recover: Developing innovative biotic symbiosis for plastic biodegradation and synthesis to solve their end of life challenges in the agriculture and food industries."**
Call: BBI-201 9-S02-R3

Letter of Support

There are huge challenges with the currently available end of life treatment of the plastics used in the agriculture and food industries. Recover will help providing novel biotechnological solutions applying microorganisms, enzymes and insects to degrade conventional plastic packaging and agricultural films waste streams but also result in new feedstock for the bio-based industries. The process will be optimized and performed either ex-situ in composting reactors (preferred route for 11011-recyclable plastics from municipal solid waste or for agricultural films that can be collected easily) or i n-situ i n the case of bioremediating soil pollution, by e.g. mulching films, further to contributing to the long-term removal of non-biodegradable polymers from the environment. In the former case, the project will also deliver chitosan extracted from the vectorized insects after their use, which can be turned as a new source of bioplastics used in our target sectors, as well as in new fertilizers valorizing in cascade any by-products of the process.

BIO CAMPO as one of the Spanish leader organic farming producers confirms its interest in joining the Advisory Board that will be established under the Recover project. Based on our knowledge and market experience, we can contribute to the project and we are interested to follow the relevant project results (remotely and joining selected project meetings) in conditions that will be agreed in a Non Disclosure Agreement.

We strongly believe in the positive impact of this project and we are also willing to inform our network about the existence and outcomes of the project and support other dissemination activities.

Signed at Pilar de la Horadada (Spain), 26-07-2019,
Name: David Samper Martinez, CEO

BIO CAMPO, S.L.U.
C.I.F.: B-30692339
Ind. Los Urreas - C/. San Luis, 46 - Bajo
Tl. 968 33 84 10 - Fax: 968 10 27 91
30730 SAN JAVIER (MURCIA)

BIO CAMPO S.L.U.
CIF: B-30692339
C/ San Luis, 46 Poligoro Industrial Los Urreas
30730 San Javier. Murcia
E-mail: info@biocampo.es
Teléfono: 968 335 410

To the attention of the Coordinator:
ASA JA, Jose Sanchez

Re: Letter of Support for "**Recover: Developing innovative biotic symbiosis for plastic biodegradation and synthesis to solve their end of life challenges in the agriculture and food industries.**"
Call: BBI-201 9-S02-R3

Letter of Support

There are huge challenges with the currently available end of life treatment of the plastics used in the agriculture and food industries. Recover will help providing novel biotechnological solutions applying microorganisms, enzymes and insects to degrade conventional plastic packaging and agricultural films waste streams but also result in new feedstock for the bio-based industries. The process will be optimized and performed either ex-situ in composting reactors (preferred route for 11011-recyclable plastics from municipal solid waste or for agricultural films that can be collected easily) or i n-situ i n the case of bioremediating soil pollution, by e.g. mulching films, further to contributing to the long-term removal of non-biodegradable polymers from the environment. In the former case, the project will also deliver chitosan extracted from the vectorized insects after their use, which can be turned as a new source of bioplastics used in our target sectors, as well as in new fertilizers valorizing in cascade any by-products of the process.

COMPOSTINGREEN S.L. herein confirms its interest in joining the Advisory Board that will be established under the Recover project. Based on our knowledge and market experience, we can contribute to the project and we are interested to follow the relevant project results (remotely and joining selected project meetings) in conditions that will be agreed in a Non Disclosure Agreement.

We strongly believe in the positive impact of this project and we are also willing to inform our network about the existence and outcomes of the project and support other dissemination activities.

Signed at Orihuela (Spain), 26-07-201 9,
Name: Mauricio Ivan Martinez Fernandez, CEO

Compostingreen, S.L.
CIF: B-54703749
Parque Científico y Empresarial Univ. Miguel Hernández de Elche. Avda. de la Universidad s/n
Edificio Quorum III, 03202 Elche, Alicante
E-mail: ventas@compostingreen.com
Teléfono: 650029618 Sitio web: www.compostingreen.com

CARBIOLICE
Biodegradability is the future

To the attention of the Coordinator:
ASAJA, Jose Sanchez
3th of September 2019

Re: Letter of Support for "Recover: Developing innovative biotic symbiosis for plastic biodegradation and synthesis to solve their end of life challenges in the agriculture and food industries."
Call: BBI-2019-SO2-R3

Letter of Intent

There are huge challenges with the currently available end of life treatment of the plastics used in the agriculture and food industries. Recover will help providing novel biotechnological solutions applying microorganisms, enzymes and insects to degrade conventional plastic packaging and agricultural films waste streams but also result in new feedstocks for the bio-based industries. The process will be optimized and performed either ex-situ in composting reactors (preferred route for non-recyclable plastics from municipal solid waste or for agricultural films that can be collected easily) or in-situ in the case of bioremediating soil pollution, by e.g. mulching films, further to contributing to the long-term removal of non-biodegradable polymers from the environment. In the former case, the project will also deliver chitosan extracted from the vectorized insects after their use which can in turn be a new source of bioplastics used in our target sectors, as well as in new fertilizers valorizing in cascade any by-products of the process.

CARBIOLICE herein confirms its interest in joining the Advisory Board that will be established under the Recover project especially in terms of participating in joint clustering initiative. We also have a special interest in collaborating in the dissemination activities, definition of market requirements. Based on our knowledge and market experience, we can contribute to the project, and we are interested to follow the relevant project results (remotely and joining selected project meetings) in conditions that will be agreed in a Non Disclosure Agreement.

We strongly believe in the positive impact of this project and we are also willing to inform our network about the existence and outcomes of the project and support other dissemination activities.

Nadia AUCLAIR,
CEO of CARBIOLICE
Signed at RIOM on 3/09/2019

MetGen Oy
VAT: FI 2075584-9
Rakentajantie 26
20780 Kaarina, Finland
www.metgen.com

To the attention of the Coordinator:
ASAJA, Jose Sanchez
Date: 20.8.2019

Re: Letter of Support for "Recover: Developing innovative biotic symbiosis for plastic biodegradation and synthesis to solve their end of life challenges in the agriculture and food industries."
Call: BBI-2019-SO2-R3

Letter of Intent

There are huge challenges with the currently available end of life treatment of the plastics used in the agriculture and food industries. Recover will help providing novel biotechnological solutions applying microorganisms, enzymes and insects to degrade conventional plastic packaging and agricultural films waste streams but also result in new feedstocks for the bio-based industries. The process will be optimized and performed either ex-situ in composting reactors (preferred route for non-recyclable plastics from municipal solid waste or for agricultural films that can be collected easily) or in-situ in the case of bioremediating soil pollution, by e.g. mulching films, further to contributing to the long-term removal of non-biodegradable polymers from the environment. In the former case, the project will also deliver chitosan extracted from the vectorized insects after their use which can in turn be a new source of bioplastics used in our target sectors, as well as in new fertilizers valorizing in cascade any by-products of the process.

MetGen Oy herein confirms its tentative interest in joining the Advisory Board that will be established under the Recover project especially in terms of participating in joint clustering initiative. We also have a special interest in collaborating in the dissemination activities, definition of market requirements. Based on our knowledge and market experience, we can contribute to the project, and we are interested to follow the relevant project results (remotely and joining selected project meetings) in conditions that will be agreed in a Non Disclosure Agreement.

We strongly believe in the positive impact of this project and we are also willing to inform our network about the existence and outcomes of the project and support other dissemination activities.

Signed at **Kaarina** on **20th of August, 2019**,
Name: **Alex Michine, Chief Executive Officer**
Signature:

METGEN
The Art of Biocatalysis

REPSOL

To the attention of the Coordinator:
ASAJA, Jose Sanchez
Date: 02-09-2019

Re: Letter of Support for "Recover: Developing innovative biotic symbiosis for plastic biodegradation and synthesis to solve their end of life challenges in the agriculture and food industries."
Call: BBI-2019-SO2-R3

Letter of Intent

There are huge challenges with the currently available end of life treatment of the plastics used in the agriculture and food industries. Recover will help providing novel biotechnological solutions applying microorganisms, enzymes and insects to degrade conventional plastic packaging and agricultural films waste streams but also result in new feedstocks for the bio-based industries. The process will be optimized and performed either ex-situ in composting reactors (preferred route for non-recyclable plastics from municipal solid waste or for agricultural films that can be collected easily) or in-situ in the case of bioremediating soil pollution, by e.g. mulching films, further to contributing to the long-term removal of non-biodegradable polymers from the environment. In the former case, the project will also deliver chitosan extracted from the vectorized insects after their use which can in turn be a new source of bioplastics used in our target sectors, as well as in new fertilizers valorizing in cascade any by-products of the process.

Repsol SA herein confirms its interest in joining the Advisory Board that will be established under the Recover project especially in terms of participating in joint clustering initiative. The role would be mainly theoretical guidance and provision of raw materials, market requirements, materials specifications, and possibly a role in the value chain of enzymes exploitation. Based on our knowledge and market experience, we can contribute to the project, and we are interested to follow the relevant project results (remotely and joining selected project meetings) in conditions that will be agreed in a Non Disclosure Agreement.

No legally binding obligations on the parties will be created, implied, or inferred by mere signature of the present document. Any eventual commitment to financial or technical contributions to the project will have to be based on a future Agreement to be agreed in separate document between the involved parties.

I hereby sign this Letter of Intent,

Signed in Madrid, on September 3, 2019
Tomas Malango, D Experimentation and Technology Support
Corp. Div. Technology and Corporate Venturing

ALGAENERGY

To the attention of the Coordinator:
ASAJA, Jose Sanchez
Date: 2th Sept 2019

Re: Letter of Support for "RECOVER: Developing innovative biotic symbiosis for plastic biodegradation and synthesis to solve their end of life challenges in the agriculture and food industries."

Call: BBI-2019-SO2-R3

Letter of Intent

There are huge challenges with the currently available end of life treatment of the plastics used in the agriculture and food industries. Recover will help providing novel biotechnological solutions applying microorganisms, enzymes and insects to degrade conventional plastic packaging and agricultural films waste streams but also result in new feedstocks for the bio-based industries. The process will be optimized and performed either ex-situ in composting reactors (preferred route for non-recyclable plastics from municipal solid waste or for agricultural films that can be collected easily) or in-situ in the case of bio-remediating soil pollution, by e.g. mulching films, further to contributing to the long-term removal of non-biodegradable polymers from the environment. In the former case, the project will also deliver chitosan extracted from the vectored insects after their use which can in turn be a new source of bioplastics used in our target sectors, as well as in new fertilizers valorizing in cascade any by-products of the process.

ALGAENERGY, S.A. (BIC MEMBER associate) herein confirms its interest in joining the Advisory Board that will be established under the RECOVER project specially in terms of participating in joint clustering initiative and further bio-based projects in which micro-algae could complete one or more parts of the development of technical areas. We also have a special interest in advising in the CO2 producing and fixing, biomass production, valorization procedures, dissemination activities and the definition of market requirements. Based on our knowledge and market experience, we can contribute to the project, and we are interested to follow the relevant project results (remotely and joining selected project meetings) in conditions that will be agreed in a Non- Disclosure Agreement.

We strongly believe in the positive impact of this project and we are also willing to inform our network about the existence and outcomes of the project and support other dissemination activities.

ALGAENERGY
Avda. de Europa nº 19
Parque Empresarial de "La Moraleja"
28108 Alcobendas (Madrid)
Tel: (34) 91 490 20 20 - Fax: (34) 91 490 47 94
www.algaenergy.es

Signed at Alcobendas, Madrid, 4th September 2019

Augusto Rodríguez-Villa Matons, President, AlgaEnergy

PACKAGING CLUSTER

To the attention of the Coordinator:
ASAJA, Jose Sanchez
Date: 26th July 2019

Re: Letter of Support for "Recover: Developing innovative biotic symbiosis for plastic biodegradation and synthesis to solve their end of life challenges in the agriculture and food industries."
Call: BBI-2019-SO2-R3

Letter of Intent

There are huge challenges with the currently available end of life treatment of the plastics used in the agriculture and food industries. Recover will help providing novel biotechnological solutions applying microorganisms, enzymes and insects to degrade conventional plastic packaging and agricultural films waste streams but also result in new feedstocks for the bio-based industries. The process will be optimized and performed either ex-situ in composting reactors (preferred route for non-recyclable plastics from municipal solid waste or for agricultural films that can be collected easily) or in-situ in the case of bioremediating soil pollution, by e.g. mulching films, further to contributing to the long-term removal of non-biodegradable polymers from the environment. In the former case, the project will also deliver chitosan extracted from the vectorized insects after their use which can in turn be a new source of bioplastics used in our target sectors, as well as in new fertilizers valorizing in cascade any by-products of the process.

PACKAGING CLUSTER herein confirms its interest in joining the Advisory Board that will be established under the Recover project especially in terms of participating in joint clustering initiative. We also have a special interest in collaborating in the dissemination activities, definition of market requirements. Based on our knowledge and market experience, we can contribute to the project, and we are interested to follow the relevant project results (remotely and joining selected project meetings) in conditions that will be agreed in a Non Disclosure Agreement.

We strongly believe in the positive impact of this project and we are also willing to inform our network about the existence and outcomes of the project and support other dissemination activities.

Signed at Terrassa (Barcelona) on 26th July 2019,
Name: Artur Costa, President

33896504R ARTUR COSTA (R: G65920035)

Firmado digitalmente por 33896504R ARTUR COSTA (R: G65920035)
Fecha: 2019.07.26 08:17:53 +02'00'

FCC FOMENTO DE CONSTRUCCIONES Y CONTRATAS, S.A.

To the attention of the Coordinator:
ASAJA, Jose Sanchez
Date: 26-07-2019

Re: Letter of Support for "Recover: Developing innovative biotic symbiosis for plastic biodegradation and synthesis to solve their end of life challenges in the agriculture and food industries."
Call: BBI-2019-SO2-R3

Letter of Support

There are huge challenges with the currently available end of life treatment of the plastics used in the agriculture and food industries. Recover will help providing novel biotechnological solutions applying microorganisms, enzymes and insects to degrade conventional plastic packaging and agricultural films waste streams but also result in new feedstocks for the bio-based industries. The process will be optimized and performed either ex-situ in composting reactors (preferred route for non-recyclable plastics from municipal solid waste or for agricultural films that can be collected easily) or in-situ in the case of bioremediating soil pollution, by e.g. mulching films, further to contributing to the long-term removal of non-biodegradable polymers from the environment. In the former case, the project will also deliver chitosan extracted from the vectorized insects after their use which can in turn be a new source of bioplastics used in our target sectors, as well as in new fertilizers valorizing in cascade any by-products of the process.

FOMENTO DE CONSTRUCCIONES Y CONTRATAS S.A. herein confirms its interest in joining the Advisory Board that will be established under the Recover project. Based on our knowledge and market experience, we can contribute to the project, and we are interested to follow the relevant project results (remotely and joining selected project meetings) in conditions that will be agreed in a Non Disclosure Agreement.

We strongly believe in the positive impact of this project and we are also willing to inform our network about the existence and outcomes of the project and support other dissemination activities.

Signed at Madrid on 26-07-2019,
Name: Sylvain Cortes.

FCC FOMENTO DE CONTRUCCIONES Y CONTRATAS S.A.

CENTRO LOMBRICOLTURA TOSCANO

To the attention of the Coordinator:
ASAJA, Jose Sanchez
Date: 03/09/2019

Re: Letter of Support for "Recover: Developing innovative biotic symbiosis for plastic biodegradation and synthesis to solve their end of life challenges in the agriculture and food industries."
Call: BBI-2019-SO2-R3

Letter of Intent

There are huge challenges with the currently available end of life treatment of the plastics used in the agriculture and food industries. Recover will help providing novel biotechnological solutions applying microorganisms, enzymes and insects to degrade conventional plastic packaging and agricultural films waste streams but also result in new feedstocks for the bio-based industries. The process will be optimized and performed either ex-situ in composting reactors (preferred route for non-recyclable plastics from municipal solid waste or for agricultural films that can be collected easily) or in-situ in the case of bioremediating soil pollution, by e.g. mulching films, further to contributing to the long-term removal of non-biodegradable polymers from the environment. In the former case, the project will also deliver chitosan extracted from the vectorized insects after their use which can in turn be a new source of bioplastics used in our target sectors, as well as in new fertilizers valorizing in cascade any by-products of the process.

Centro Lombricoltura Toscano herein confirms its interest in joining the Advisory Board that will be established under the Recover project especially in terms of participating in joint clustering initiative. We also have a special interest in collaborating in the dissemination activities, definition of market requirements. Based on our knowledge and market experience, we can contribute to the project, and we are interested to follow the relevant project results (remotely and joining selected project meetings) in conditions that will be agreed in a Non Disclosure Agreement.

We strongly believe in the positive impact of this project and we are also willing to inform our network about the existence and outcomes of the project and support other dissemination activities.

Signed at **San Giuliano Terme (Pisa)** on **03/09/2019**,
Name: **Marco Calcaprina – Commercial Mangager**
Signature

Centro Lombricoltura Toscano soc agr.semplice
Sede legale: via De Amicis, 114/A 56017 San Giuliano Terme (Pisa)
Tel 050 7916631 P.IVA e CF 02109680500 codice sdi KRRH6B9
Mail: info@lombricolturaclt.it clt@pec.agritel.it Web: www.lombricolturaclt.it

To the attention of the Coordinator:

ASAJA, Jose Sanchez

03 September 2019

Regarding: Letter of Support for "Recover: Developing innovative biotic symbiosis for plastic biodegradation and synthesis to solve their end of life challenges in the agriculture and food industries."

Call: BBI-2019-SO2-R3

Letter of Intent

There are huge challenges with the currently available end of life treatment of the plastics used in the agriculture and food industries. Recover will help providing novel biotechnological solutions applying microorganisms, enzymes and insects to degrade conventional plastic packaging and agricultural films waste streams but also result in new feedstocks for the bio-based industries. The process will be optimized and performed either ex-situ in composting reactors (preferred route for non-recyclable plastics from municipal solid waste or for agricultural films that can be collected easily) or in-situ in the case of bioremediating soil pollution, by e.g. mulching films, further to contributing to the long-term removal of non-biodegradable polymers from the environment. In the former case, the project will also deliver chitosan extracted from the vectorized insects after their use which can in turn be a new source of bioplastics used in our target sectors, as well as in new fertilizers valorizing in cascade any by-products of the process.

ISWA (International Solid Waste Association) herein confirms its interest in joining the Advisory Board that will be established under the Recover project especially in terms of participating in joint clustering initiative. We also have a special interest in collaborating in the dissemination activities as well as the definition of market requirements serving the purpose of reduced plastics pollution. Based on our knowledge and market experience, we can contribute to the project, and we are interested to follow the relevant project results (remotely and joining selected project meetings) in conditions that will be agreed in a Non-Disclosure Agreement.

We strongly believe in the positive impact of this project and we are also willing to inform our network about the existence and outcomes of the project and support other dissemination activities.

Signed at **Vienna** on **03rd of September 2019**,

Name: **Arne Ragossnig, Managing Director**

Signature:

International Solid Waste Association
ISWA General Secretariat
Auerspergstrasse 15/41
A-1080 Vienna, Austria
www.iswa.org

sotrafa
Transformación de Materias Plásticas
Global Solutions

To the attention of the Coordinator:
UAL, Maria Jose Lopez
11/03/2020

Re: Letter of Support for "Recover: Development of innovative biotic symbiosis for plastic biodegradation and synthesis to solve their end of life challenges in the agriculture and food industries."
Call: BBI-2019-SO2-R3

Letter of Intent

There are huge challenges with the currently available end of life treatment of the plastics used in the agriculture and food industries. Recover will help providing novel biotechnological solutions applying microorganisms, enzymes and insects to degrade conventional plastic packaging and agricultural films waste streams but also result in new feedstocks for the bio-based industries. The process will be optimized and performed either ex-situ in composting reactors (preferred route for non-recyclable plastics from municipal solid waste or for agricultural films that can be collected easily) or in-situ in the case of bioremediating soil pollution, by e.g. mulching films, further to contributing to the long-term removal of non-biodegradable polymers from the environment. In the former case, the project will also deliver chitosan extracted from the vectorized insects after their use which can in turn be a new source of bioplastics used in our target sectors, as well as in new fertilizers valorizing in cascade any by-products of the process.

Sotrafa herein confirms its interest in joining the Advisory Board that will be established under the Recover project especially in terms of participating in joint clustering initiative. We will also provide samples of mulching films to valorise in the project, as well as test the application of chitosan on active mulching films for the partners to validate the effect on the growth on crops. Based on our knowledge and market experience, we can contribute to the project, and we are interested to follow the relevant project results (remotely and joining selected project meetings) in conditions that will be agreed in a Non Disclosure Agreement.

We strongly believe in the positive impact of this project and we are also willing to inform our network about the existence and outcomes of the project and support other dissemination activities.

Almeria 11/03/2020 ,
Carlos López García
R&D, Quality and Environment Manager
Signature

SOTRAFA, S. A.
A-28705846
Paraje Cartabona, nº 12
SANTA MARÍA DEL ÁGUILA
04710 - El Ejido (Almería)

copa*cogeca
european farmers european agri-cooperatives

CC(20)438:1 –BM/va

Brussels, 23 January 2020

For the attention of the RECOVER Project Coordinator

Dear Mr Jose-Fernando Sánchez,

Due to a miscommunication regarding Copa and Cogeca's involvement in the RECOVER project, we would like to ask you to withdraw our organization from the consortium. This includes relinquishing all responsibility of tasks and financial allocations relating to the project.

Copa and Cogeca recognize the important potential of the RECOVER project to our sector developing innovative biotic symbiosis between microorganisms and insects for the Agro-food Plastic Wastes biodegradation from our soils. This is particularly important for the European agricultural and food industries that cannot avoid the use of plastics in order to satisfy market requirements. This project will introduce new ways of circularity in the use and management of plastics materials used for agricultural practices and food packaging, fostering the sectorial competitiveness as well as reducing the environmental impact of this material.

In addition, we commit to be part of the Advisory Board of the RECOVER project and will communicate and disseminate the project's objectives and results through our internal communications channel.

Yours sincerely,

Fernando Castillo
Administrative Director

Copa - Cogeca | European Farmers European Agri-Cooperatives
61, Rue de Trèves | B - 1040 Bruxelles | www.copa-cogeca.eu
EU Transparency Register Number | Copa 44856881231-49 | Cogeca 09586631237-74